



UTM
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System Documentation (SD)

Project (Version 4)

SELF HELP SYSTEM

MINDWELL ORGANIZATION

Date : 16/1/2024

Faculty of Computing

Prepared by: <Team 8>

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Revision Page

a. **Overview**

The current version of the Software Design (SD) focuses on the development of a Self-Help System designed to support mental health. The system includes an AI-powered chatbot for self-diagnosis, gamification for user engagement, personalized toolkits, and progress tracking. The design incorporates functional and non-functional requirements, emphasizing usability, data privacy, and accessibility. The document details the Agile development process, including requirements gathering, design, development, testing, and deployment.

b. **Target Audience**

- Primary Users: Individuals seeking accessible mental health support tools for self-diagnosis, self-help, and progress tracking.
- Professionals: Psychologists and mental health consultants who may provide additional guidance through the platform to share their knowledge.
- Stakeholders: Developers, designers, and organizations aiming to create or fund mental health solutions.

c. **Project Team Members**

Member Name	Role	Task	Status
Nurul Athirah Syafiqah binti Mohd Razali	Recorder	Record the results of discussion.	Complete
Koo Xuan	Moderator	Monitor group progress.	Complete
Muhammad Hafiz Bin Mohd Shaharuddin	Accuracy Checker & Reporter	Check and compile the assignment.	Complete
Phavanee Katriya Phon - Amnuaisuk	Skeptic	Give queries and comment on the assignment.	Complete

d. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Nurul Athirah Syafiqah binti Mohd Razali	Completed A1 initial project with introduction, existing systems, proposed system, software process model, project schedule and references.	07/11/2024

1.0	Koo Xuan	Completed updated project A1 with corrections, NABC, existing systems and chosen process model.	09/11/2024
2.0	Muhammad Hafiz	Completed initial project A2 with system features, use cases and use case diagram.	21/11/2024
2.0	Phavanee Katriya Phon-Amnuaisuk	Completed A2 project with corrections and chapter 2: specific requirements.	14/12/2024
3.0	Nurul Athirah Syafiqah binti Mohd Razali	Completed initial project A3 with initial 4+1 views.	23/12/2024
3.0	Koo Xuan	Updated A3 with chapter 3, 4, 5, 6 and 7.	3/1/2025
4.0	Muhammad Hafiz	Completed project A4 with chapter 8 and 9.	16/1/2025

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1. Introduction

1.1 Purpose

This System Documentation (SD) describes the specific requirements, system architectural design, components, data design, user interface design, requirements matrix, test cases and appendices of the MindWell Self-Help System. The primary purpose of this document is to delineate the system's objectives which provide a comprehensive documentation of the design, architecture, and functionalities of the system, supporting accessible mental health management through modular, user-centered features. Besides, it facilitates effective communication to ensure clarity and consistency among all project stakeholders, including developers, designers, and end-user representatives. It also helps guide system implementation, ensuring alignment with the project's scope and objectives.

The intended audience for the SD includes the development team to ensure a clear understanding of the system specifications, design architecture, and feature requirements for effective implementation. Besides, stakeholders include mental health organizations, investors, and product owners, to provide insight into the project's scope, goals, and potential impact. End-user representatives such as psychologists, mental health consultants, and advocacy groups are also included to validate the system's usability, accessibility, and alignment with user needs and industry standards.

This document also serves as a reference for future enhancements and maintenance, ensuring the system remains adaptable to evolving requirements while addressing its core objectives of promoting accessible mental health management.

1.2 Scope

The software product is MindWell Self-Help System which is designed to support mental health management through a user-friendly, features-rich platform. This system aims to provide accessible, cost effective tools for tailored self-help resources.

The system will provide personalized self-help by providing knowledge units and recommending them to relevant users. The knowledge units aim to increase the mental health literacy of users. Besides that, the app will also provide users with a customizable toolbox of techniques, including breathing exercises, guided journaling and additional resources. These techniques can be added to their toolbox after being exposed to the knowledge units available. Moreover, there will be user engagement through gamification in the system which promotes regular interaction by implementing a system of points, badges, and personalized avatars. Last but not least, the system's secure and private user experience ensures robust data privacy and compliance with regulations to protect sensitive user information. However, the system will not replace professional mental health services or clinical diagnoses. The system will also not serve as an alternative to emergency services or crisis intervention.

Through this MindWell Self-Help System, It empowers users with tools for self-assessment and skill-building. Besides, it encourages participation and commitment to mental health practices by incorporating gamified features. The system also enhances user confidence by providing accessible, non-intrusive mental health support.

1.3 Definitions, Acronyms and Abbreviation

AI: Artificial Intelligence. The technology employed in the MindWell system is designed to analyze user inputs, perform self-diagnosis, and provide recommendations for users.

CBI: Cognitive Behavior Inventory. A tool used to evaluate cognitive and behavioral patterns associated with mental health conditions.

DASS: Depression, Anxiety, and Stress Scales. A diagnostic tool used for assessing the severity of depression, anxiety, and stress.

MH: Mental Health. Relates to the emotional, psychological, and social well-being of an individual.

PHQ-9: Patient Health Questionnaire-9. A widely-used diagnostic tool for assessing the severity of depression.

WHODAS: World Health Organization Disability Assessment Schedule. A tool that assesses health and disability across multiple domains of functioning.

1.4 References

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1.5 Overview

This System Documentation (SD) offers a comprehensive overview of the design, functionalities, and operational structure of the MindWell Self-Help & Diagnostic System. It is organized to guide all stakeholders, including developers, designers, and mental health professionals, in understanding and implementing the system effectively.

The Software Design (SD) document is structured into several key sections: Introduction, specific requirements, system architectural design, detailed component descriptions, data design, user interface design, requirements matrix, test cases, and appendices. The Introduction serves to establish the document's purpose, scope, essential definitions, references, and a general overview of the SD. Following this, the specific requirements section enumerates the detailed features of the system, including user personas, user stories, and constraints related to performance and design. The system architectural design section articulates the selected architectural style, the rationale behind it, and the component model. The detailed description of components includes a comprehensive package diagram along with in-depth explanations of each component and subsystem. The data design section delineates the data structure, featuring a data dictionary and descriptions of significant data elements. The user interface design section provides an overview of the user interface, enhanced by screen images and interaction flows. The requirements matrix ensures traceability by mapping the system's requirements to its objectives. The test cases section outlines the testing methodology, detailing specific test cases for each subsystem. Finally, the appendices contain supplementary information, such as a traceability matrix and other pertinent documentation.

The Software Documentation (SD) is structured in a logical sequence to facilitate easy navigation and comprehension for all stakeholders. Each section builds upon the previous one, starting with high-level concepts and progressively delving into detailed technical descriptions. Subsections within each chapter provide specific details to support the main topic, ensuring clarity and thorough coverage of all aspects of the system. Visual elements, including diagrams and tables, are incorporated to supplement the written content and improve comprehension. The structured arrangement guarantees that the SD functions as a significant resource during both the development and ongoing maintenance of the MindWell system.

2. Specific Requirements

This section outlines all the key features and functions of the system in detail. It describes what the system does, how it responds to user actions, and the inputs and outputs involved. The goal is to ensure that everyone from developers to testers has a clear understanding of what the system needs to deliver and how it will behave. Each requirement is presented in a way that's easy to follow and focused on real-world use.

1.1 Persona

The system is designed to serve three main types of users, each with unique needs and goals. University students, who are often tech-savvy but face academic stress, use the system to access mental health resources and practical strategies to manage their challenges. Young professionals, dealing with burnout and emotional strain, rely on the system for quick, personalized tools and exercises to manage their well-being privately and conveniently. Lastly, mental health advocates or administrators focus on creating, managing, and improving content to ensure the resources are accurate, relevant, and beneficial for the broader community. By understanding these user types, the system delivers tailored features that meet their specific needs, making it accessible, effective, and user-friendly for everyone.

2.1.1 Persona 1 (University Student /Actor 1)

User Need

Amira is a psychology student struggling with academic stress. She wants to learn practical ways to manage her mental health and apply those techniques to cope better with challenges.

User Stories

As a university student, I want to browse, bookmark, and rate knowledge units, so I can revisit helpful strategies during stressful times and contribute to the app's recommendations.

Journey:

Amira registers for the app, logs in, and creates her profile (UC01, UC02, UC03). She browses through knowledge units and bookmarks helpful content about time management and stress relief (UC06, UC07). After applying the strategies, she rates the units to provide feedback for future users (UC08).

2.1.2 Persona 2 (Junior Executive / Actor 2)

User Need

Daniel is a corporate worker experiencing burnout. He's looking for personalized mental health exercises and tools to help him manage his emotional well-being privately

User Stories

As a young professional, I want to chat with the AI chatbot and customize tools in my toolbox, so I can address my mental health needs without external judgment.

Journey:

Daniel registers for the app, logs in, and sets up his profile to reflect his needs (UC01, UC02, UC03). He interacts with the AI chatbot to describe his feelings and get tailored mental health exercises (UC09). He adds suggested exercises to his toolbox, evolves them as he practices, and removes outdated ones (UC09, UC10, UC11). By using these tools, he earns points and spends them to unlock premium resources (UC12, UC13).

2.1.3 Persona 3 (Mental Health Advocate / Actor 3)

User Need

Sarah works as a counselor at an NGO and wants to use the app to create culturally relevant knowledge units, monitor diagnostic trends, and ensure the app stays valuable for her community.

User Stories

As a mental health advocate, I want to add knowledge units and resolve flagged issues, so I can ensure the app provides accurate and inclusive resources.

Journey:

Sarah logs in as an admin to add culturally relevant knowledge units about mindfulness and self-care (UC02, UC15, UC18). She resolves flagged issues from users, ensuring content accuracy (UC16). She monitors diagnostic trends to understand the mental health challenges users face and adjusts the content accordingly (UC17).

2.2 System Features

The Self-Help & Diagnosis System is a mobile-based platform designed to assist users in managing their mental health. It offers a comprehensive suite of features, including educational resources, self-help tools and gamification, accessible across both Android and iOS platforms. The system leverages a cloud-based database to ensure real-time data synchronization, enabling seamless user experiences and secure data storage.

The system features are illustrated in Figure 2.2.1 below. The detailed description of each module and functions is tabulated in Table 2.2.1.

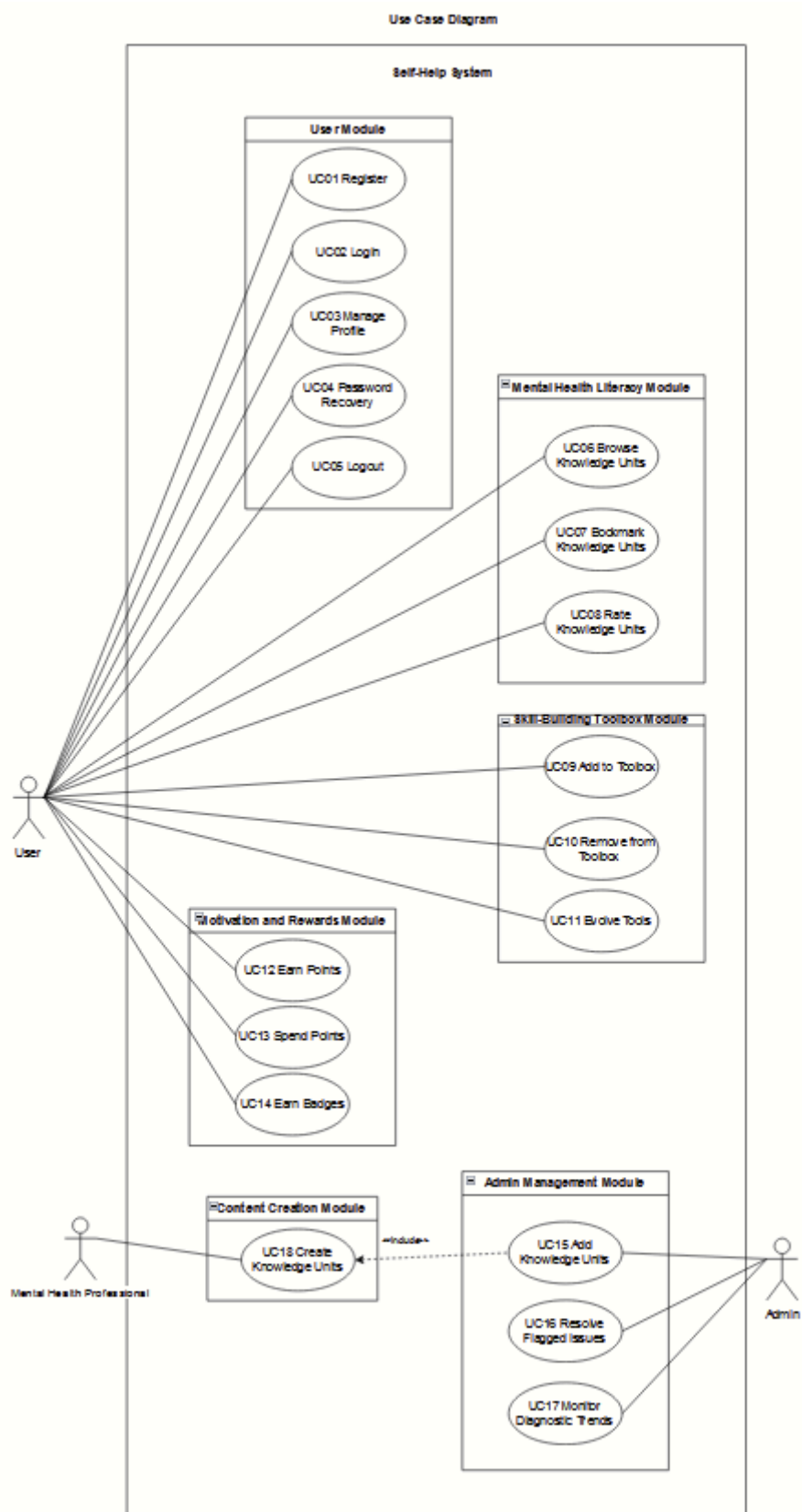


Figure 2.2.1: Use Case Diagram for Self-Help System

Link to full image: draw.io

Table 2.2.1: Description of Module and Functions for <Self Help System>

Module	Function	Description
User Management Module	UC01 - Register	This use case allows users to register an account using the mobile application.
	UC02 - Login	This use case allows users to log in to their account using the mobile application..
	UC03 - Manage Profile	This use case allows users to manage their user profile using the mobile application.
	UC04 - Password Recover	This use case allows users to recover the password to their account if they lose it through the mobile application.
	UC05 - Log Out	This use case allows users to log out of their account while using the mobile application.
Mental Health Literacy Module	UC06 - Browse Knowledge Units	This use case allows users to browse lessons created by mental health professionals on the mobile application.
	UC07 - Bookmark Knowledge Units	This use case allows users to bookmark knowledge units they find useful on the mobile app.
	UC08 - Rate Knowledge Units	This use case allows users to rate knowledge units on the mobile app.
Skill-Building Toolbox Module	UC09 - Add to Toolbox	This use case allows users to add tools that they feel are useful into their toolbox through the lesson that they take.
	UC10 - Remove from Toolbox	This use case allows users to remove tools that they do not resonate with from their toolbox.
	UC11 - Evolve Tools	This use case allows the system to introduce difficulty and variety to the tools that a patient uses.
Motivation and Rewards Module	UC12 - Earn Points	This use case allows users to check in or use the tools in their toolbox or finish a lesson to earn points.
	UC13 - Spend Points	This use case allows users to customize their avatar to spend points.
	UC14 - Earn Badges	This use case allows users to earn badges based on milestones and streaks that they achieve.
Admin Management Module	UC15 - Add Knowledge Units	This use case allows administrators to add lessons or knowledge units for users.
	UC16 - Resolve Flagged Issues	This use case allows administrators to resolve any issues related to the lessons or knowledge units.
	UC17 - Monitor Diagnostic Trends	This use case allows administrators to monitor diagnostic trends and create more relevant knowledge units.
Content Creation Module	UC18 - Create Knowledge Units	This use case allows mental health professionals to create relevant lessons or knowledge units for users.

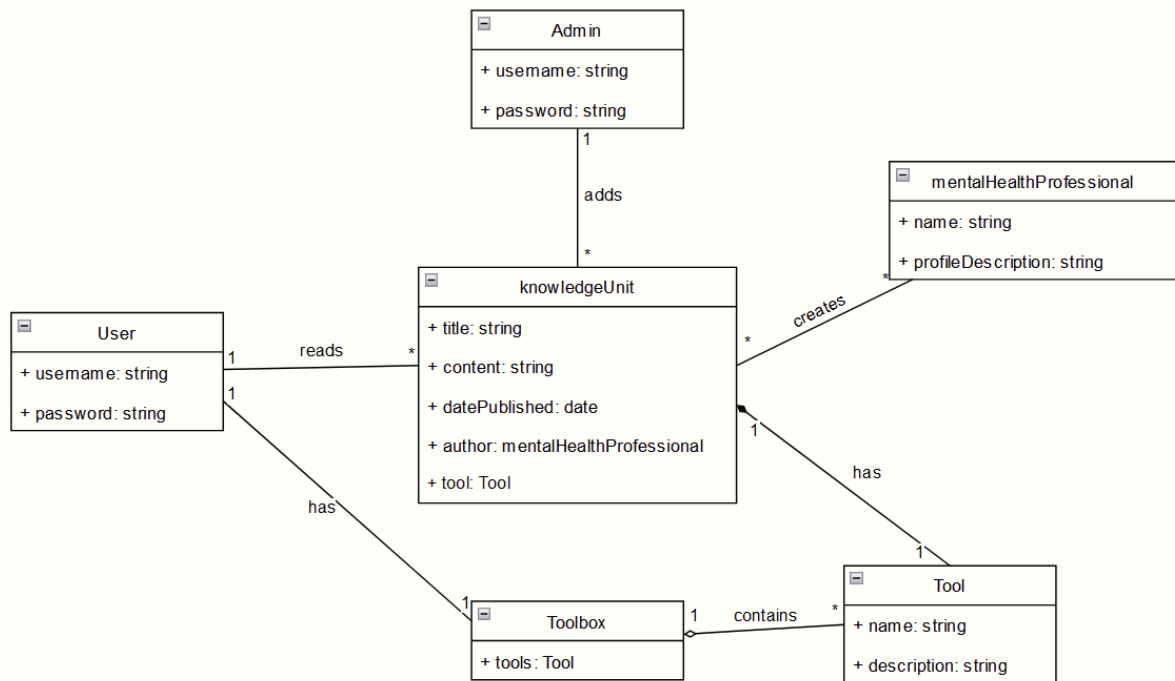


Figure 2.2.2: Domain Model for <Self Help System>

Mental Health System: Knowledge Unit Interaction

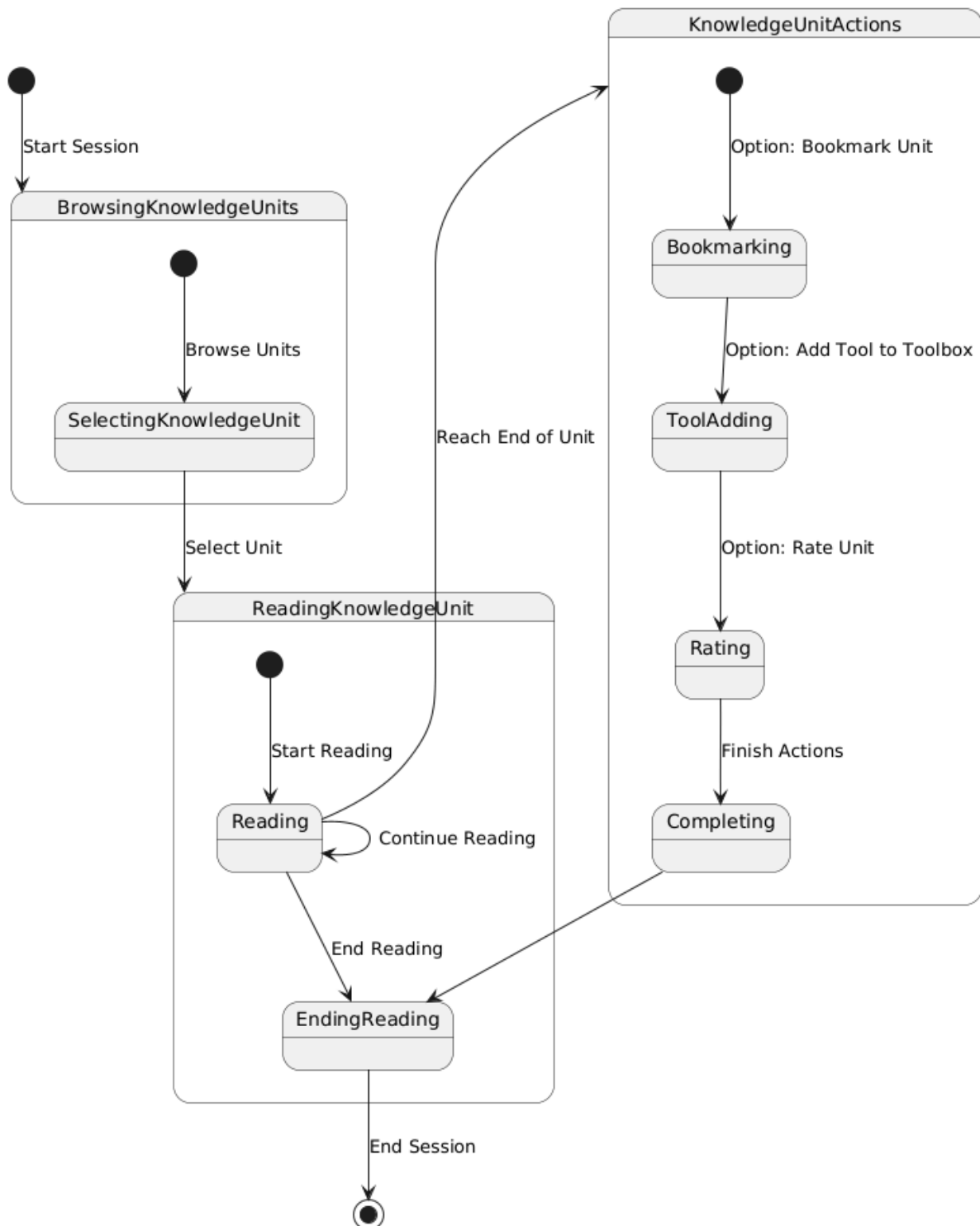


Figure 2.2.3: State Machine Diagram for Knowledge Unit Subsystem

2.3 Launch Phase

The product backlog for the system was organized into **5 key sprints** to build the MindWell Self-Help System efficiently. **Sprint 1** focused on basic user management, including registering, logging in, and profile management. **Sprint 2** introduced knowledge units where users can browse, bookmark, and rate mental health resources. In **Sprint 3**, the toolbox module was developed, allowing users to add, remove, and evolve tools for personalized mental health care. **Sprint 4** added gamification features to boost user engagement with points, badges, and rewards. Finally, **Sprint 5** implemented the admin module, enabling administrators to add knowledge units, resolve issues, and monitor diagnostic trends to keep the system relevant and accurate.

Table 2.3.1: Table showing sprint phases for <Self Help System>

Sprint	Team members assigned
<i>Sprint 1: User Management Basics</i> <i>Assigned User Story: Supports foundational user actions for all personas.</i> Use Cases: UC01: Register UC02: Login UC03: Manage Profile UC04: Password Recover UC05: Log Out	Hafiz (H), Koo Xuan (K), Athirah (A)
<i>Sprint 2: Knowledge Units</i> <i>Assigned User Story: Persona 1 (Amira): Learning strategies and applying them by browsing, bookmarking, and rating units.</i> Use Cases: UC06: Browse Knowledge Units	Hafiz (H), Phavanee (P), Koo Xuan (K)

UC07: Bookmark Knowledge Units UC08: Rate Knowledge Units	
Sprint 3: Toolbox Module Assigned User Story: Persona 2 (Daniel): Customizing and evolving tools to meet personal mental health needs. Use Cases: - UC09: Add to Toolbox - UC10: Remove from Toolbox - UC11: Evolve Tools	Phavanee (P), Koo Xuan (K), Athirah (A)
Sprint 4: Gamification Assigned User Story: Persona 2 (Daniel): Motivating user engagement with points and badges. Use Cases: - UC12: Earn Points - UC13: Spend Points - UC14: Earn Badges	Hafiz (H), Koo Xuan (K), Athirah (A)
Sprint 5: Admin Module Assigned User Story: Persona 3 (Sarah): Managing knowledge units, resolving issues, and monitoring diagnostic trends Use Cases: - UC15: Add Knowledge Units - UC16: Resolve Flagged Issues - UC17: Monitor Diagnostic Trends	Phavanee (P), Hafiz (H)

UC18: Create Knowledge Units	
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2.4 User Story Details

2.4.1 US001: User Story <User Story 1>

Table 2.4.1.1: User Story Description for <Use Case : Register>

User story: Register
ID: US001
User Story Description As a user I want to be able to register the system So that I can log in and access the system
Actor : User
Preconditions: 1. User opens the mobile application.
Flow of events: 1. User opens the mobile application and navigates to the registration page. 2. User inputs personal details, including full name, email address, password. 3. User agrees to the terms and conditions by checking the provided box. 4. User submits the registration form. 5. System validate the input data 5.1 system ensures all mandatory fields are filled 5.2 verifies the email format is correct 5.3 confirms the password meet required criteria 6. System checks if the email is already registered 6.1 System prompts the user to log in or recover their password 7. The system creates the user account and stores it in the database. 8. System sends a confirmation email or message to the registered email address. 9. User sees a success message confirming the registration process.
Alternative flow : 1. Invalid input during registration 1.1 System highlights the invalid fields 1.2 User is prompted to correct the errors and resubmit 2. Email already exists in the system 2.1 System displays an error message: "Email is already registered." 2.2 User is directed to the login page or password recovery option.
Postconditions: 1. User account is stored securely in the database. 2. Users are redirected to the login page.
Exception flow:

1. No internet connection
 - 1.1 System displays an error message : “No internet connection. Please check your network and try again.
 - 1.2 User is unable to proceed with registration until the connection is restored.
2. Server error during registration
 - 2.1 System displays an error message: “Unable to process your registration at this time. Please try again later.”
 - 2.2 Registration is aborted, and the user is prompted to retry later.

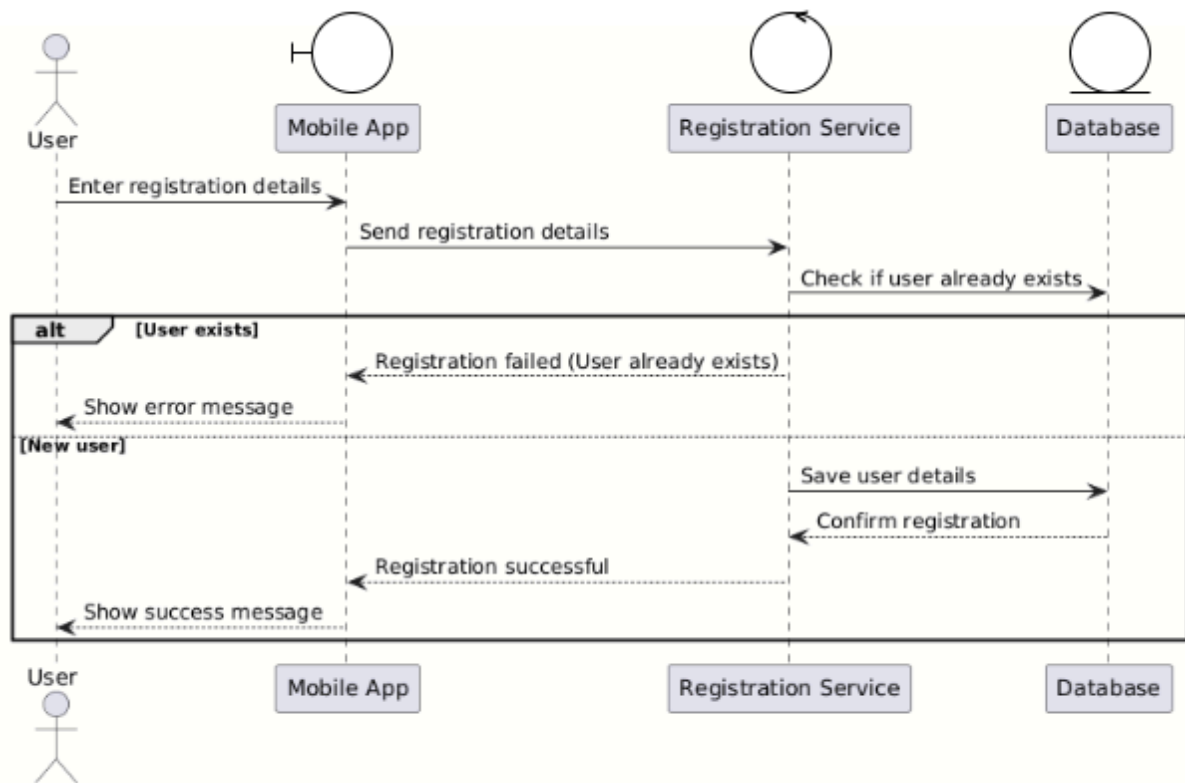


Figure 2.4.1.1: Sequence Diagram for <User Story/Scenario UC01>

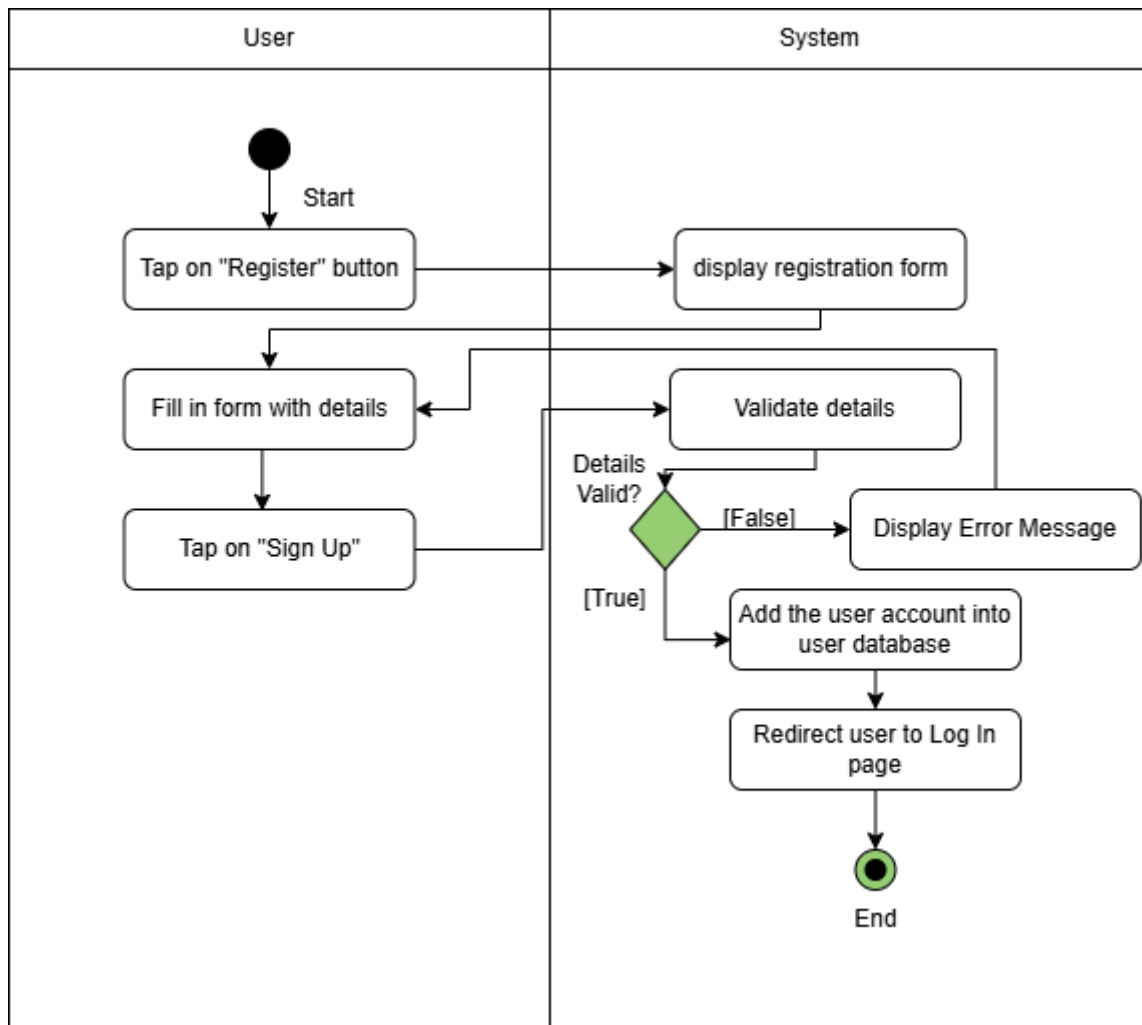


Figure 2.4.1.2: Activity Diagram for <<User Story/Scenario UC01>>

2.4.2 US002: User Story <User Story 2>

Table 2.4.2.1: User Story Description for <Use Case : Login>

User story: Login
ID: US002
User Story Description As an user I want to log in to the system So that I can access my account and the system's features.
Actor : User
Preconditions : 1. User has successfully registered an account in the system.
Flow of events: 1. User opens the mobile application and navigates to the login page. 2. User enters their registered email address and password. 3. User clicks the "Login" button. 4. System verifies the entered email and password 4.1 System matches the email and password against the database records. 5. System logs the user into their account. 6. User is redirected to the dashboard or homepage. 7. System displays a welcome message or dashboard with personalized content.
Alternative flow : 1. Incorrect email and password entered 1.1 System displays an error message: "Invalid email or password. Please try again." 1.2 Users are prompted to re-enter their credentials. 2. Forgot password 2.1 User clicks the "Forgot Password" link. 2.2 System redirects the user to the password recovery page. 2.3 User follows the password recovery process (covered in UC04).
Postcondition: 1. User is authenticated and logged into their account. 2. System starts a user session and enables access to all account features.
Other Conditions: 1. Passwords are case-sensitive.
Exception flow: 1. No internet connection 1.1 System displays an error message: "Unable to connect. Please check your internet connection." 1.2 Users cannot proceed with login until the connection is restored.

2. System timeout or server error:
 - 2.1 System displays an error message: "Service temporarily unavailable. Please try again later."
 - 2.2 User is unable to log in until the issue is resolved.

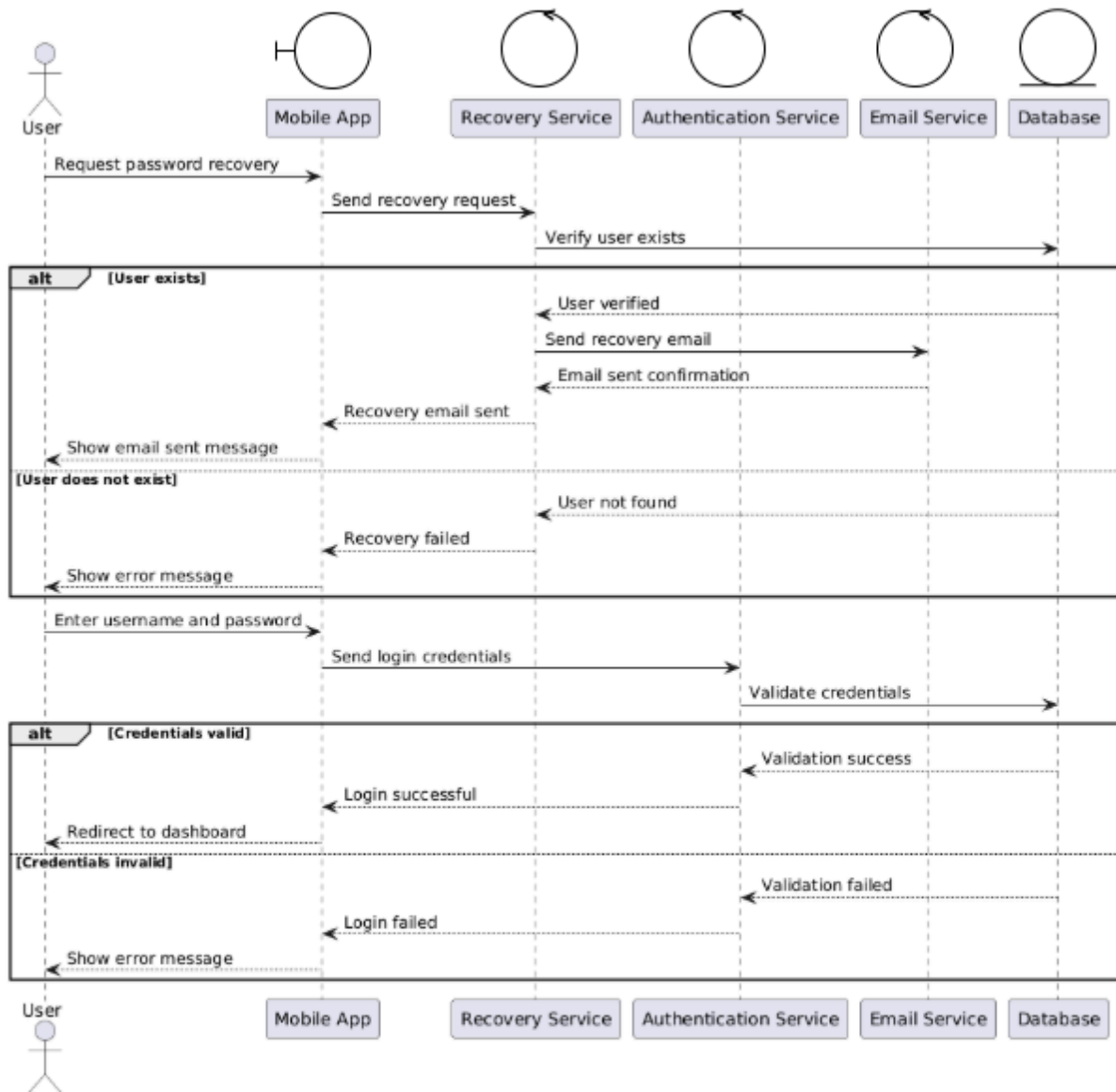


Figure 2.4.2.2: Sequence Diagram for <User Story/Scenario UC02 & UC04>

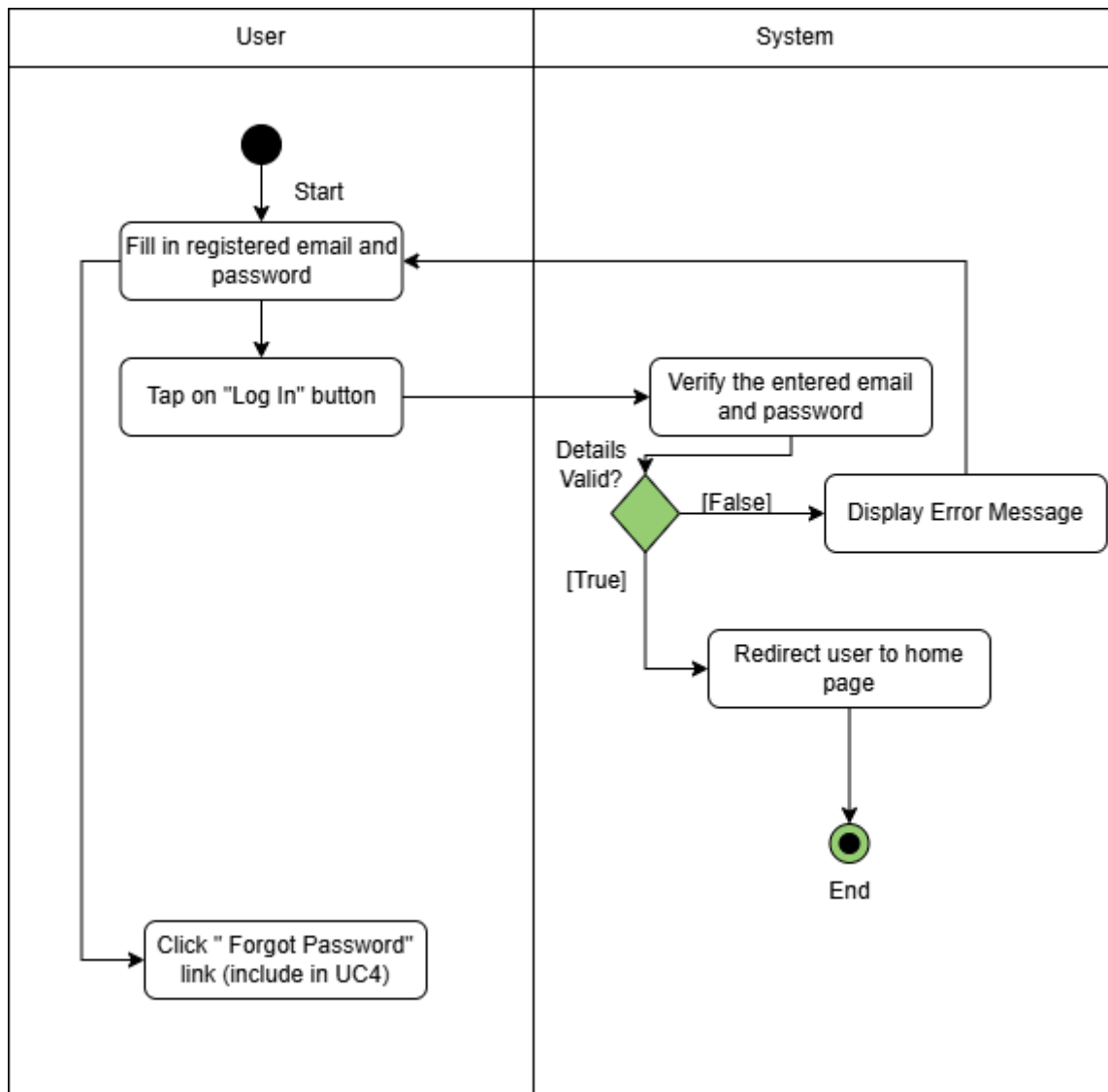


Figure 2.4.2.3: Activity Diagram for <<User Story/Scenario UC02 >>

2.4.3 US003: User Story <User Story 3>

Table 2.4.3.1: User Story Description for < Use Case : Manage Profile>

User story: Manage Profile
ID: US003
User Story Description As a user I want to manage my profile details. So that I can keep my information updated and relevant.
Actor : User
Preconditions: 1. User is logged into their account.
Flow of events: 1. User navigates to the “Manage Profile” section from the dashboard. 2. Users view their existing profile details (e.g., name, email, profile picture). 3. User selects the specific field(s) they wish to edit: 3.1 Updates personal details such as name, email, or contact number. 3.2 Change their profile picture by uploading a new image. 3.3 Modifies notification preferences or other optional settings. 4. User clicks the “Save Changes” button. 5. System validates the updated information: 5.1 Ensures mandatory fields are not left empty. 5.2 Checks for valid formats (e.g., email address). 6. If validation succeeds, the system updates the profile in the database. 7. System displays a confirmation message: “Profile updated successfully.”
Alternative flow : 1. Invalid input during profile update 1.1 System highlights the incorrect fields (e.g., invalid email format). 1.2 User is prompted to correct the errors and resubmit. 2. User cancels profile update 2.1 User clicks the “Cancel” button. 2.2 No changes are saved, and the user is returned to the previous screen.
Postcondition : 1. Updated profile details are stored securely in the database. 2. System reflects the changes in real-time on the user dashboard.
Exception flow: 1. No internet connection: 1.1 System displays an error message: “Unable to connect. Please check your internet connection.” 1.2 Users cannot save changes until the connection is restored. 2. System timeout or server error:

2.1 System displays an error message: "Unable to update profile at this time. Please try again later."
2.2 Changes are not saved, and the user is prompted to retry.

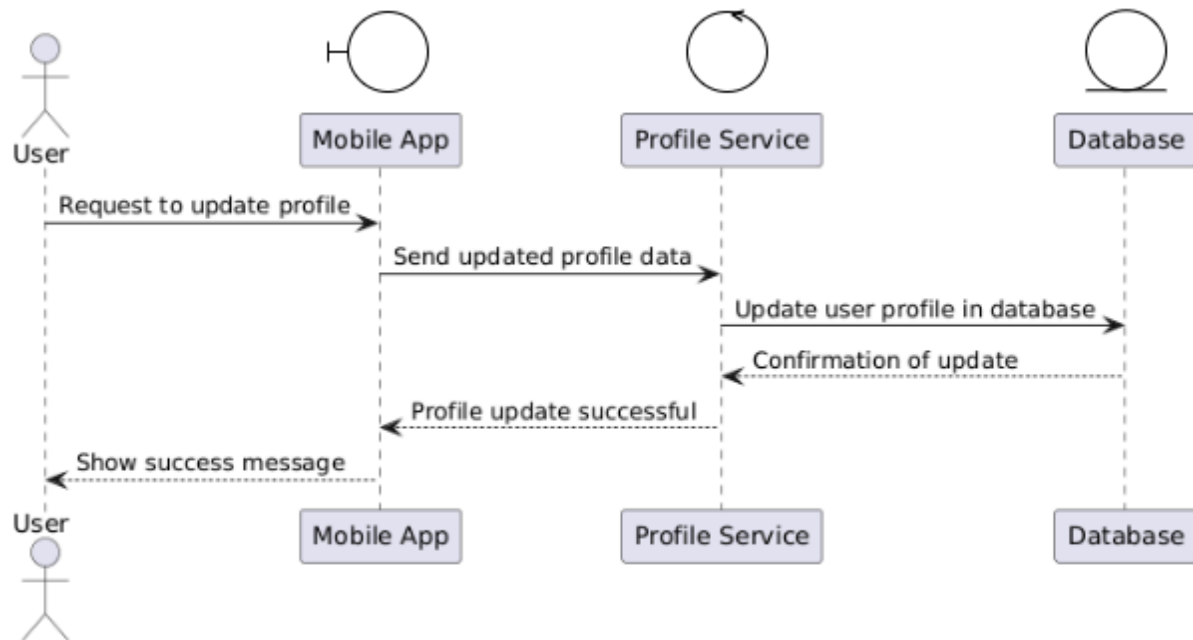


Figure 2.4.3.2: Sequence Diagram for Manage Profile

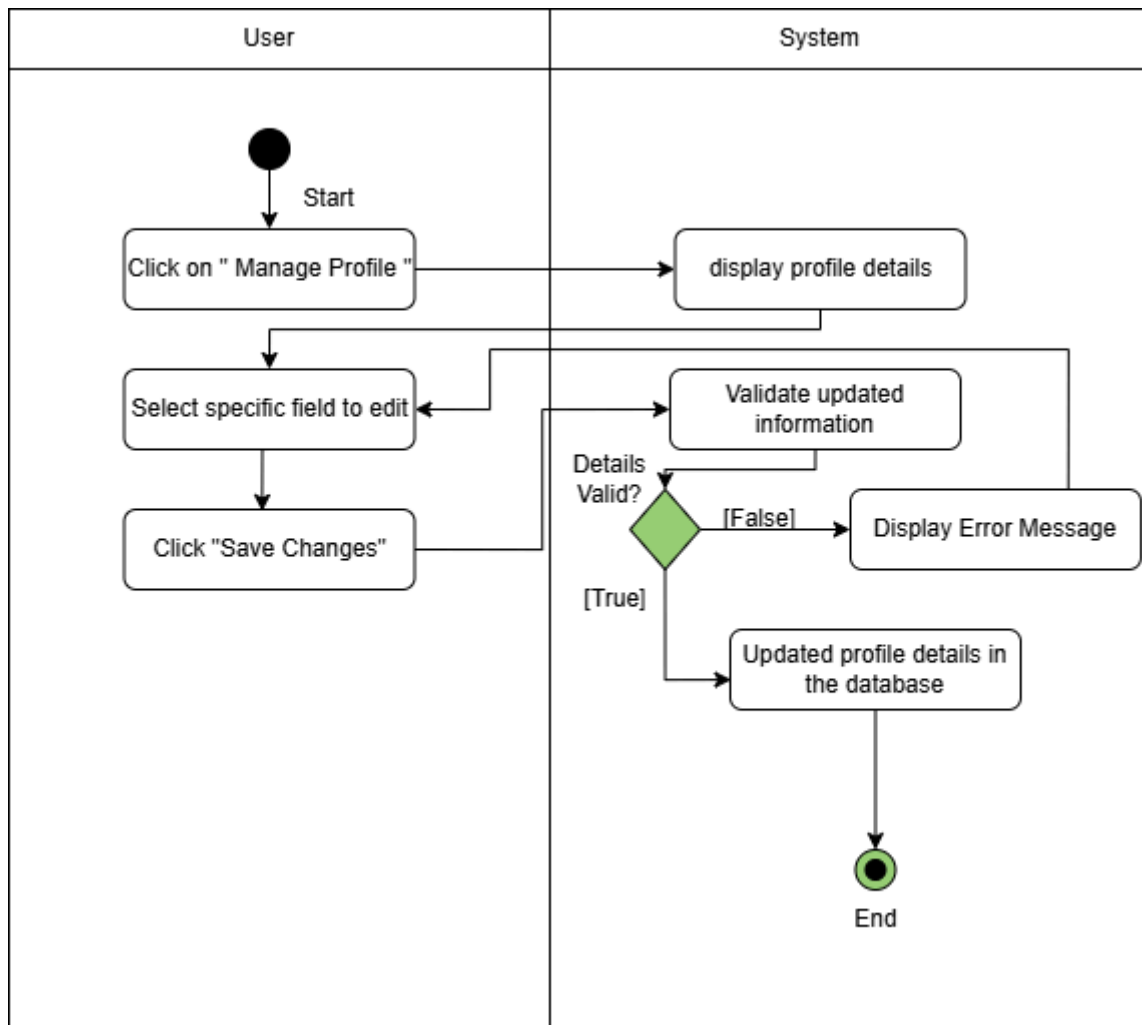


Figure 2.4.3.3: Activity Diagram for Manage Profile

2.4.4 US004 User Story <User Story 4>

Table 2.4.4.1: User Story Description for < Use Case : Password Recover >

User story: Password Recover
ID: US004
User Story Description As a user I want to recover my password if I forget it. So that I can regain access to my account.
Actor : User
Preconditions: 1. User has an existing account in the system.
Flow of events: 1. User clicks the “Forgot Password” link on the login page. 2. System prompts the user to enter their registered email address. 3. User provides their email address and clicks the “Submit” button. 4. System validates the provided email address 4.1 System ensures the email is registered in the database. 5. If the email is valid, the system sends a password recovery email or code to the user’s registered email address. 6. User receives the email and follows the instructions provided 6.1 Clicks the link to reset their password or enters the recovery code in the application. 7. User is redirected to a password reset page. 8. User creates a new password, adhering to the required security standards. 9. User confirms the new password and submits the form. 10. System updates the password in the database and displays a success message: “Your password has been reset successfully.”
Alternative flow : 1. Invalid email address entered 1.1 System displays an error message: “Email address not found. Please try again.” 1.2 User is prompted to re-enter a valid email. 2. User cancels the recovery process 2.1 User exits the process, and no action is taken.
Postcondition: 1. The user’s password is updated in the database. 2. Users can log in using the new password.
Other Conditions : 1. Password reset links or codes are valid for a limited time. 2. New password must meet the security requirements.
Exception flow:

1. Email delivery failure
 - 1.1 System displays an error message: "Unable to send password recovery email. Please try again later."
 - 1.2 User is prompted to retry.
2. No internet connection
 - 2.1 System displays an error message: "Unable to connect. Please check your internet connection."
 - 2.2 User cannot proceed with password recovery until the connection is restored.
3. Password reset link expired
 - 3.1 System displays an error message: "The password reset link has expired. Please initiate a new password recovery process."

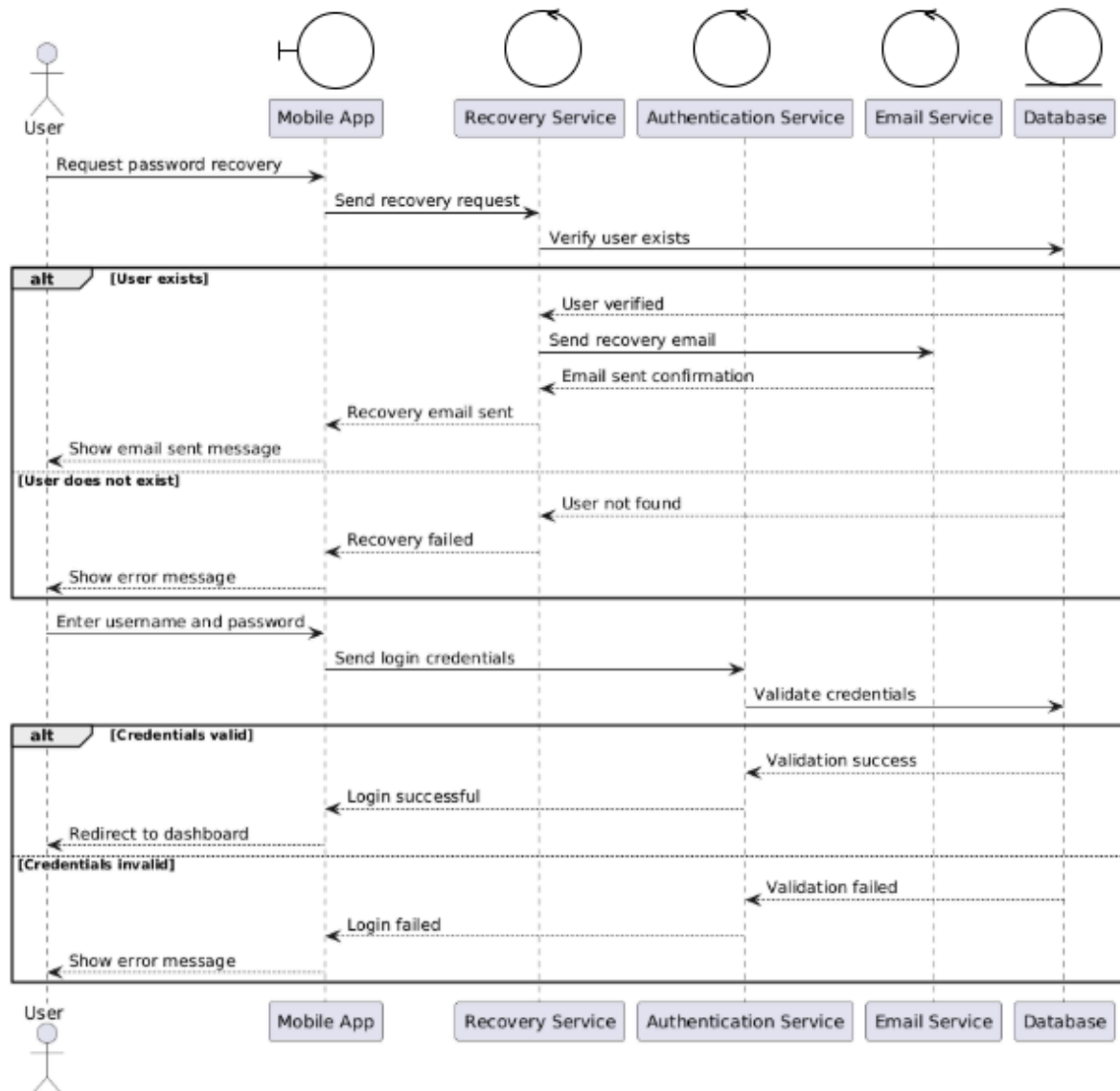


Figure 2.4.4.2: Sequence Diagram for Password Recover

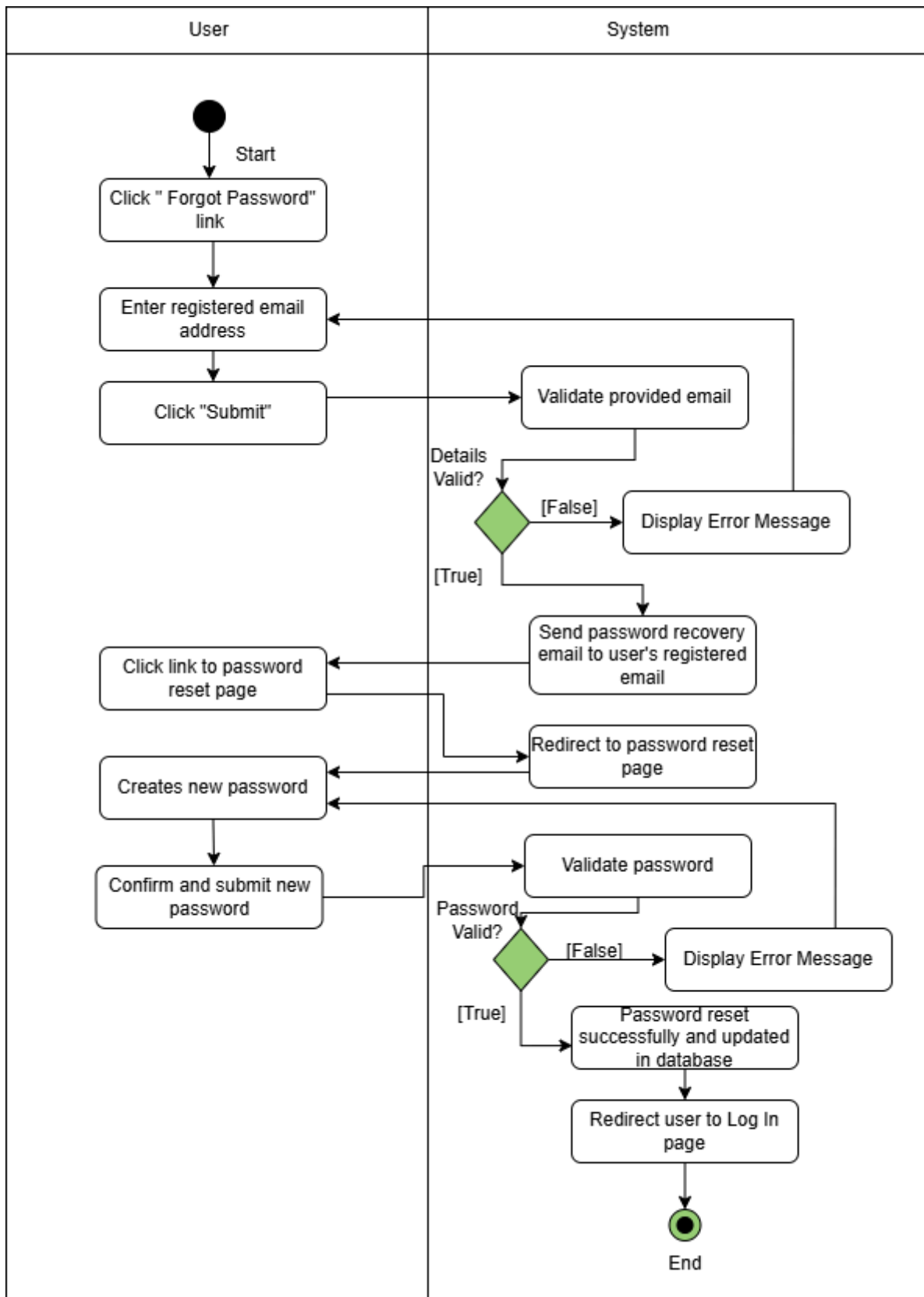


Figure 2.4.4.3: Activity Diagram for Password Recover

2.4.5 US005 User Story <User Story 5>

Table 2.4.5.1: User Story Description for < Use Case : Log Out>

User story: Log Out
ID: US005
User Story Description As a user I want to log out of the system. So that I can ensure my account is secure when I am not using it.
Actor : User
Preconditions : 1. User is logged into their account.
Flow of events: 1. User navigates to the menu or settings section of the application. 2. User selects the “Log Out” option. 3. System prompts the user for confirmation: “Are you sure you want to log out?” 4. User confirms the log-out action. 5. System terminates the user session and clears temporary data associated with the session. 6. User is redirected to the login page or home screen. 7. System displays a confirmation message: “You have successfully logged out.”
Alternative flow : 1. User cancels the log-out action: 1.1 User chooses “Cancel” on the confirmation prompt. 1.2 System retains the user’s session, and no changes are made.
Postcondition : 1. User session is terminated.
Other Conditions : 1. Logging out must ensure that sensitive data is securely deleted. 2. If the user was in the middle of an action, the system should prompt them to save changes before logging out.
Exception flow: 1. System error during log out 1.1 System displays an error message: “Unable to log out at this time. Please try again later.” 1.2 User session remains active until the issue is resolved. 2. No internet connection (if relevant) 2.1 System stores the log-out request locally and terminates the session when connectivity is restored.

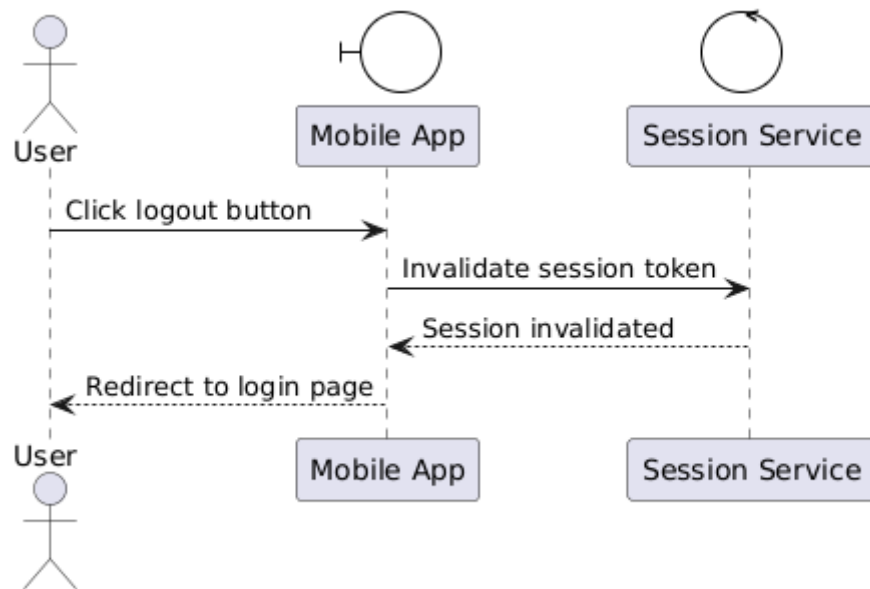


Figure 2.4.5.2: Sequence Diagram for Log Out

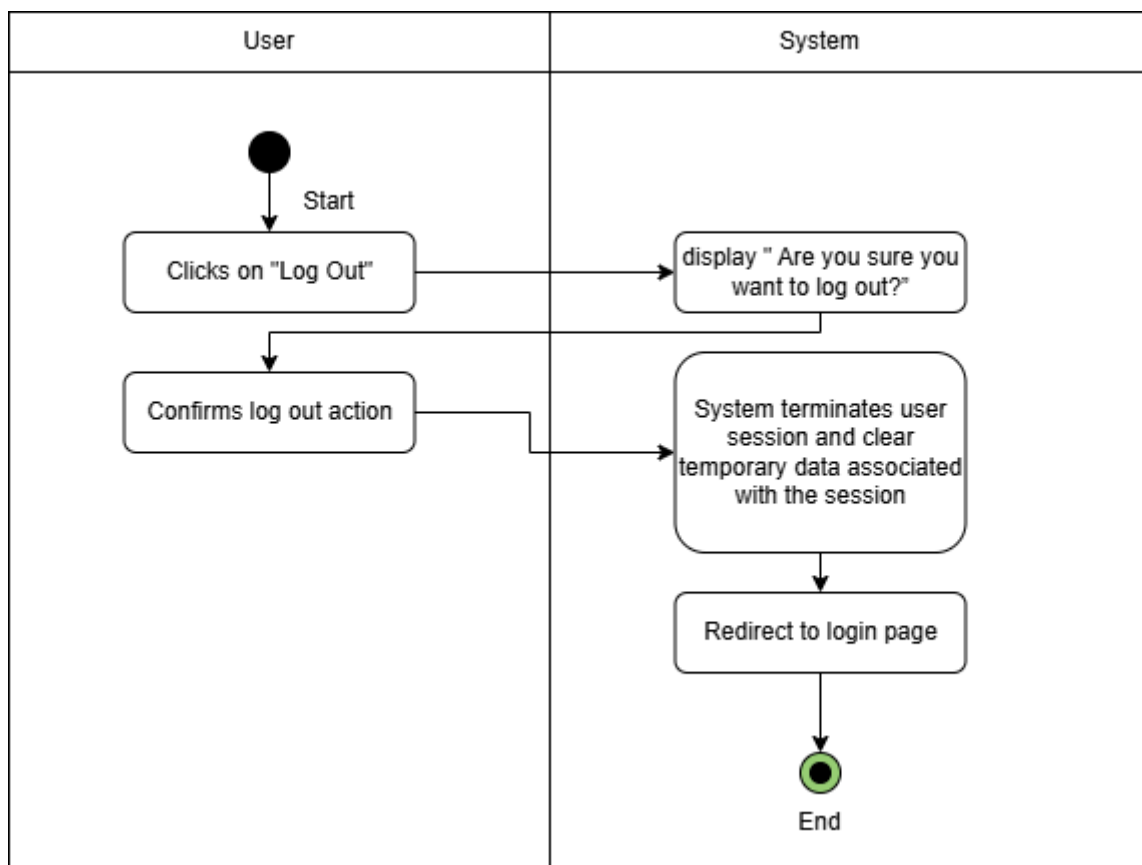


Figure 2.4.5.3: Activity Diagram for Log Out

2.4.6 US006 User Story <User Story 6>

Table 2.4.6.1: User Story Description for Browse Knowledge Units

User story: Browse Knowledge Units
ID: US006
User Story Description As a user I want to browse available knowledge units. So that I can access lessons and resources related to mental health.
Actor : User
Preconditions : 1. User is logged into their account. 2. The application is connected to the internet
Flow of events: 1. User navigates to the “Knowledge Units” section from the dashboard or main menu. 2. System displays a list of knowledge units, categorized by topics (e.g., stress management, mindfulness, coping strategies). 3. User scrolls through the list or uses filters/search functionality to find specific knowledge units. 4. User selects a knowledge unit to view details, such as: 4.1 Title and description of the unit. 4.2 Content type (e.g., article, video, guided exercise). 5. User clicks on the selected unit to access the full content. 6. System loads the content and displays it in a readable/viewable format.
Alternative flow : 1. No knowledge units available 1.1 System displays a message: “No knowledge units are currently available. Please check back later.” 1.2 User is redirected to the dashboard or previous screen. 2. User cancels browsing 2.1 User exits the “Knowledge Units” section and returns to the dashboard.
Postcondition : 1. User successfully accesses and views a knowledge unit. 2. System records user interactions for recommendations or analytics.
Exception flow: 1. No internet connection 1.1 System displays an error message: “Unable to load knowledge units. Please check your internet connection.” 1.2 User cannot browse knowledge units until connectivity is restored. 2. System timeout or server error 2.1 System displays an error message: “Unable to fetch knowledge units at this time. Please try again later.”

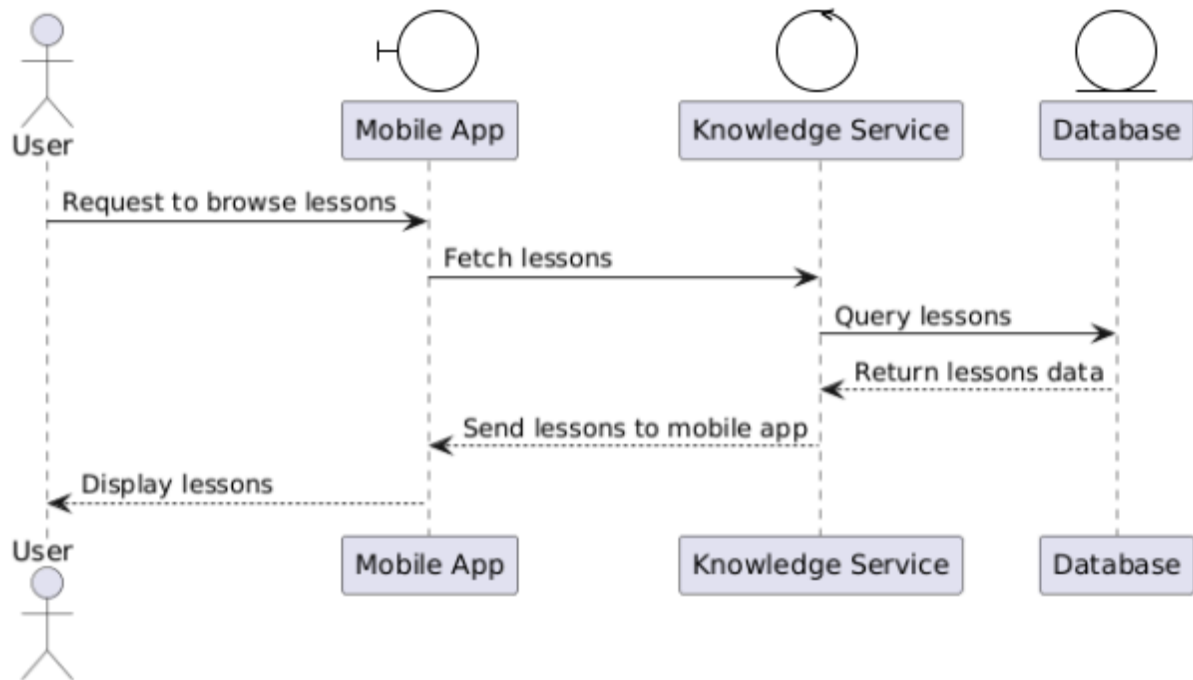


Figure 2.4.6.2: Sequence Diagram for Browse Knowledge Units

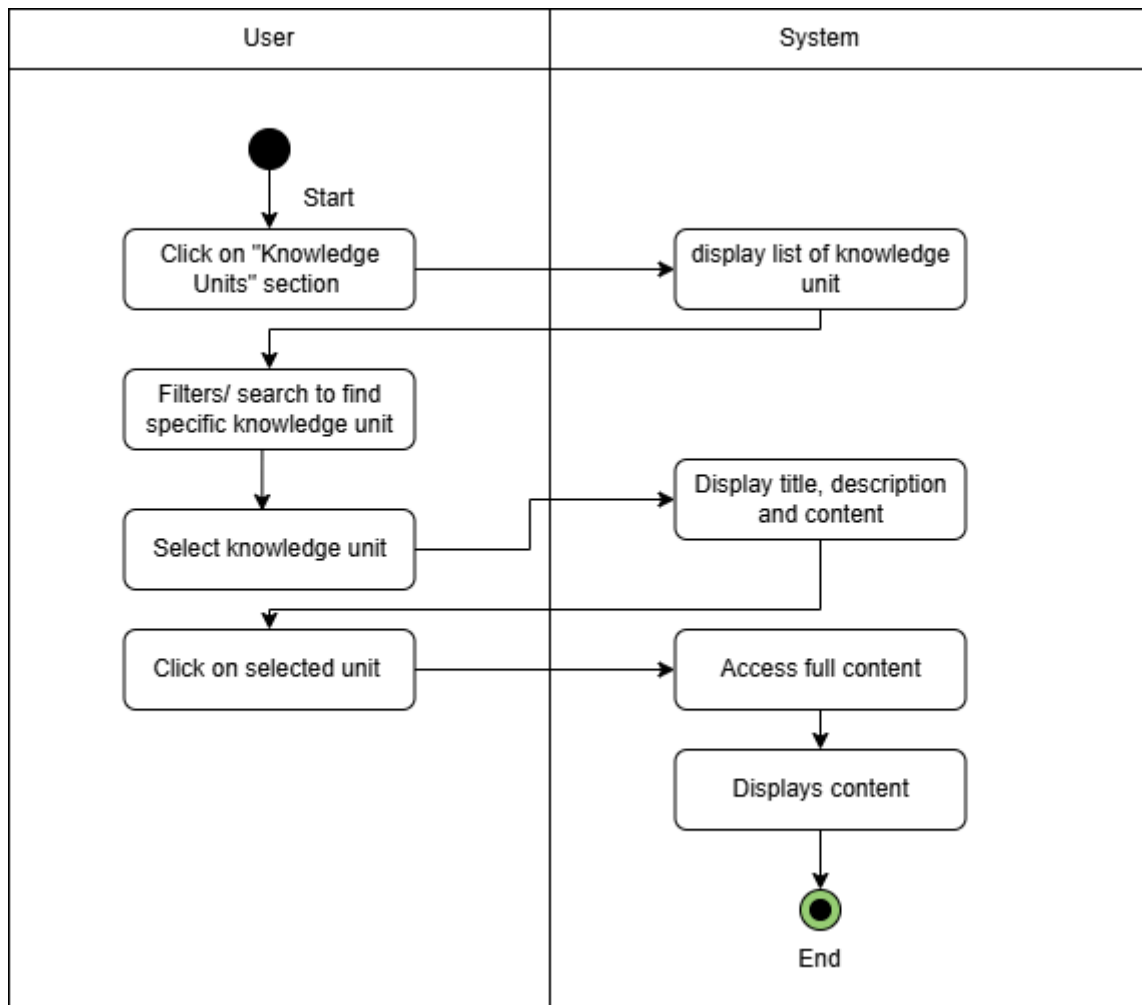


Figure 2.4.6.3: Activity Diagram for Browse Knowledge Units

2.4.7 US007 User Story <User Story 7>

Table 2.4.7.1: User Story Description for < Use Case : Bookmark Knowledge Units>

User story: Bookmark Knowledge Units
ID: US007
User Story Description As a user I want to bookmark knowledge units. So that I can save and easily access the units that are important to me later.
Actor : User
Preconditions: 1. User is logged into their account. 2. The knowledge units section is accessible, and content is available for browsing.
Flow of events: 1. User navigates to the “Knowledge Units” section. 2. User selects a specific knowledge unit from the displayed list or search results. 3. System displays details of the selected knowledge unit. 4. User clicks the “Bookmark” button/icon associated with the knowledge unit. 5. System marks the unit as bookmarked and saves it under the user’s account. 6. System provides confirmation feedback, and displays “Knowledge unit bookmarked successfully.” 7. The bookmarked unit is added to the “Bookmarked Units” section for quick access.
Alternative flow : 1. User un-bookmarks a knowledge unit 1.1 User clicks the “Unbookmark” button/icon on a previously bookmarked unit. 1.2 System removes the unit from the user’s bookmarks and updates the status. 1.3 System provides confirmation feedback, e.g., “Knowledge unit removed from bookmarks.”
Postcondition : 1. The knowledge unit is saved under the user’s bookmarks. 2. Users can access the bookmarked knowledge units from a dedicated section.
Other Conditions : 1. System ensures that bookmarking functionality works seamlessly across devices. 2. Bookmarked units remain accessible even if the user logs out and logs back in. 3.
Exception flow:

1. No internet connection
 - 1.1 System displays an error message: "Unable to bookmark knowledge unit. Please check your internet connection."
 - 1.2 The unit is not bookmarked until connectivity is restored.

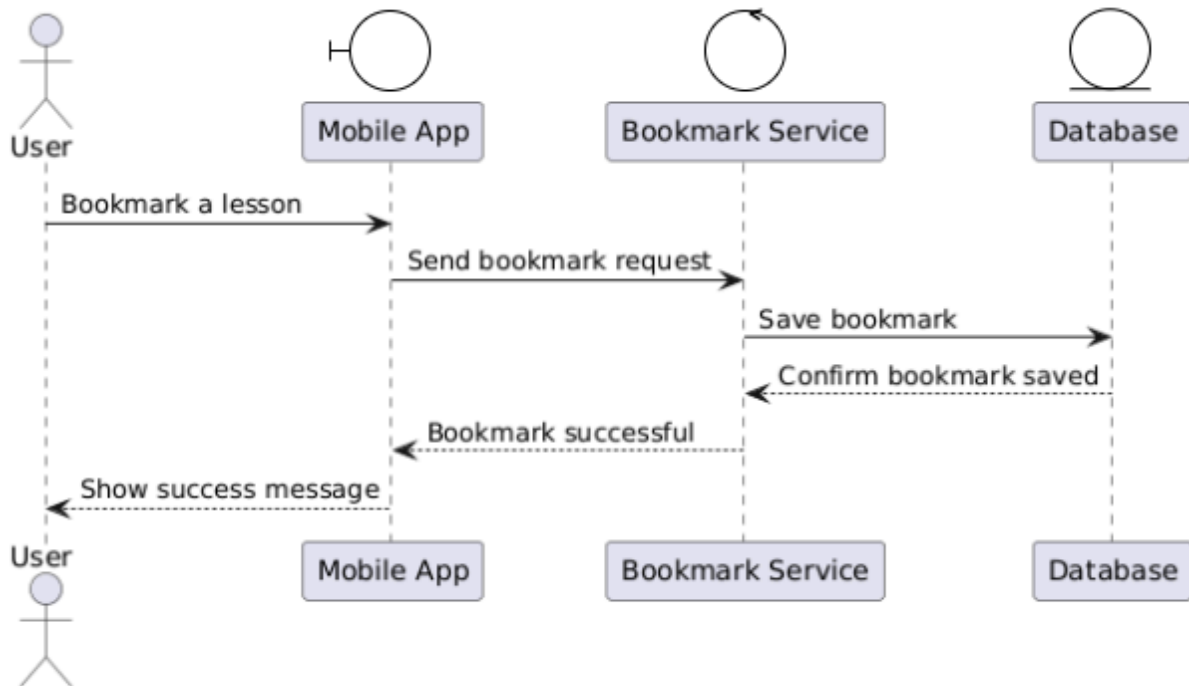


Figure 2.4.7.2: Sequence Diagram for Bookmark Knowledge Units

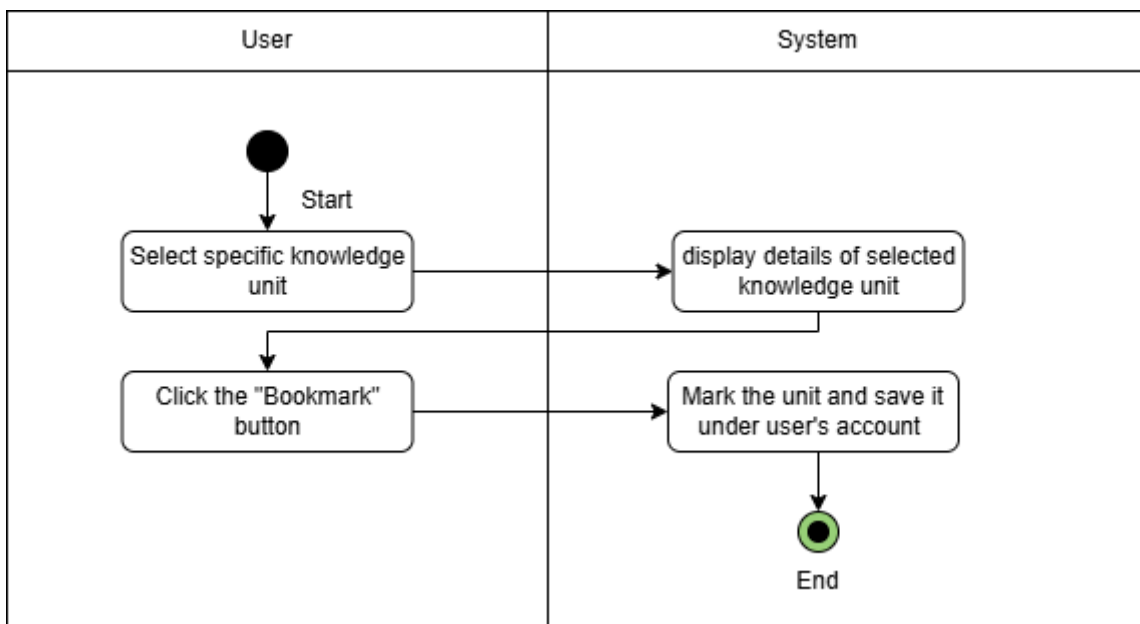


Figure 2.4.7.3: Activity Diagram for Bookmark Knowledge Units

2.4.8 US008 User Story <User Story 8>

Table 2.4.8.1: User Story Description for < Use Case : Rate Knowledge Units >

User story: Rate Knowledge Units
ID: US008
User Story Description As a user I want to rate knowledge units. So that I can provide feedback on the content and help improve recommendations for myself and other users.
Actor : User
Preconditions: 1. User is logged into their account. 2. The knowledge units section is accessible, and content is available for viewing.
Flow of events: 1. User navigates to the “Knowledge Units” section. 2. User selects and opens a specific knowledge unit. 3. System displays the content of the knowledge unit along with a rating feature (e.g., a star-based system or like/dislike buttons). 4. User selects a rating (e.g., 1–5 stars or thumbs up/thumbs down). 5. System records the user’s rating and saves it to the database. 6. System displays confirmation feedback, e.g., “Your rating has been submitted.”
Alternative flow : 1. User changes their rating 1.1 User navigates back to the rated knowledge unit. 1.2 User selects a new rating. 1.3 System updates the rating in the database and provides confirmation feedback, e.g., “Your rating has been updated.” 2. User cancels rating 2.1 User exits the knowledge unit without providing a rating. 2.2 No rating is recorded.
Postcondition : 1. The rating is stored in the database and associated with the specific knowledge unit. 2. The system updates its analytics or recommendation engine based on the user’s rating
Other Conditions: 1. User can only rate a knowledge unit once but can update their rating later.
Exception flow: 1. No internet connection

- 1.1 System displays an error message: "Unable to submit your rating. Please check your internet connection."
1.2 The rating is not saved until connectivity is restored.

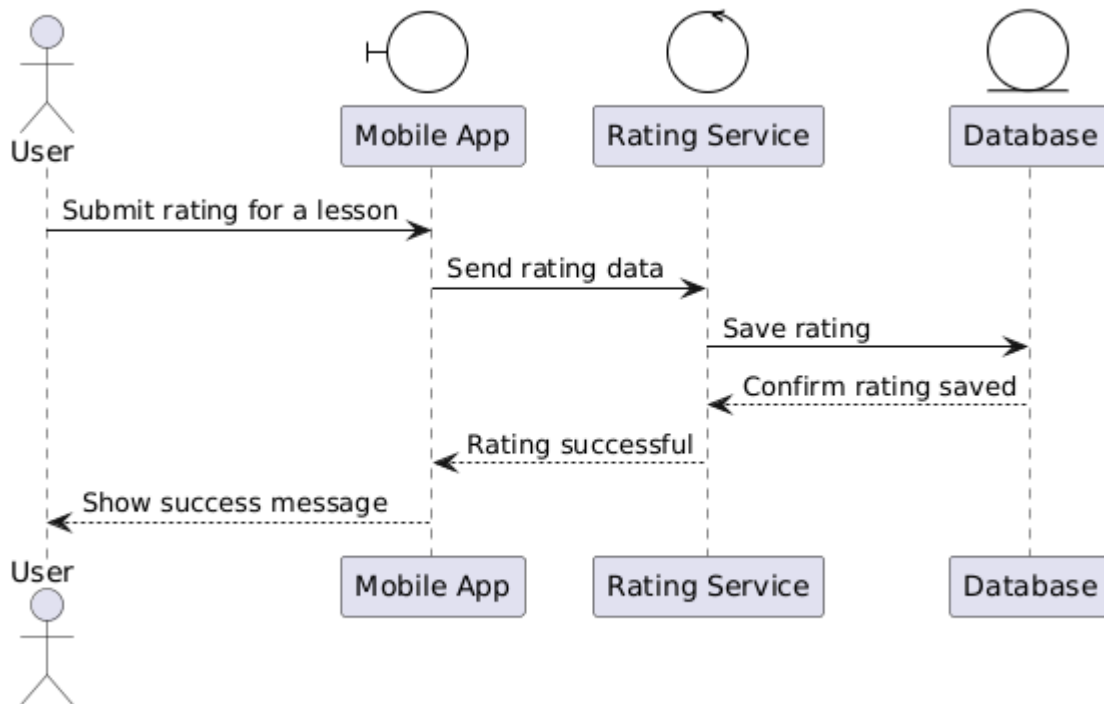


Figure 2.4.8.1: Sequence Diagram for Rate Knowledge Units

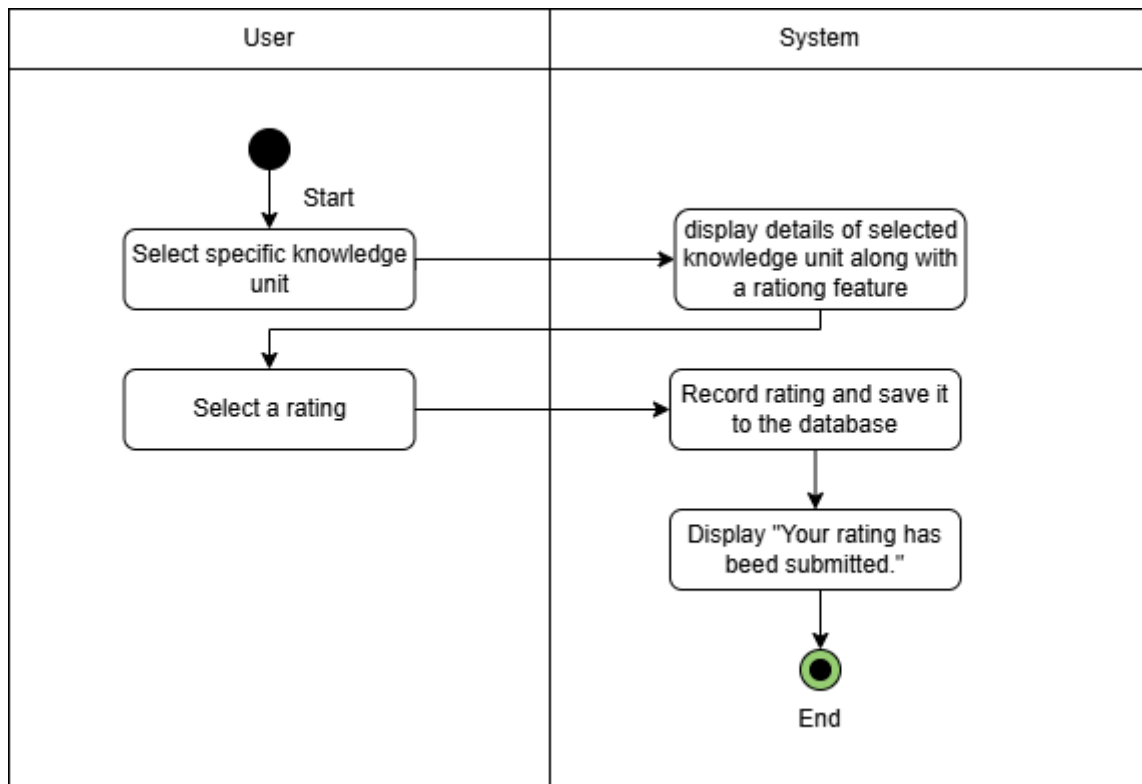


Figure 2.4.8.2: Activity Diagram for Rate Knowledge Units

2.4.9 US009 User Story <User Story 9>

Table 2.4.9.1 : User Story Description for Add to Toolbox

User story: Add to Toolbox
ID: US009
User Story Description As a user I want to be able to add useful tools to my Toolbox So that I can use them whenever I please
Preconditions: 1. User is logged into their account. 2. User is at the end of reading a knowledge unit.
Flow of events: 1. An option that allows users to add the skill that they have read about in their Toolbox is displayed on the screen. 2. Users are prompted to select the option to add the tool to their Toolbox or dismiss the message. a. If the user selects to dismiss the message, AF1 occurs. 3. The tool is added to the user's Toolbox. 4. The flow ends.
Alternative flow 1: 1. The tool is not added to the user's Toolbox.
Postconditions: 1. The tool gets added to the user's Toolbox.
Exception flow: 1. No internet connection a. System displays an error message: "Unable to add tool to Toolbox. Please check your internet connection." b. 1.2 The tool is not added until connectivity is restored.

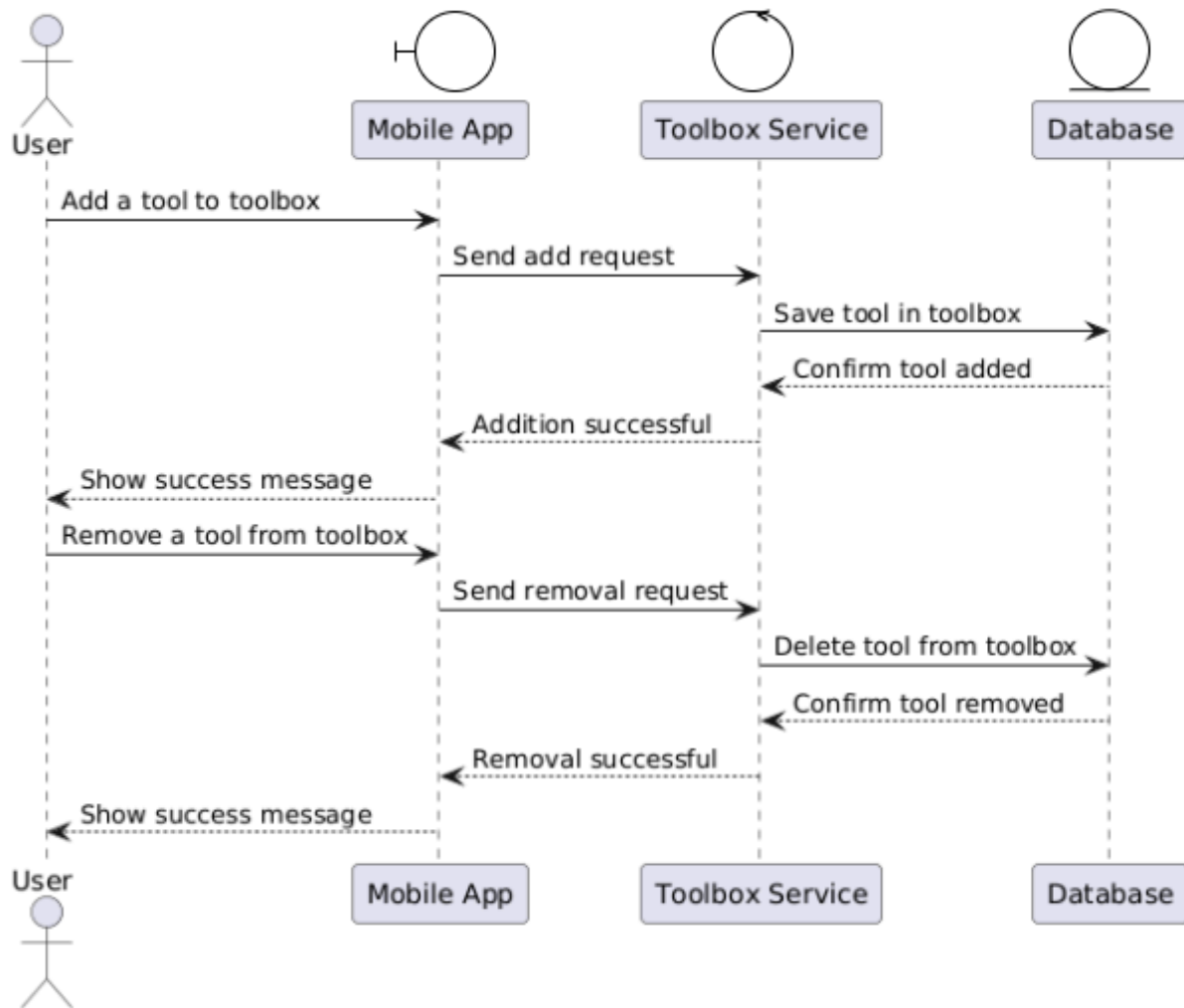


Figure 2.4.9.1 : Sequence Diagram for Add tools to Toolbox

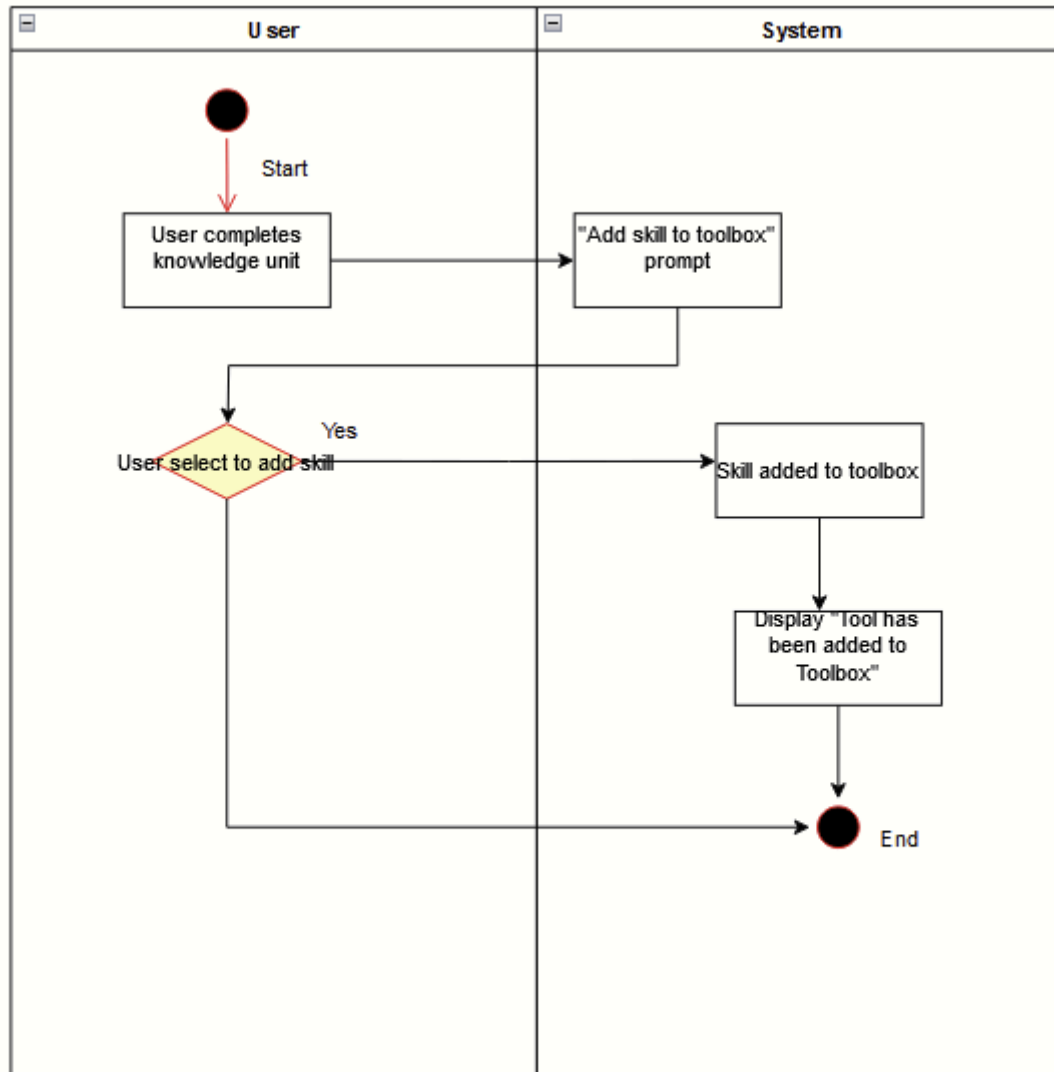


Figure 2.4.9.2 : Activity Diagram for Add tools to Toolbox

2.4.10 US010 User Story <User Story 10>

Table 2.4.10.1 : User Story Description for Remove from Toolbox

User story: Remove from Toolbox
ID: US010
User Story Description As a user I want to be able to remove tools from my Toolbox So that I can only keep tools relevant to me
Preconditions: 1. User is logged into their account. 2. User is able to access their Toolbox.
Flow of events: 1. User navigates to the "Toolbox" page. 2. User selects the "edit Toolbox" option. 3. User can remove tools that are no longer relevant to them by clicking the "delete tool" button. 4. The flow ends.
Exception flow: 1. No internet connection a. System displays an error message: "Unable to remove tool from Toolbox. Please check your internet connection." b. The tool is not removed until connectivity is restored.
Postconditions: 1. The tool is removed from the user's Toolbox.

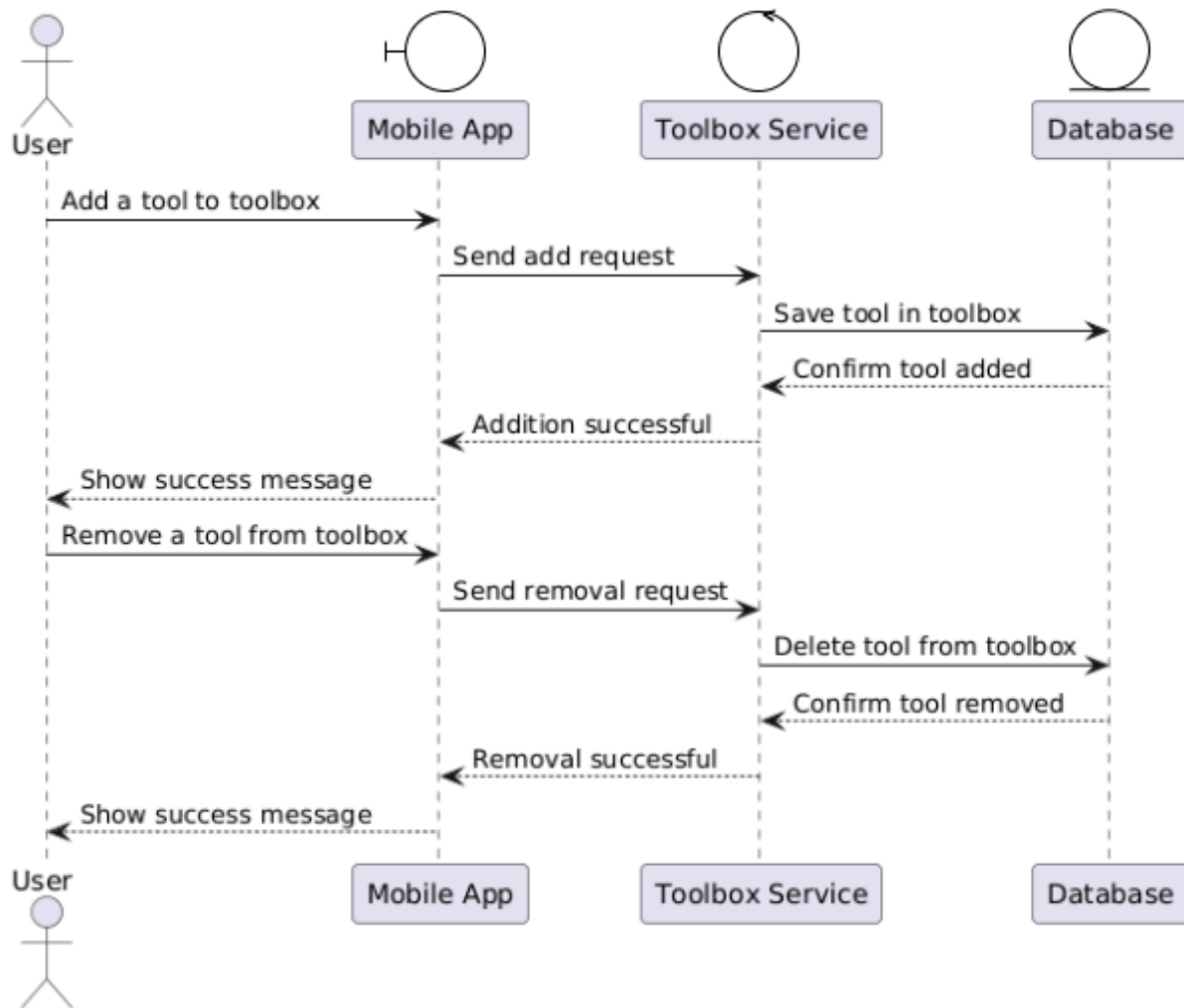


Figure 2.4.10.2: Sequence Diagram for Remove from Toolbox

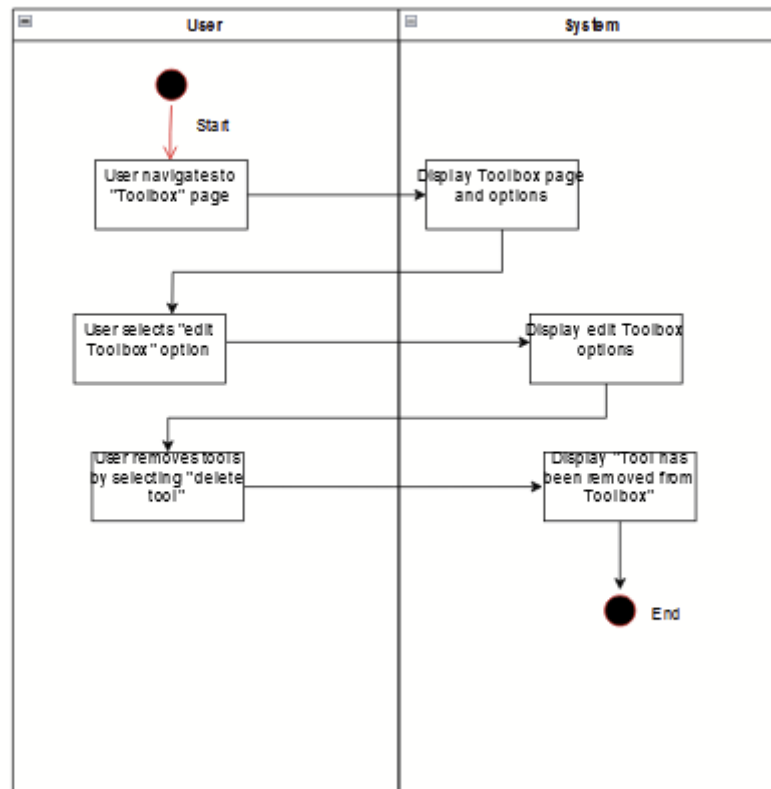


Figure 2.4.10.3: Activity Diagram for Remove from Toolbox

2.4.11 US011 User Story <User Story 11>

Table 2.4.11.1 : User Story Description for Evolve Tools

User story: Evolve Tools
ID: US011
User Story Description As a user I want to be able to evolve the tools in my Toolbox So that I can keep using tools that are suited to my skill level
Preconditions: 1. User is logged into their account. 2. User is able to access their Toolbox.
Flow of events: 1. After using a tool for a set number of times, users are prompted to evolve their tools. 2. Users can select to evolve their tools or dismiss the message. a. If users dismiss the message, AF1 occurs. 3. The users' tool has evolved. 4. The flow ends.
Alternative Flow 1: 1. The tool is not evolved.
Exception flow: 1. No internet connection c. System displays an error message: "Unable to evolve the tool. Please check your internet connection." d. The tool is not evolved until connectivity is restored.
Postconditions: 2. The tool in the user's Toolbox is evolved.

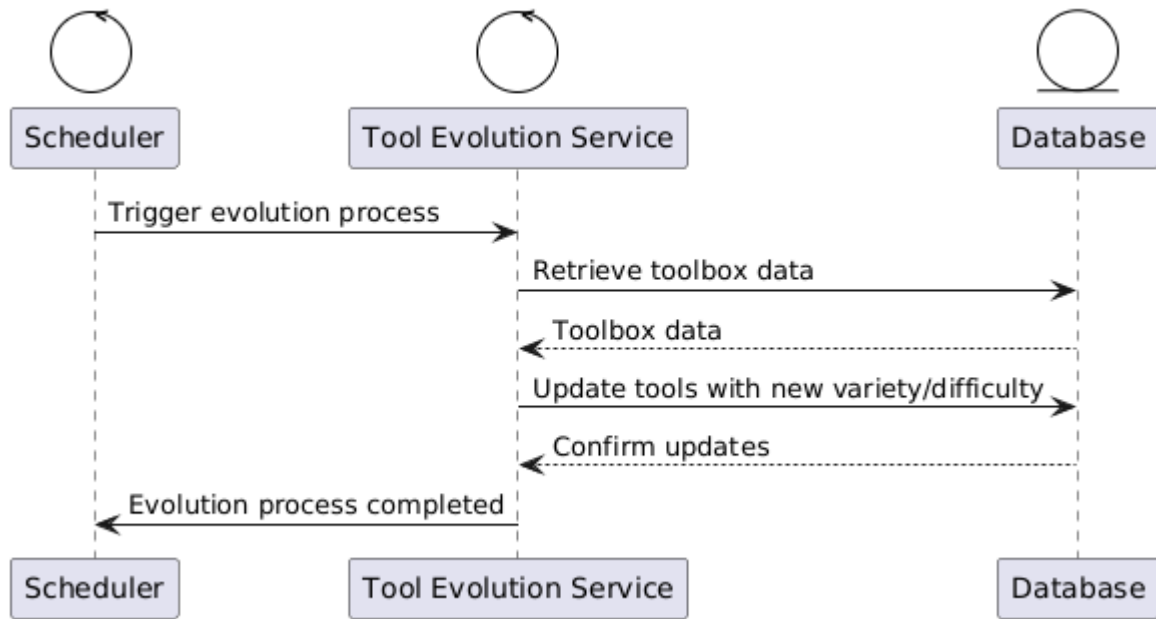


Figure 2.4.11.2 : Sequence Diagram for Evolve Tools

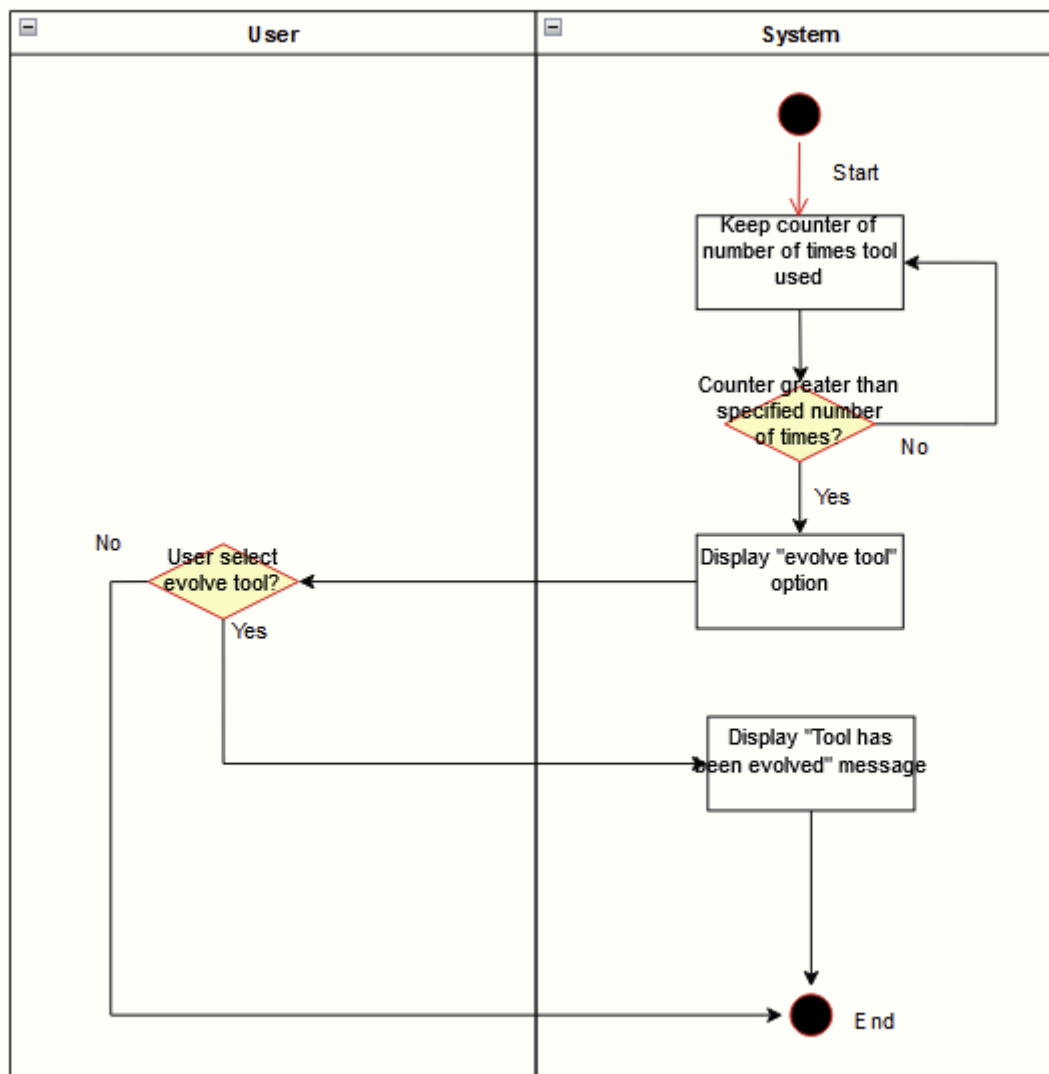


Figure 2.4.11.3 : Activity Diagram for Evolve Tools

2.4.12 US012 User Story <User Story 12>

Table 2.4.12.1 : User Story Description for Earn Points

User story: Earn Points
ID: US012
User Story Description As a user I want earn points by completing mental health activities, So that I can track my progress and unlock premium resources
Preconditions: 1. User is logged into their account 2. The activity is available and accessible
Flow of events: 1. User completes an activity through the Mobile App 2. The Mobile App triggers the point system to calculate earned points 3. The Point System sends a request to the Database to update the user's total points. 4. The Database updates the points and confirms the update to the Point System 5. The Point System sends the updated points back to the Mobile App 6. The Mobile App displays the updated points to the user
Alternative flow: 1. User completes multiple activities at once: 1.1 Points for each activity are calculated sequentially 1.2 The total updated points are displayed to the user in real-time 2. User completes an activity offline: 2.1 The Mobile App stores the completed activity locally 2.2 Points are synchronized with the Database once an internet connection is restored
Postconditions: 1. The points are successfully stored in the Database 2. The updated points are visible on the user interface
Other conditions: 1. Points are calculated based on predefined rules for different activities (For example : lessons = 10 points, check-ins = 5 points) 2. Points are only rewarded for successfully completed activities
Exception flow: 1. Database update failure : 1.1 The Point System retries the update process up to 3 times 1.2 If unsuccessful, the system logs the failure and displays an error message: "Unable to update your points, Please try again later." 1.3 Points are queued for later update when the system is restored 2. No internet connection : 2.1 System stores the activity locally 2.2 Points are updated once the connection is re-established

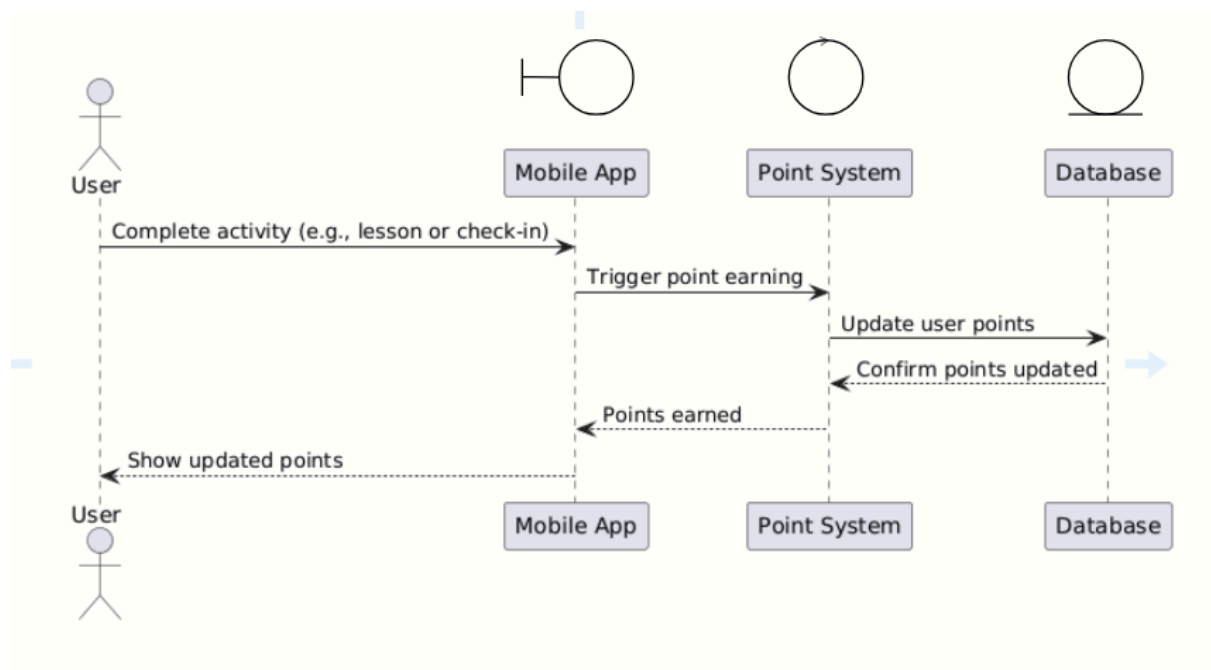


Figure 2.4.12.2 : Sequence Diagram for Earn Points

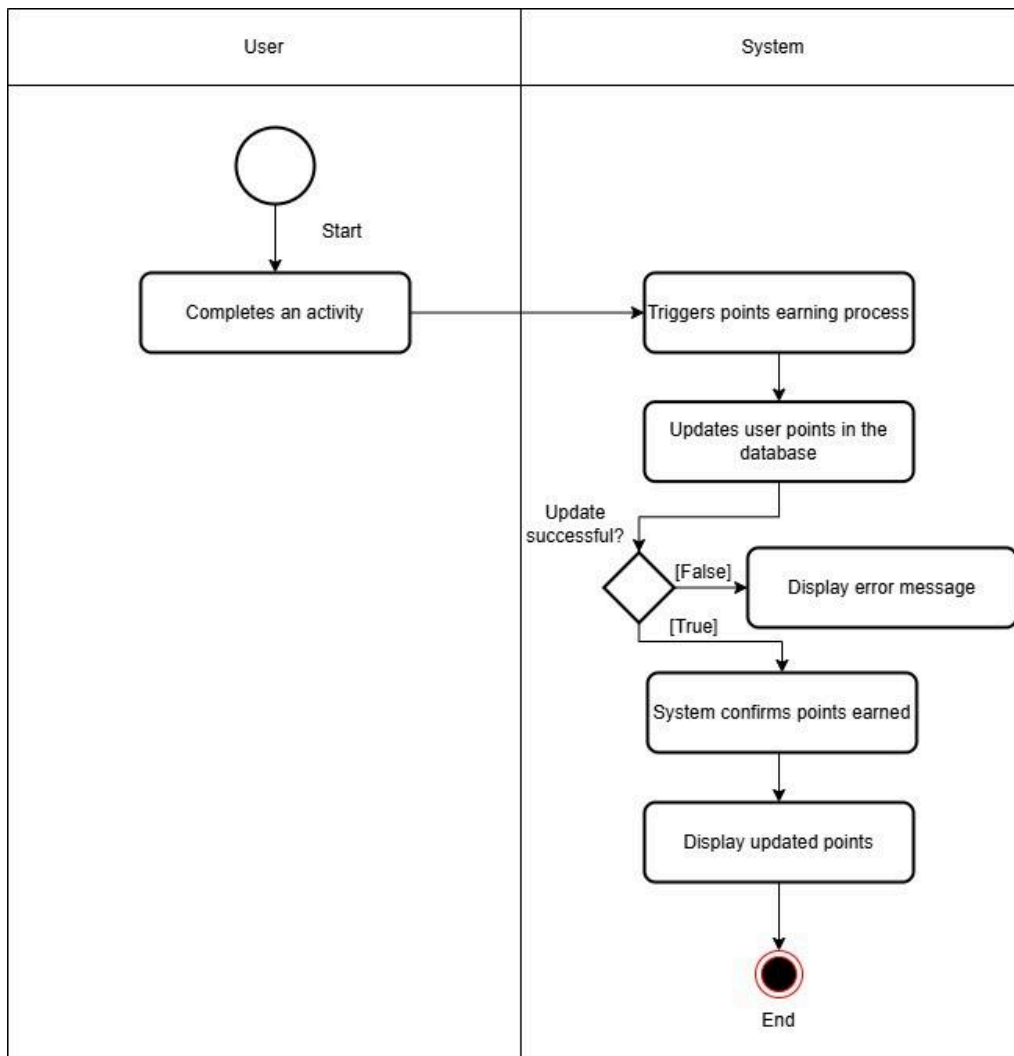


Figure 2.4.12.3 : Activity Diagram for Earn Points

2.4.13 US013 User Story <User Story 13>

Table 2.4.13.1: User Story Description for Spend Points

User story: Spend Points
ID: US013
User Story Description As a user I want use my earned points to customize my avatar or unlock premium features, So that I can personalize my experience and make the app more engaging
Preconditions: 1. User is logged into their account 2. User has sufficient points in their balance 3. Customization options are available
Flow of events: 1. User selects the option to customize the avatar or unlock a premium feature via the Mobile App 2. The Mobile App checks the user's points balance by interacting with the Point System 3. The Point System retrieves the user's points from the Database a. If the user has insufficient points, AF1 occurs 4. If the user has sufficient points : a. The Point System deducts the required points b. The Avatar Service applies the customization and updates it in the Database 5. The Mobile App displays the updated avatar or premium feature to user
Alternative flow 1 : 1. The Point System determines that the user does not have enough points 2. The Mobile App displays an error message : "You do not have enough points to complete this action"
Alternative flow 2: 1. User exits the customization screen before confirming 2. No points are deducted, and no changes are applied
Exception flow: 1. Database access failure: 1.1 The system displays an error message: "Unable to retrieve points balance. Please try again later." 1.2 No points are deducted, and the process is halted. 2. Avatar service failure: 2.1 The Avatar service fails to apply the customization. 2.2 System rolls back the points deduction and notifies the user: "Customization failed. Points have not been deducted." 3. No Internet connection::

- 3.1 System displays an error message: “Unable to process your request. Please check your internet connection.”
- 3.2 Customization is not applied, and points remain unaffected.

Postconditions:

1. The user’s points balance is updated in the Database.
2. The customized avatar or unlocked feature is successfully applied and visible on the user interface.

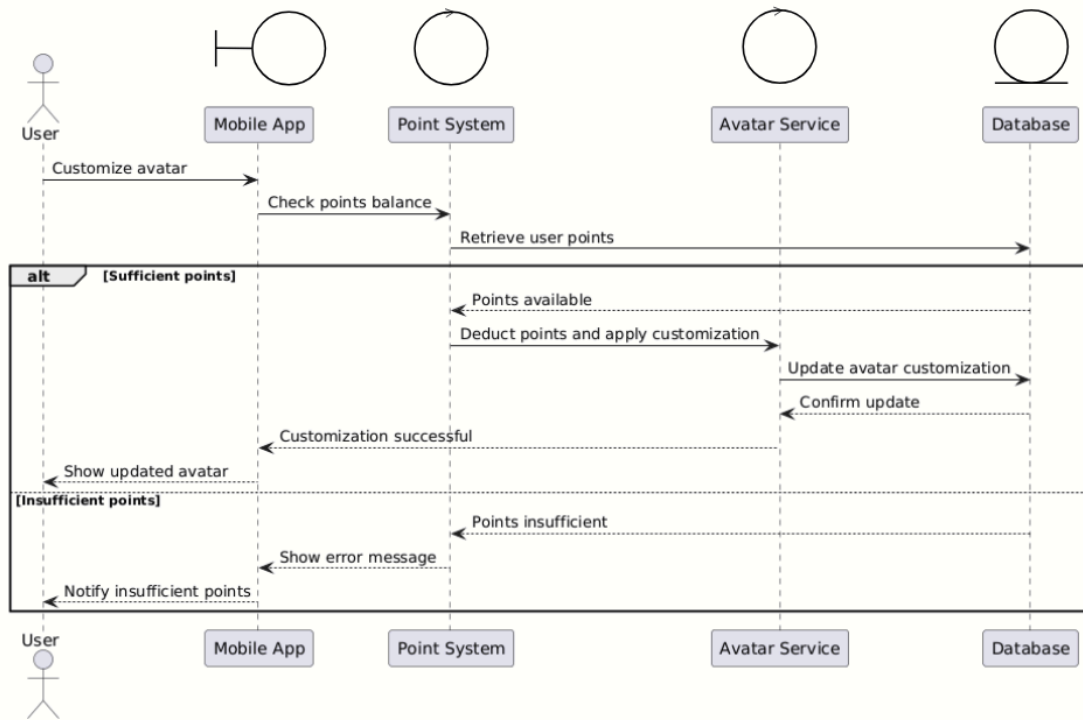


Figure 2.4.13.1: Sequence Diagram for Spend Points

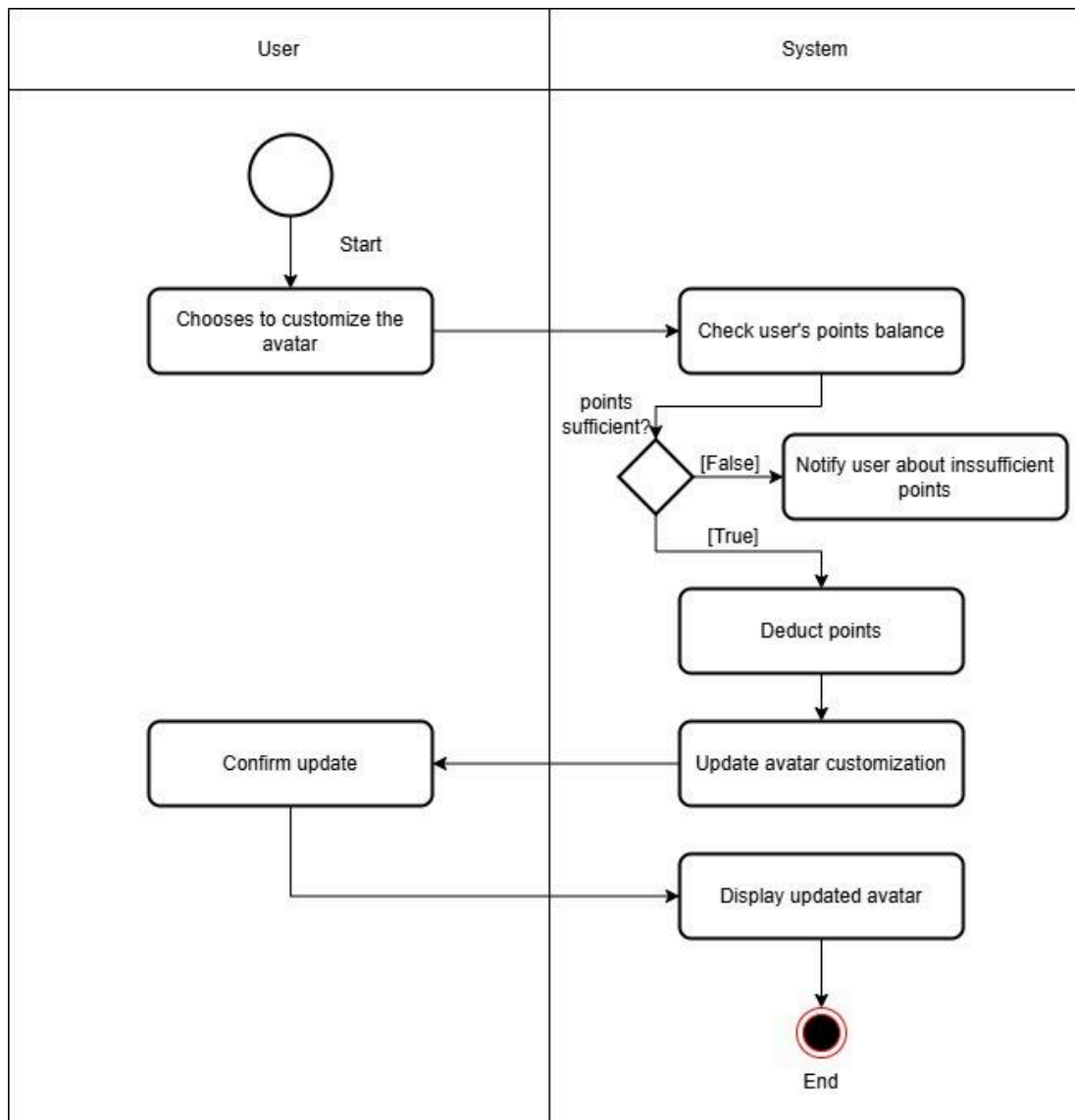


Figure 2.4.13.2: Activity Diagram for Spend Points

2.4.14 US014 User Story <User Story 14>

Table 2.4.14.1: User Story Description for Earn Badges

User story: Earn Badges
ID: US014
User Story Description As a user I want to be awarded badges upon completing specific milestones, So that I can track my achievements and feel motivated to reach new goals
Preconditions: 1. User is logged into their account 2. Milestones and achievements are defines in the system
Flow of events: 1. The Achievement Tracker checks the completion of a milestone 2. The Badge Service verifies the user's achievements from the Database a. If no milestones are achieved, AF1 occurs 3. If a milestone is achieved : a. The Badge Service verifies the milestone b. The Badge Service awards the badge to the user c. The Badge Service notifies the Achievement Tracker that the badge has awarded 4. The Achievement Tracker confirms and displays the awarded badge to the user
Alternative flow 1: 1. The Badge Service determines that no achievements have been completed. 2. The Badge Service notifies the Achievement Tracker that no badge will be awarded.
Exception flow: 1. Database Access Failure: 1.1 The Badge Service cannot verify achievements due to a database error. 1.2 System displays an error message: "Unable to verify achievements. Please try again later." 2. No Internet Connection: 2.1 System displays an error message: "Unable to connect. Please check your internet connection." 2.2 Badge verification is halted until connectivity is restored.
Postconditions: 1. If a milestone is achieved, the awarded badge is stored in the Database and displayed to the user. 2. If no milestone is achieved, no changes are made.

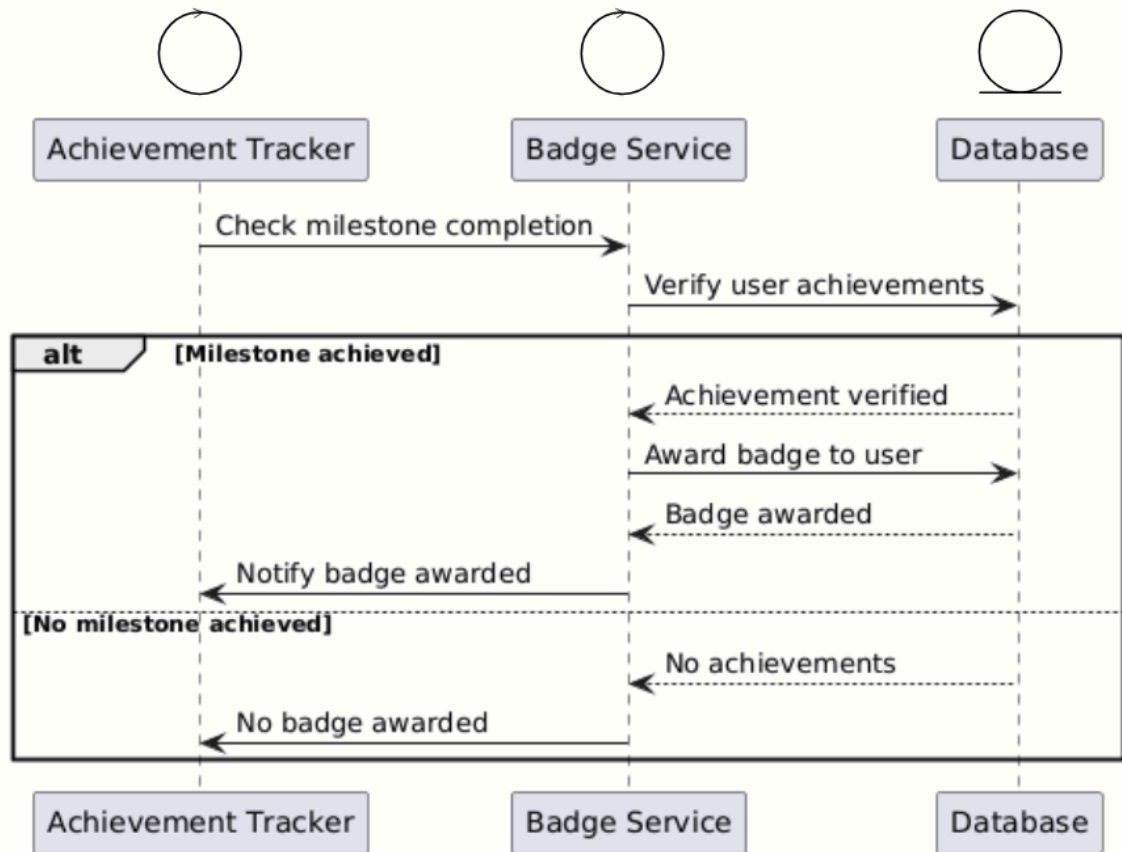


Figure 2.4.14.1: Sequence Diagram for Earn Badges

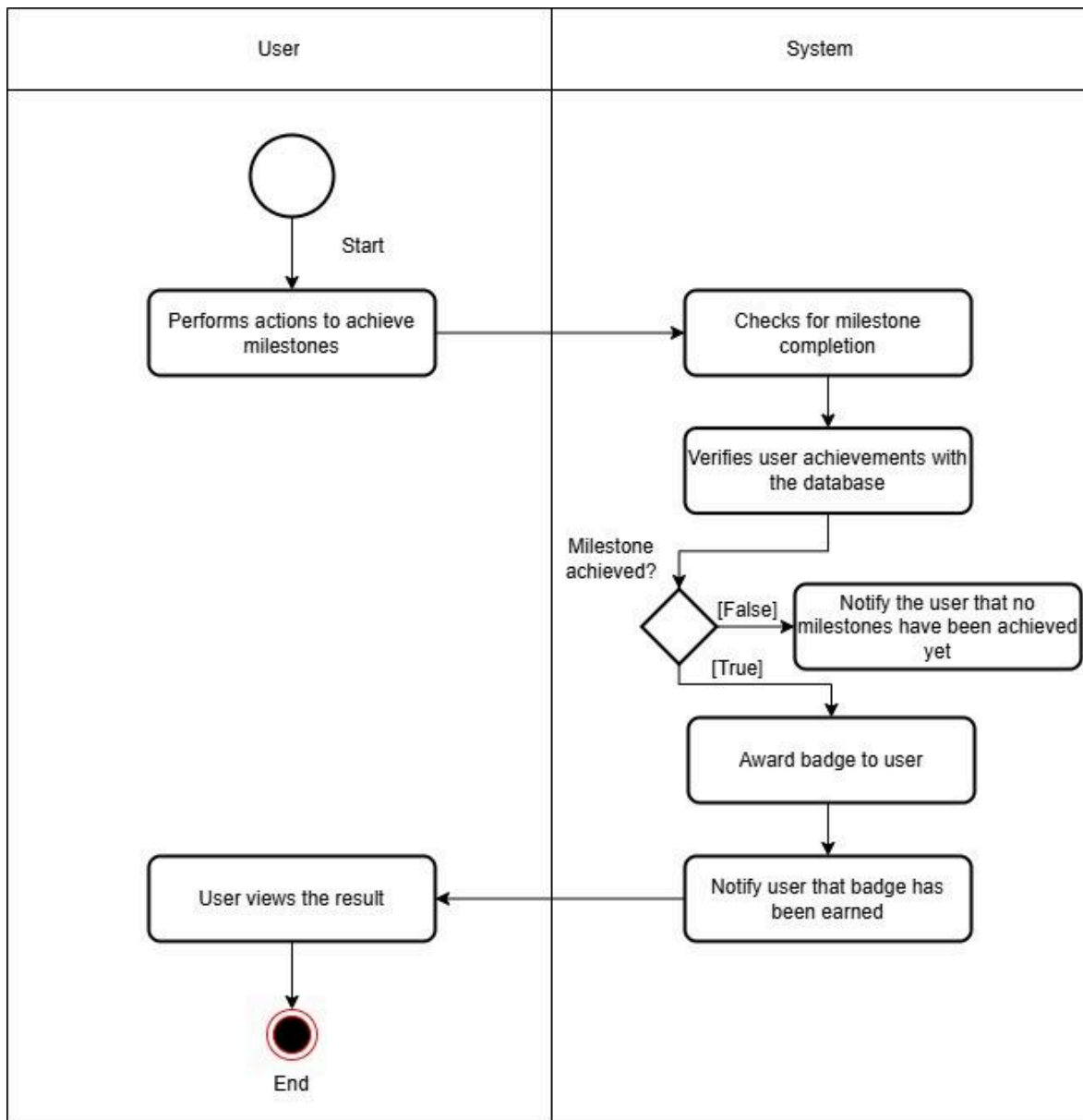


Figure 2.4.14.2: Activity Diagram for Earn Badges

2.4.15 US015 User Story <User Story 15>

Table 2.4.15.1: User Story Description for <Use Case : Add Knowledge Unit>

User story: Add Knowledge Unit
ID: US015
User Story Description As an admin, I want to add new knowledge units so that I can provide relevant mental health resources to users.
Actor : Admin
Preconditions : 1. The admin is logged into the system. 2. The admin has the required permissions to add content.
Flow of events: 1. The admin navigates to the "Add Knowledge Units" section. 2. The admin inputs the knowledge unit details, such as title, description, and content. 3. The admin uploads optional files (e.g., videos or images). 4. The admin reviews and confirms the entry. 5. The system saves the knowledge unit in the database and confirms the action.
Alternative flow : 1. Missing required details: 1.1 The system highlights missing fields and prompts the admin to complete them. 2. Invalid file format: 2.1 The system rejects the upload and displays an error message.
Postcondition : 1. The new knowledge unit is stored in the database. 2. The unit is available for users to browse.
Exception flow: 1. No internet connection: 1.1 The system displays an error message and does not save the entry. 2. Server error: 2.1 The system prompts the admin to retry later.

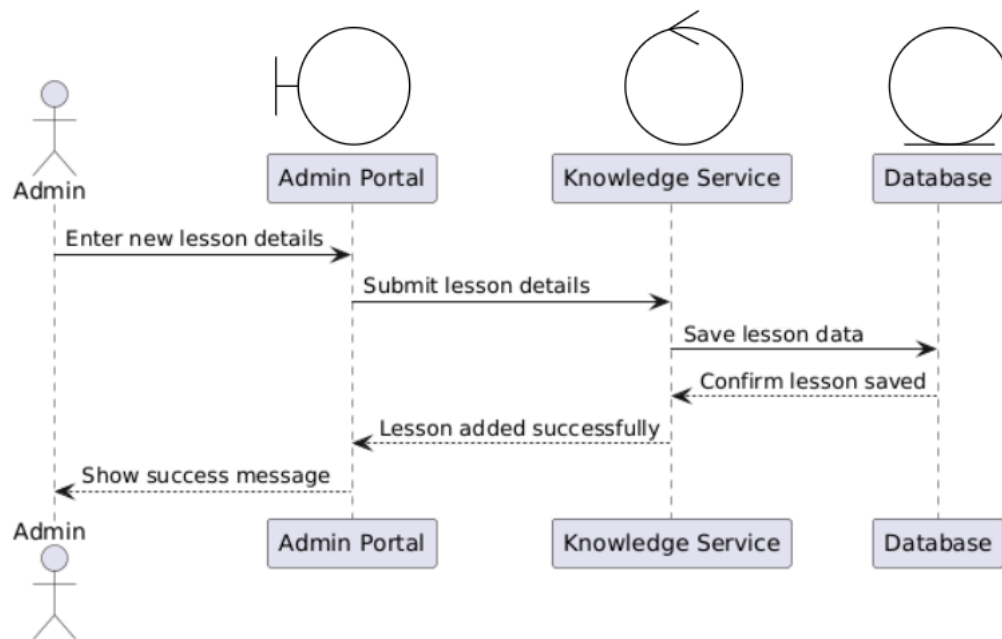


Figure 2.4.15.1: Sequence Diagram for Add Knowledge Unit

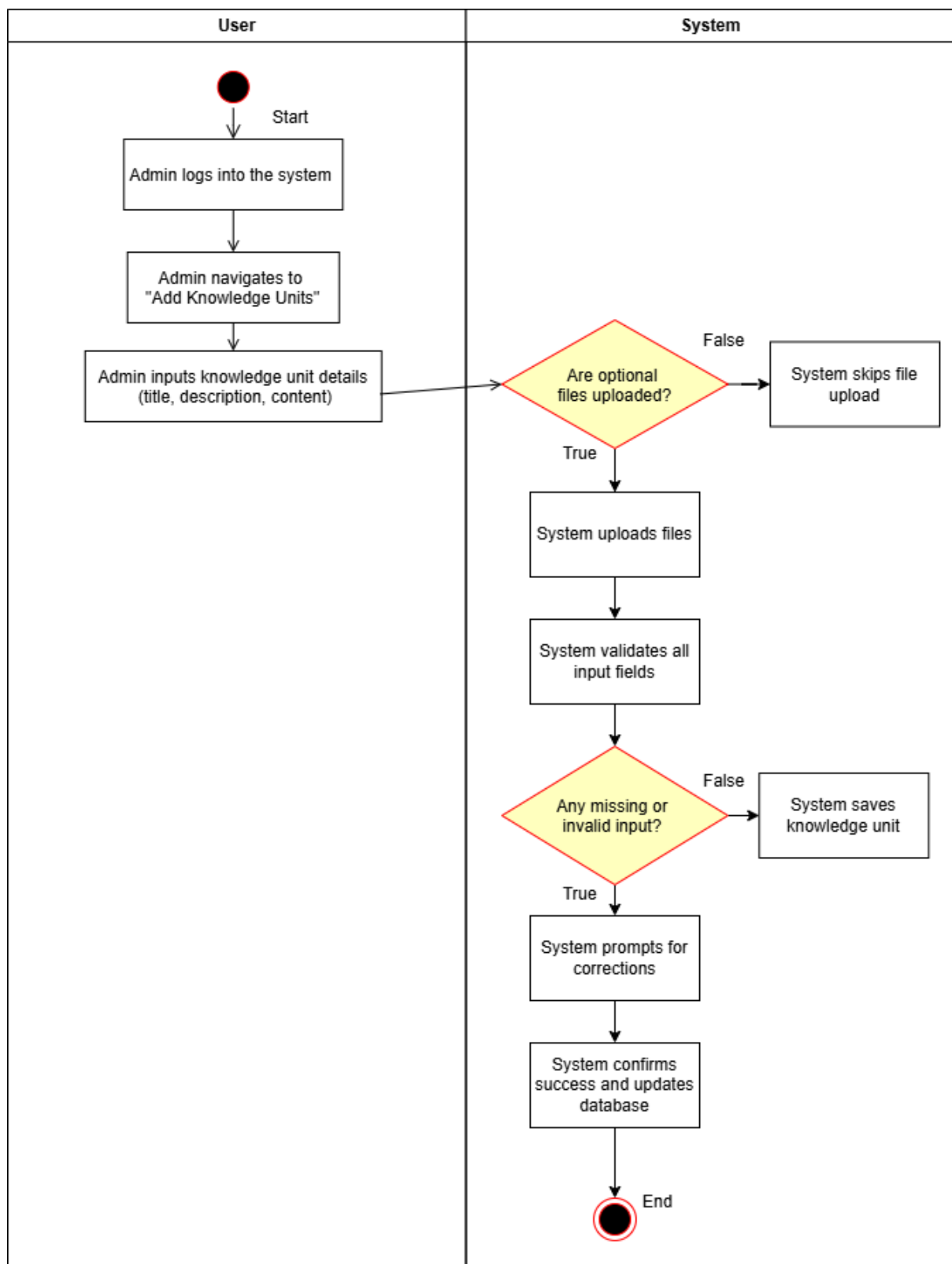


Figure 2.4.15.2: Activity Diagram for Add Knowledge Unit

2.4.16 US016 User Story <User Story 16>

Table 2.4.16.1: User Story Description for <Use Case : Resolve Flagged Issues>

User story: Resolve Flagged Issues
ID: US016
User Story Description As an admin, I want to resolve flagged issues so that I can ensure the accuracy and quality of the content.
Actor : Admin
Preconditions : 1. The admin is logged into the system. 2. The system contains flagged issues reported by users.
Flow of events: 1. The admin navigates to the "Resolve Flagged Issues" section. 2. The system displays a list of flagged issues. 3. The admin selects an issue and reviews the details. 4. The admin either edits the content or marks the issue as resolved. 5. The system updates the issue status in the database.
Alternative flow : 1. Admin dismisses an invalid flag: 1.1 The system marks the flag as "dismissed."
Postconditions : 1. The flagged issue status is updated to "resolved" or "dismissed."
Exception flow: 1. No internet connection: 1.1 The system displays an error and prevents updates. 2. System error: 2.1 The admin is prompted to retry.

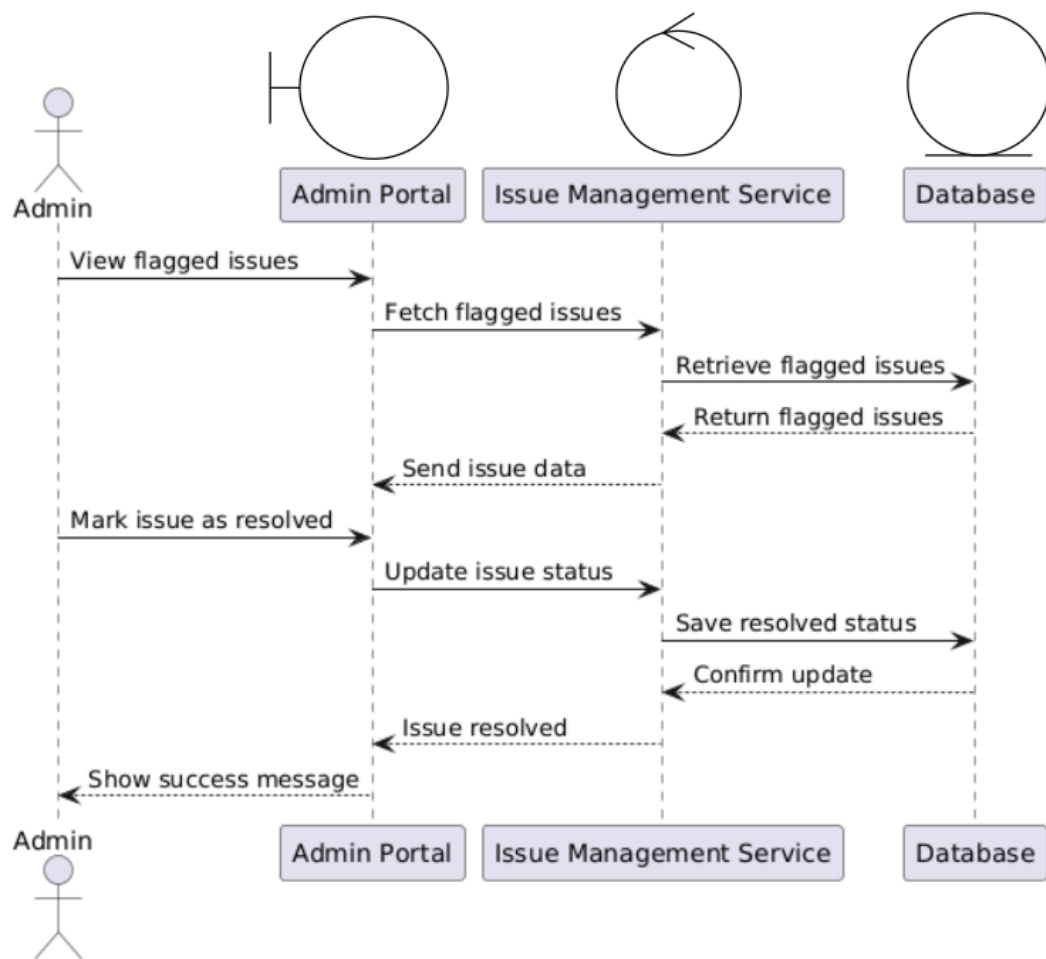


Figure 2.4.16.1: Sequence Diagram for Resolve Flagged Issues

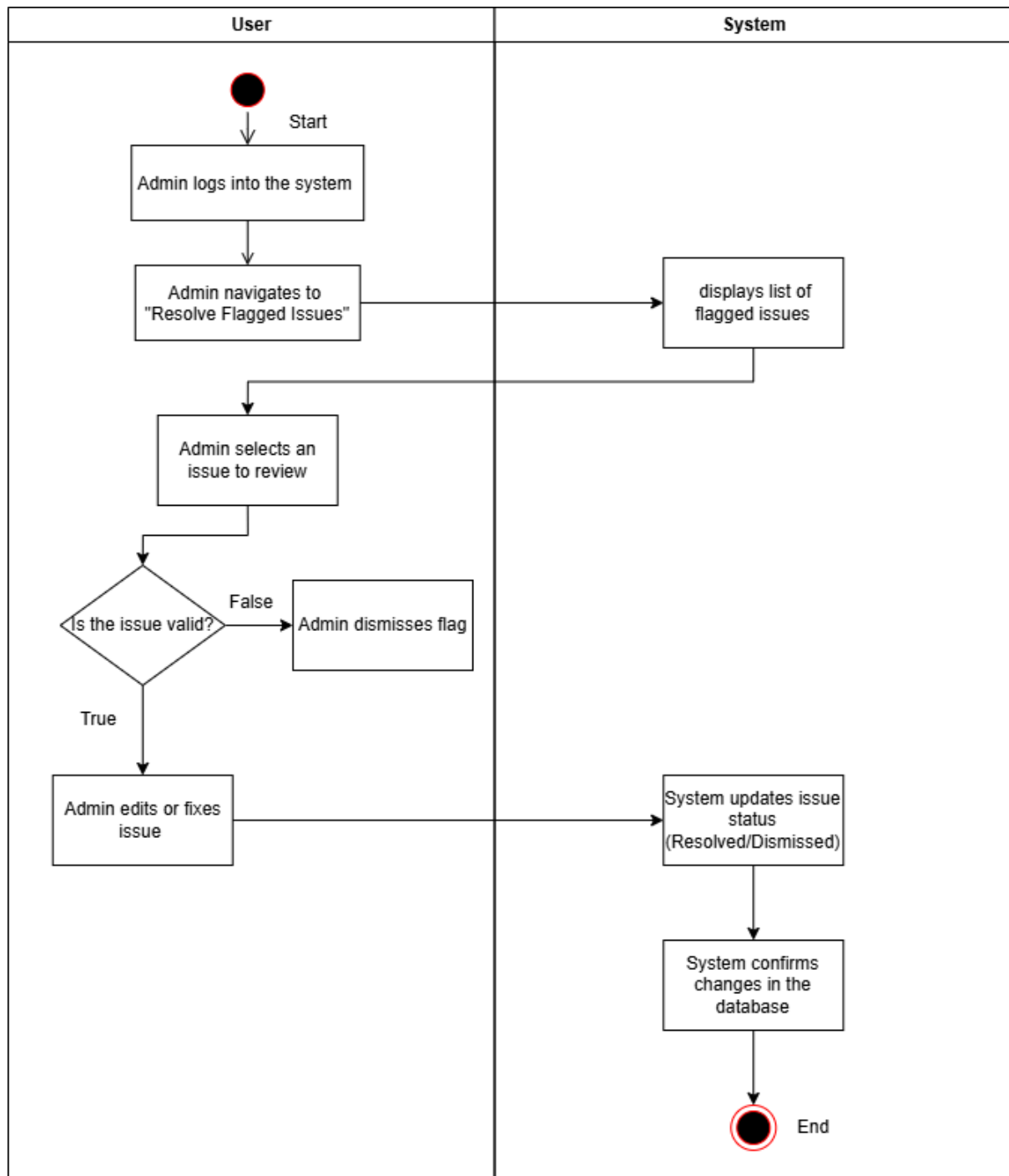


Figure 2.4.16.2: Activity Diagram for Resolve Flagged Issues

2.4.17 US017 User Story <User Story 17>

Table 2.4.17.1: User Story Description for <Use Case : Monitor Diagnostic Trends>

User story: Monitor Diagnostic Trends
ID: US017
User Story Description As an admin, I want to monitor diagnostic trends so that I can understand user behavior and update content accordingly.
Actor : Admin
Precondition : 1. The admin is logged into the system. 2. Diagnostic data is available in the system.
Flow of events: 1. The admin navigates to the "Monitor Diagnostic Trends" section. 2. The system generates a dashboard displaying user diagnostic trends. 3. The admin applies filters to analyze trends (e.g., time range, demographics). 4. The system displays insights and trends based on the selected filters.
Alternative flow : 1. No relevant data for selected filters: 1.1 The system displays a "No data available" message.
Postconditions : 1. The admin gains insights into user trends. 2. Data is used to adjust content or strategy.
Exception flow: 1. No internet connection: 1.1 The system cannot load the dashboard. 2. System error: 2.1 The admin is prompted to retry.

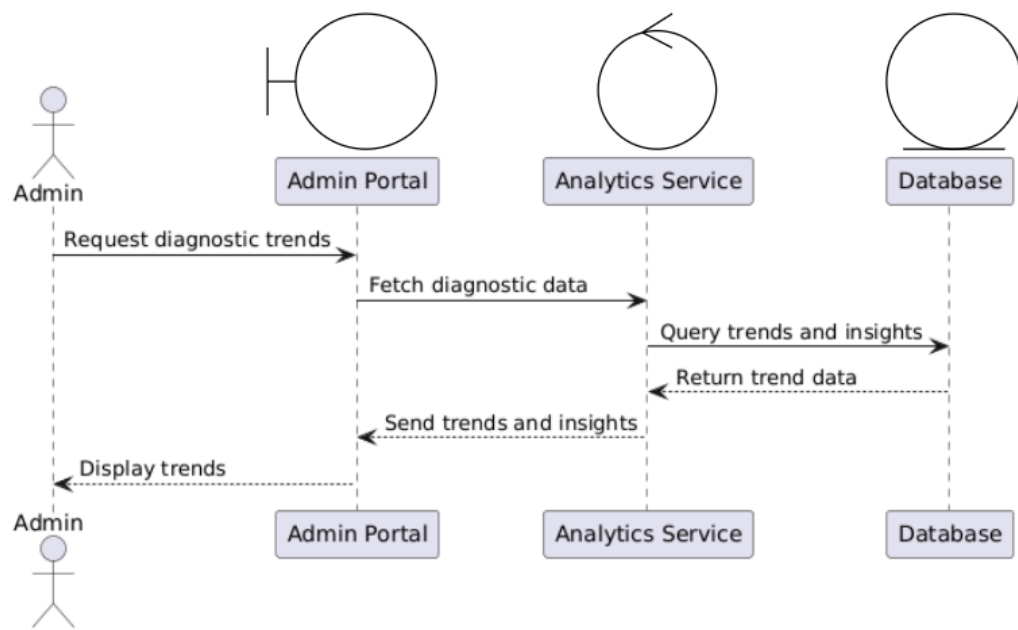


Figure 2.4.17.1: Sequence Diagram for Monitor Diagnostic Trends

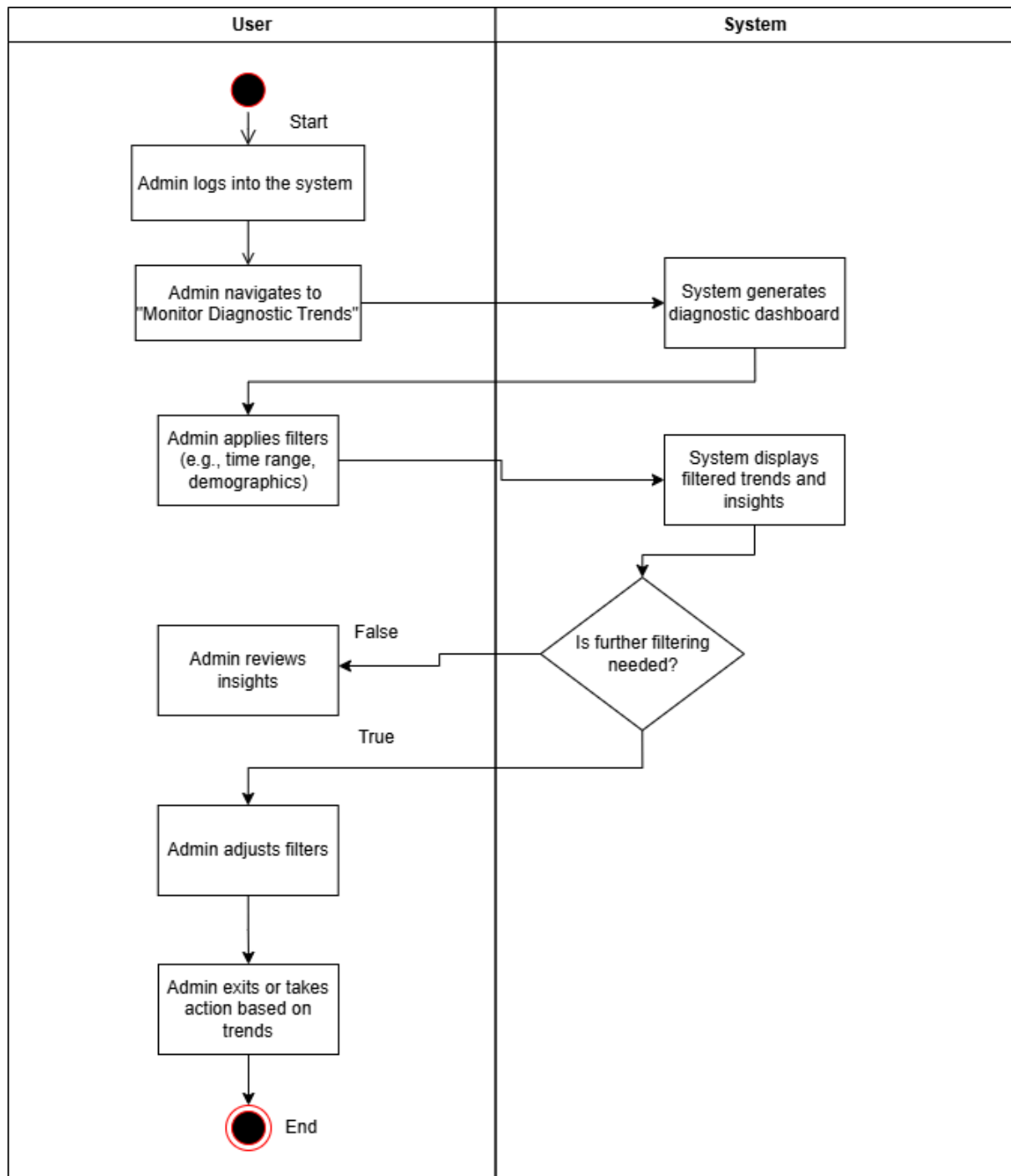


Figure 2.4.17.2: Activity Diagram for Monitor Diagnostic Trends

2.4.18 US0018 User Story <User Story 18>

Table 2.4.18.1: User Story Description for <Use Case : Create Knowledge Units >

User story:Create Knowledge Units
ID: US018
User Story Description As a mental health professional, I want to create knowledge units so that I can share accurate and valuable mental health content.
Actor : Mental Health Professional
Precondition : 1. The user is logged in as a content creator. 2. The user has the required permissions to create content.
Flow of events: 1. The user navigates to the "Create Knowledge Units" section. 2. The user inputs details such as title, description, and content. 3. The user uploads optional attachments (e.g., videos, files). 4. The user previews and submits the knowledge unit. 5. The system saves the unit and notifies the admin for approval (if required).
Alternative flow : 1. Missing required fields: 1.1 The system highlights incomplete fields and prompts corrections.
Postconditions : 1. The knowledge unit is saved in the system. 2. The unit becomes available for users after approval (if applicable).
Exception flow: 1. No internet connection: 1.1 The system prevents saving and notifies the user. 2. Server error: 2.1 The user is prompted to retry later.

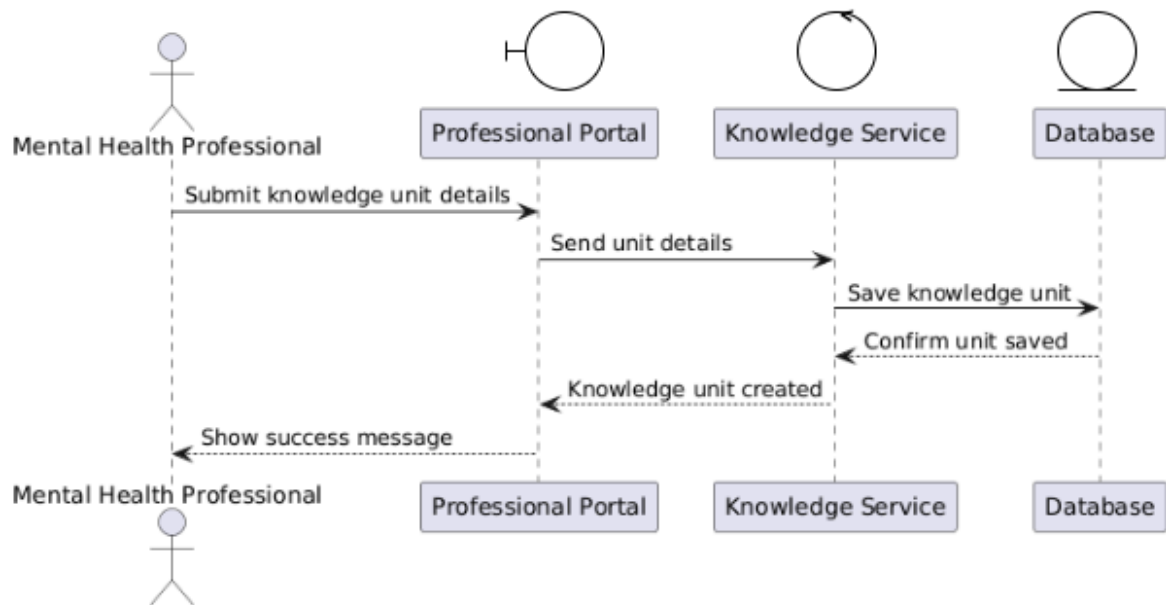


Figure 2.4.18.1: Sequence Diagram for Create Knowledge Units

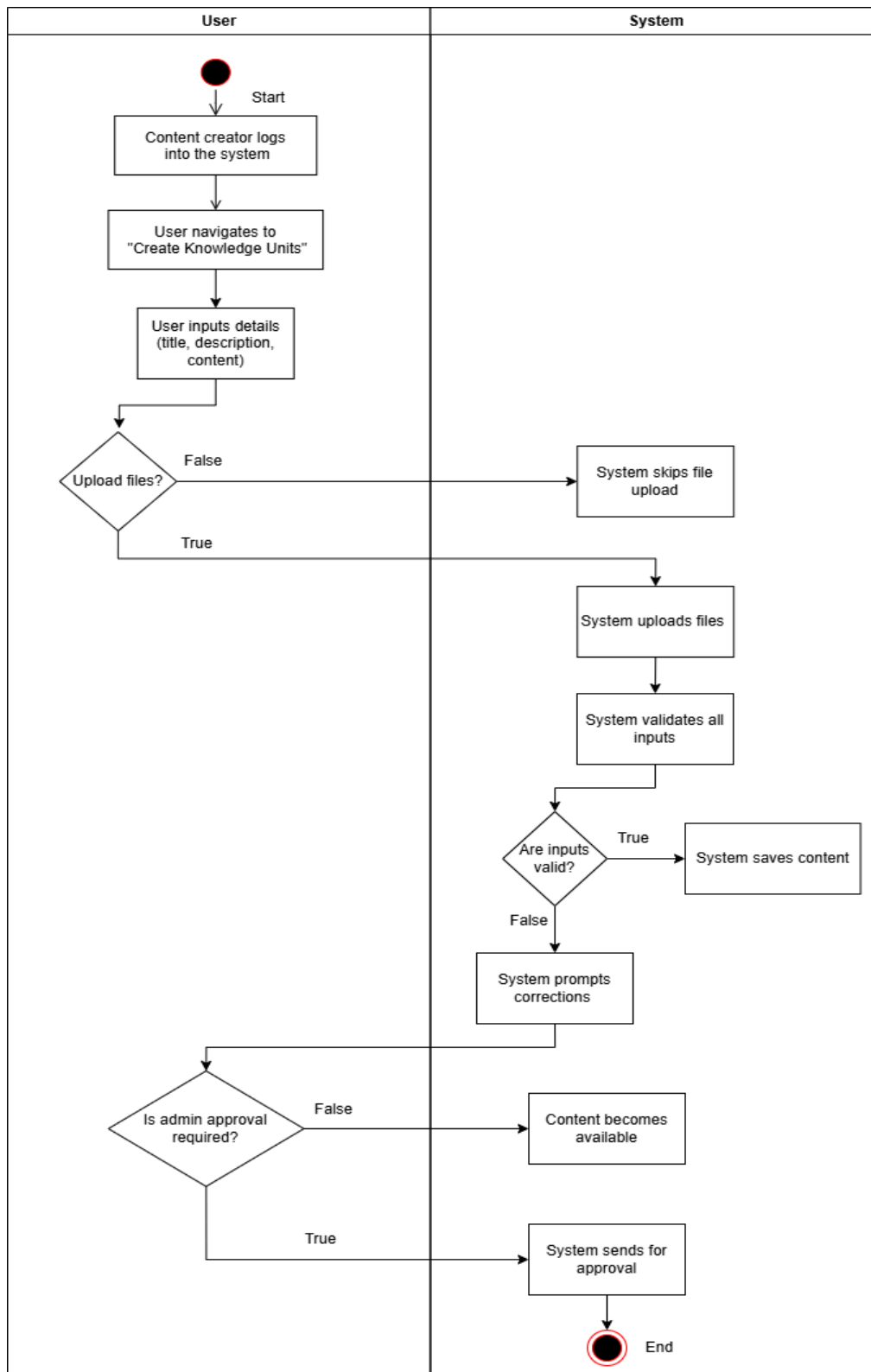


Figure 2.4.18.2: Activity Diagram for Create Knowledge Units

2.5 Performance and Other Requirements

Performance requirements :

Response time :

- The system should respond to user requests, such as chatbot queries, content browsing, and diagnostic assessments, within 2 seconds under normal conditions.
- This ensures that users can interact with the system quickly, without unnecessary delays.
- Real-time features, including Virtual Counseling and AI Chatbot interactions, must be completed with less than 1 second latency for uninterrupted communication.

Throughput :

- The system should be able to process 100 simultaneous submissions per second for features such as Diagnostic Questionnaires without any performance degradation.
- This ensures that the system can handle high levels of user engagement during peak periods.

Capacity :

- The system must support 5,000 concurrent video counseling sessions during peak usage hours without affecting performance.
- This requires an optimized architecture capable of handling real-time video communication.

Availability :

- The system must maintain an uptime of 99.9% annually, ensuring users can reliably access critical features like booking appointments and engaging in virtual counseling.
- There should be scheduled downtime for maintenance with minimal user impact.

Software system attributes :

Usability :

- The system must offer an intuitive and user-friendly interface that allows users of all technical backgrounds to easily navigate through features like the Peer Support Forum, Diagnostic Questionnaires, and Mental Health Awareness content.
- The platform must be accessible to users with varying linguistic skills, supporting multilingual interfaces.
- The system should feature clear navigation and well-labeled sections for users to quickly find what they need.

Reliability :

- The system must be fault-tolerant, ensuring that it continues to operate smoothly even in the event of hardware or software failures.
- Daily backups of user data must be conducted, with recovery possible within 30 minutes in the event of a system failure.

Maintainability :

- The system must be designed for easy maintenance, allowing for module updates, bug fixes, and feature additions without disrupting existing functionalities.

- The source code should follow industry-standard coding practices, with clear documentation to ensure ease of debugging and future enhancements.

Scalability :

- The system must be able to scale efficiently to accommodate future growth, supporting increasing numbers of concurrent users and larger datasets.
- As the user base grows, the system should handle additional load seamlessly without the need for a complete redesign.

Portability :

- The system must be compatible across multiple platforms and devices, including iOS, Android, and web browsers.
- Users should be able to access the system on any device, whether mobile or desktop.

Other requirements :

System requirements :

- The system must prioritize robust security features to protect sensitive user data, such as personal information, diagnostic results, and health data.
- All data must be encrypted both at rest (while stored) and in transit (during transmission) using industry-standard encryption protocols like AES-256 and TLS.
- User accounts must be secured with two-factor authentication (2FA) to prevent unauthorized access.
- The system must implement role-based access control (RBAC), ensuring that users (e.g., family members) cannot access sensitive data that they are not authorized to view.

Compliance requirements :

- The system must comply with relevant local and international data protection laws such as Malaysia's PDPA and the EU's GDPR to ensure legal compliance for user data handling.
- The system must ensure that all translations of materials are culturally sensitive and comply with local regulations on mental health education.

Environmental requirements :

- The platform must function in low-bandwidth environments, particularly in underserved or rural areas, with automatic fallback to text-based content when video or real-time features are not feasible due to connectivity issues.
- This ensures accessibility for users who may not have access to high-speed internet.

2.6 Design Constraints

Environmental constraints : These constraints relate to the physical and operational settings in which the system will function, including device usage conditions, connectivity, and user environments. The MindWell platform must perform effectively across diverse settings, whether used at home, in a workplace, or during commutes. To achieve this, the system requires a responsive design that adapts to various screen sizes and orientations, including smartphones, tablets, and desktops. It should also operate reliably in low-bandwidth environments to ensure accessibility for users in underserved or rural areas without interruptions. Furthermore, the system should support offline functionality for saved resources, such as bookmarked knowledge units or toolbox activities, with automatic synchronization when internet connectivity is restored. An example of environmental constraints is "The system must allow offline access to bookmarked knowledge units and synchronize data when connectivity is re-established."

Hardware and software constraints : These constraints relate to the device specifications and operating systems required for the system to function optimally. To ensure accessibility for users from various socioeconomic backgrounds, the system must operate on devices with limited specifications, such as smartphones with a minimum of 2 GB RAM and 32 GB of storage space. The system must also be compatible with major operating systems, including Android 9+, iOS 12+, and common web browsers like Chrome, Firefox, and Safari. Additionally, the system must avoid requiring high-end hardware, ensuring functionality on mid-range devices while maintaining compatibility with older versions of operating systems. For example, "The system must support devices with as little as 2 GB RAM, ensuring it remains functional on budget smartphones."

Security and privacy constraints : These constraints relate to the protection of sensitive user data and compliance with legal standards. Given the sensitive nature of mental health data, the MindWell system must adhere to the highest standards of security and privacy by encrypting all data at rest using AES-256 and data in transit using TLS protocols. It must also implement two-factor authentication (2FA) for user accounts to enhance security. The system must ensure compliance with Malaysia's Personal Data Protection Act (PDPA) and global regulations such as GDPR for user data handling. Lastly, it must offer users complete control over their data, including the ability to export or delete their records at any time. For instance, "All user data must be encrypted using AES-256 to ensure security both in storage and during transmission."

Integration and compatibility constraints : These constraints relate to the ability of the system to function alongside other tools and remain compatible across updates. To deliver a seamless user experience, the system must integrate effectively with third-party services such as payment gateways for premium features and crisis support systems for immediate assistance. It should also support APIs for future extensions, including connections to wearable health devices or productivity apps. Furthermore, the system must maintain backward compatibility during updates to prevent disruptions for existing users. For example, "The system must integrate with popular payment gateways like PayPal and Stripe for processing premium subscriptions."

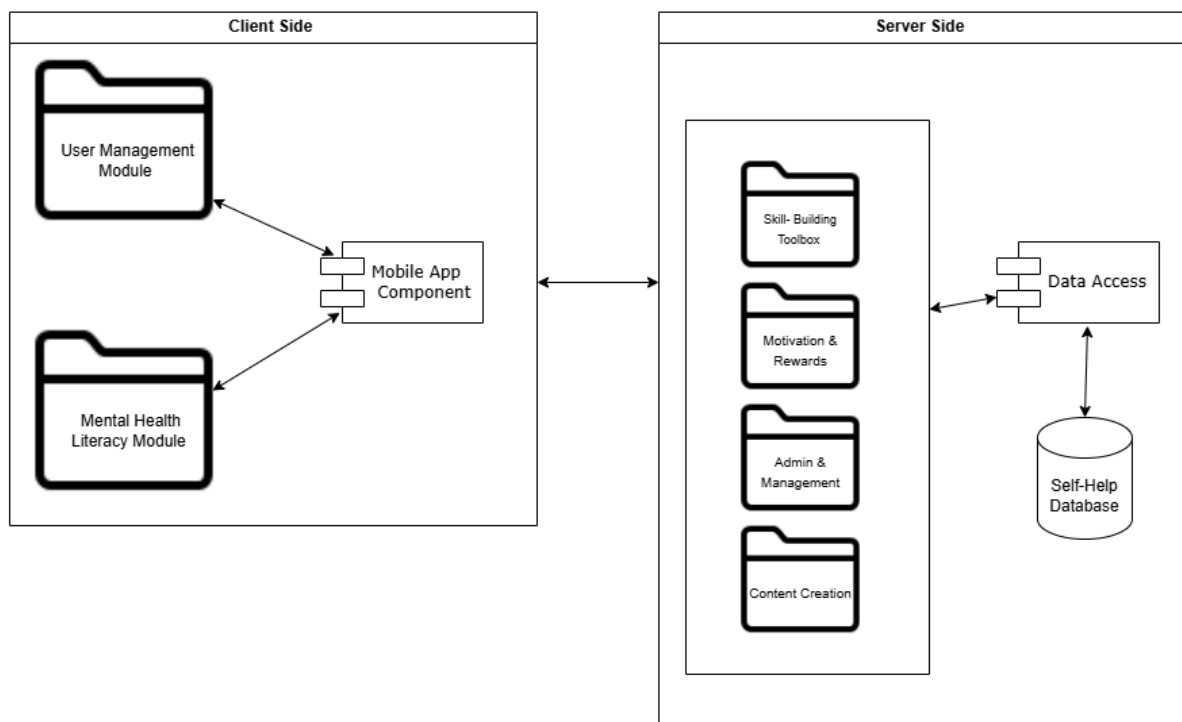
Performance constraints : These constraints relate to the system's ability to meet operational and user demand efficiently. The MindWell system must ensure high performance and responsiveness by supporting real-time chatbot interactions with a response time of under 2 seconds per query. The system should handle up to 1,000 concurrent users without compromising performance. Additionally, it must process large datasets, such as diagnostic trends or personalized content recommendations, within 5 seconds for visualization and analytics. For instance, "The system must provide chatbot responses within 2 seconds, even during peak usage periods."

3. Architectural Rationale

3.1 Architecture Style and Rationale

The architecture style that was chosen for this project is the client-server architecture style. The application on end user devices (client-side) interacts with a backend server that stores and processes data. This will ensure that user data is secure, accessible and centralized. The client-server architecture also ensures general functionality, where it is available to all clients and does not need to be implemented by all services.

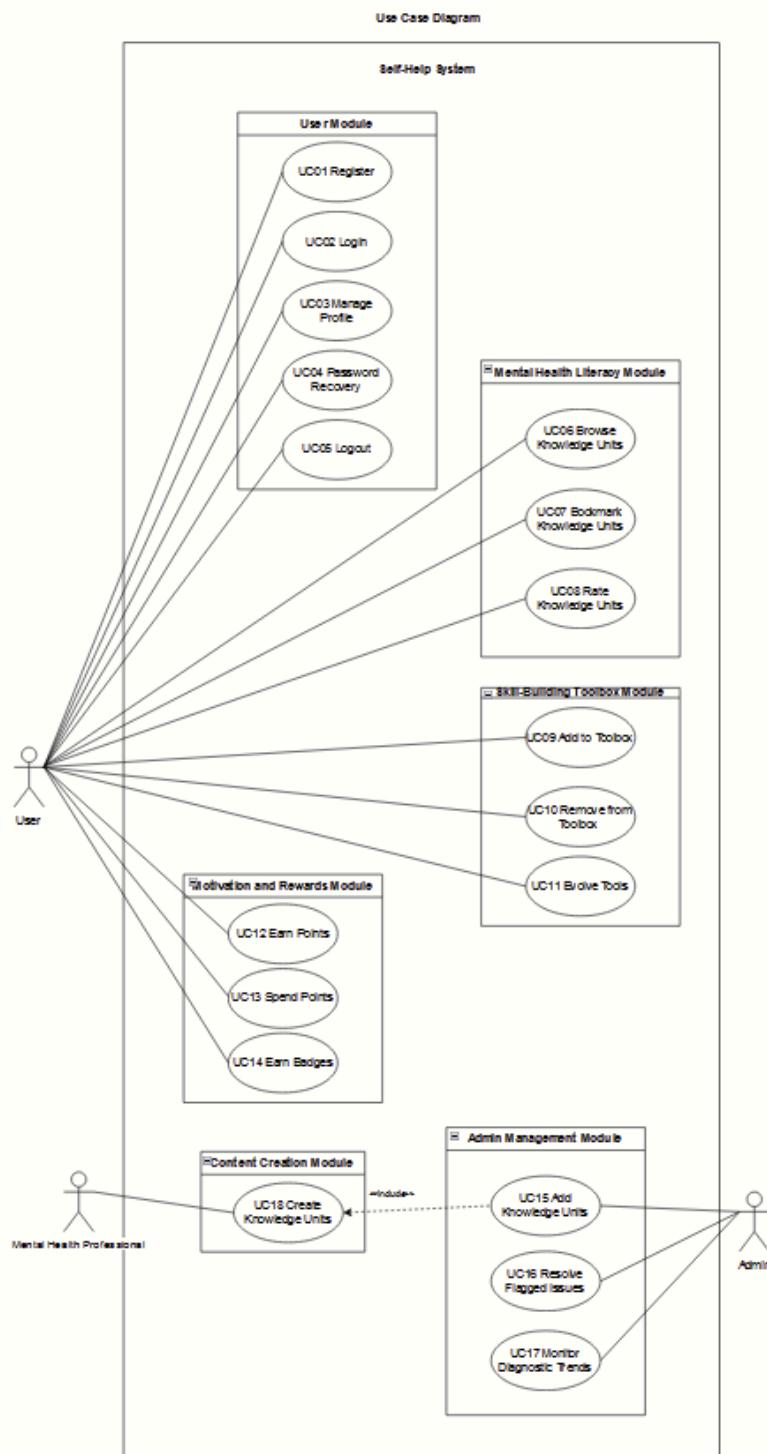
For this specific project, we go with a 'thin' client and a 'fat' server, where most of the processing and data storage occur at the server side, so that the client-side application is light and fast.



label

4. Architectural Views

4.1 Use Case View



Link to full image: draw.io

Figure 4.1.1: Use Case Diagram for Self Help System

4.2 Implementation View

The implementation view adopts a client-server architecture, with client-side modules like User Management and Mental Health Literacy, and server-side subsystems including Skill-Building Toolbox, Motivation and Rewards, Admin Management, and Content Creation. These components interact through the Self-Help Database, which serves as a central repository, ensuring modularity, scalability, and efficient client-server communication.

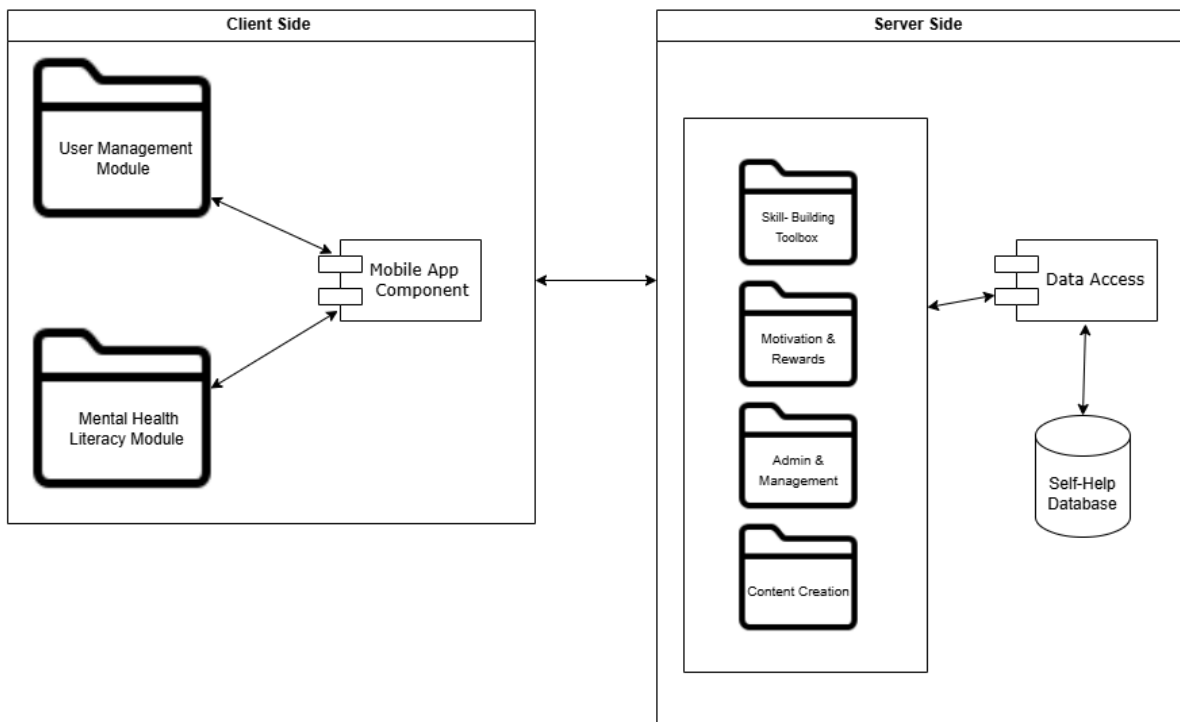


Figure 4.2: Component Diagram of <MindWell Self-Help & Diagnostic System>

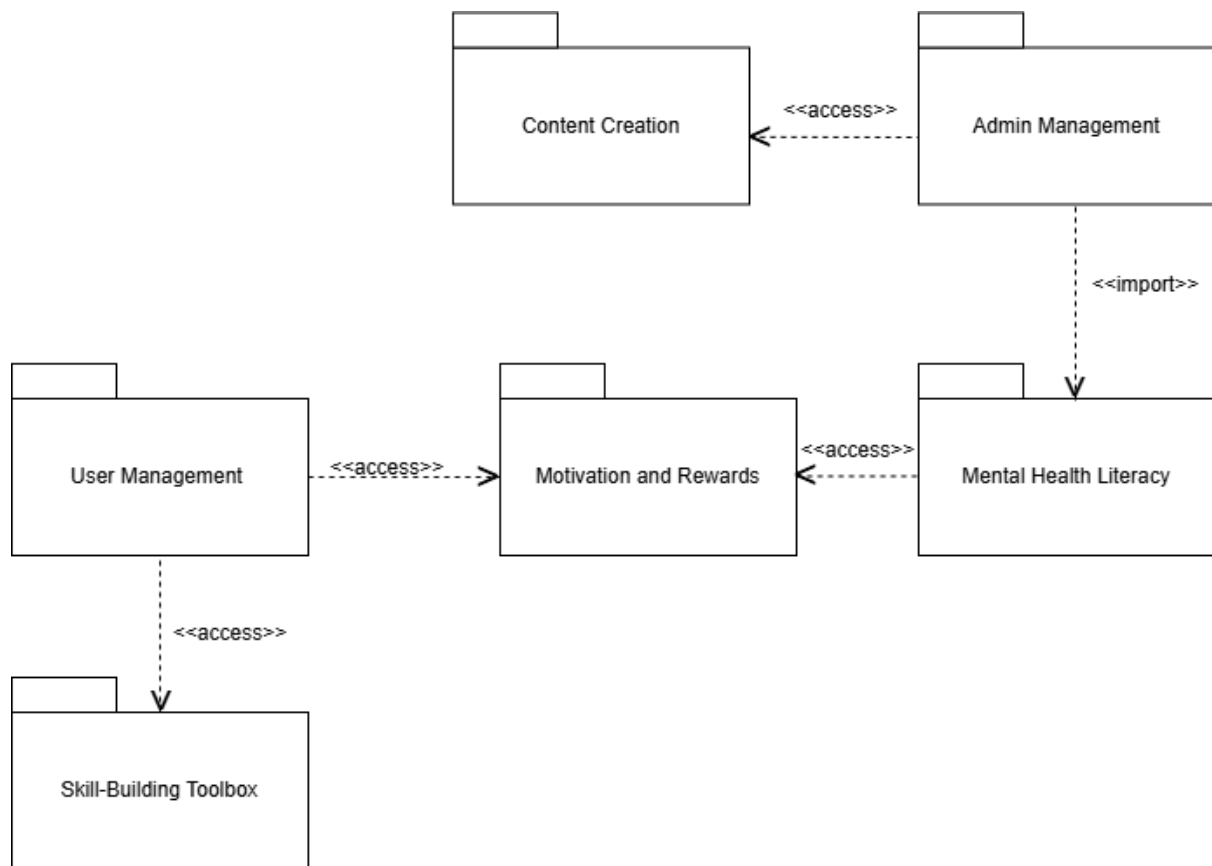


Figure 4.2.1: Package Diagram for <MindWell Self-Help & Diagnostic System>

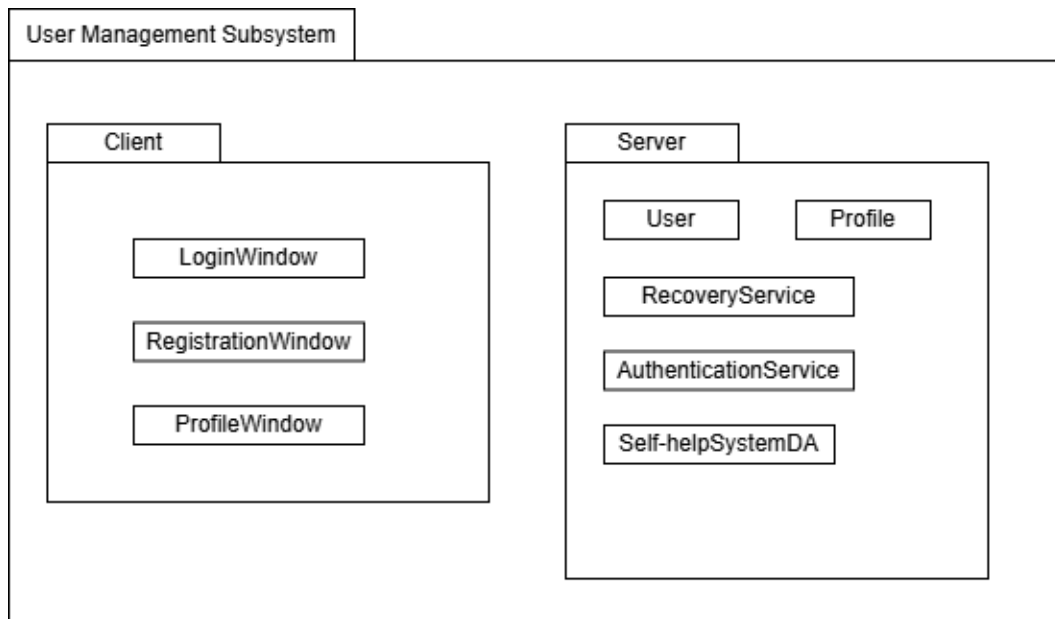


Figure 4.2.1.1: Package Diagram for <User Management> Subsystem

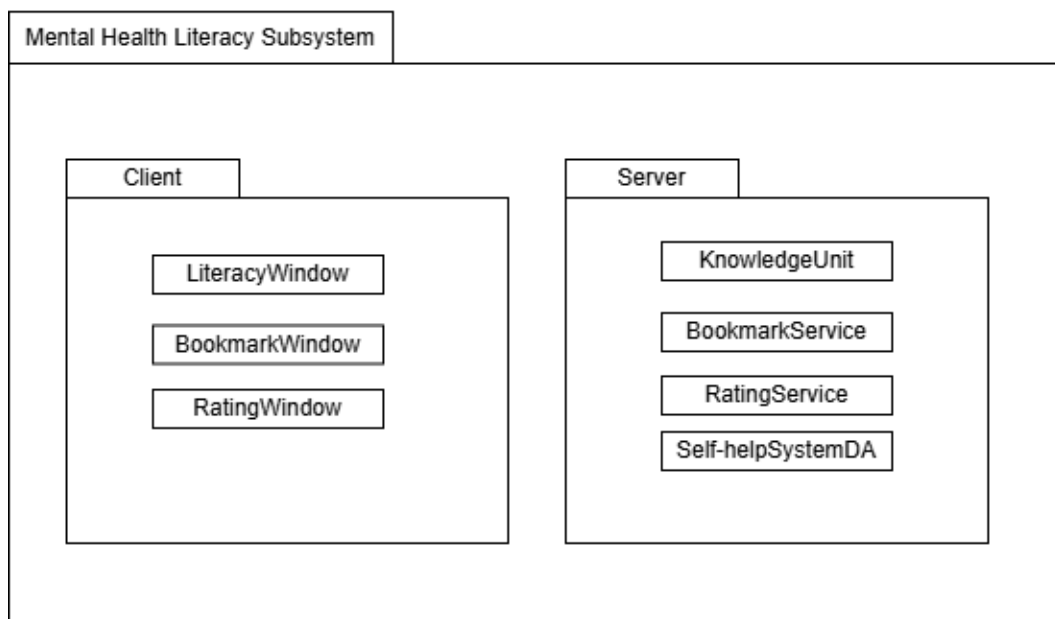


Figure 4.2.1.2: Package Diagram for <Mental Health Literacy> Subsystem

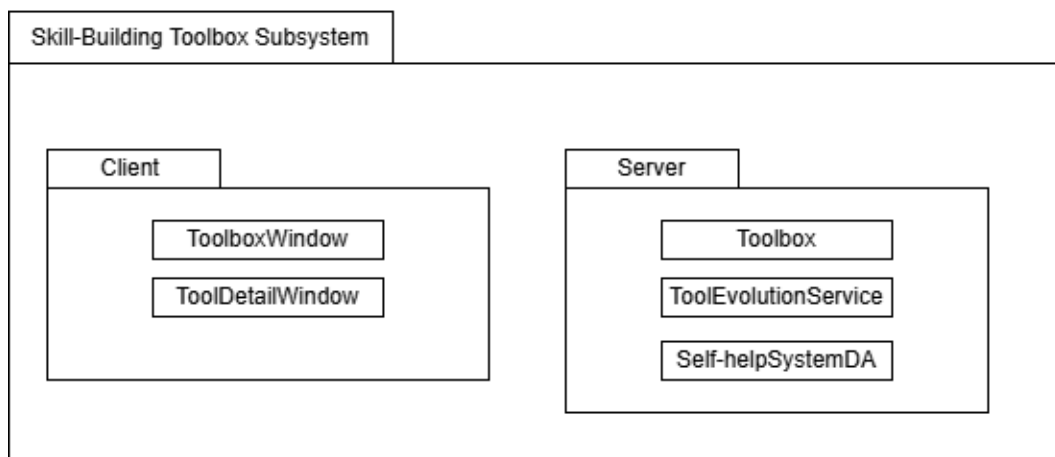


Figure 4.2.1.3: Package Diagram for <Skill-Building Toolbox> Subsystem

4.3 Logical View

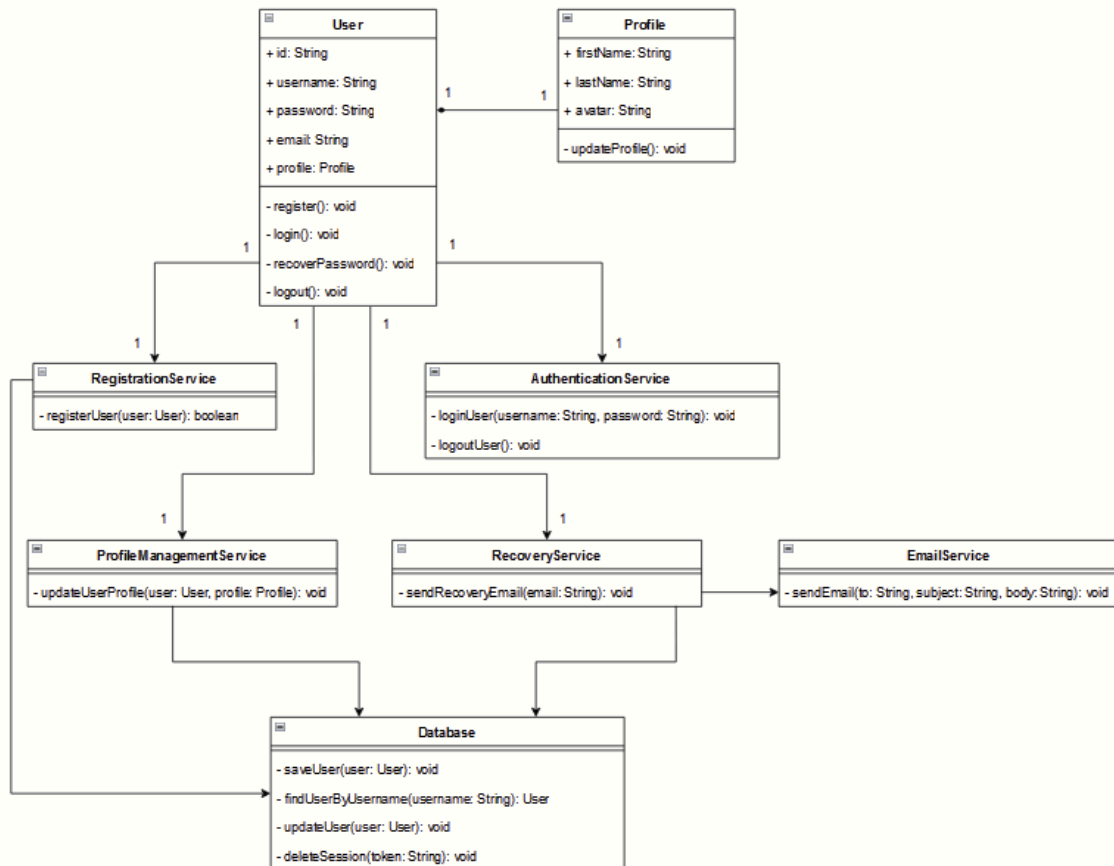


Figure 4.3.1: Class diagram for <User Management> Subsystem

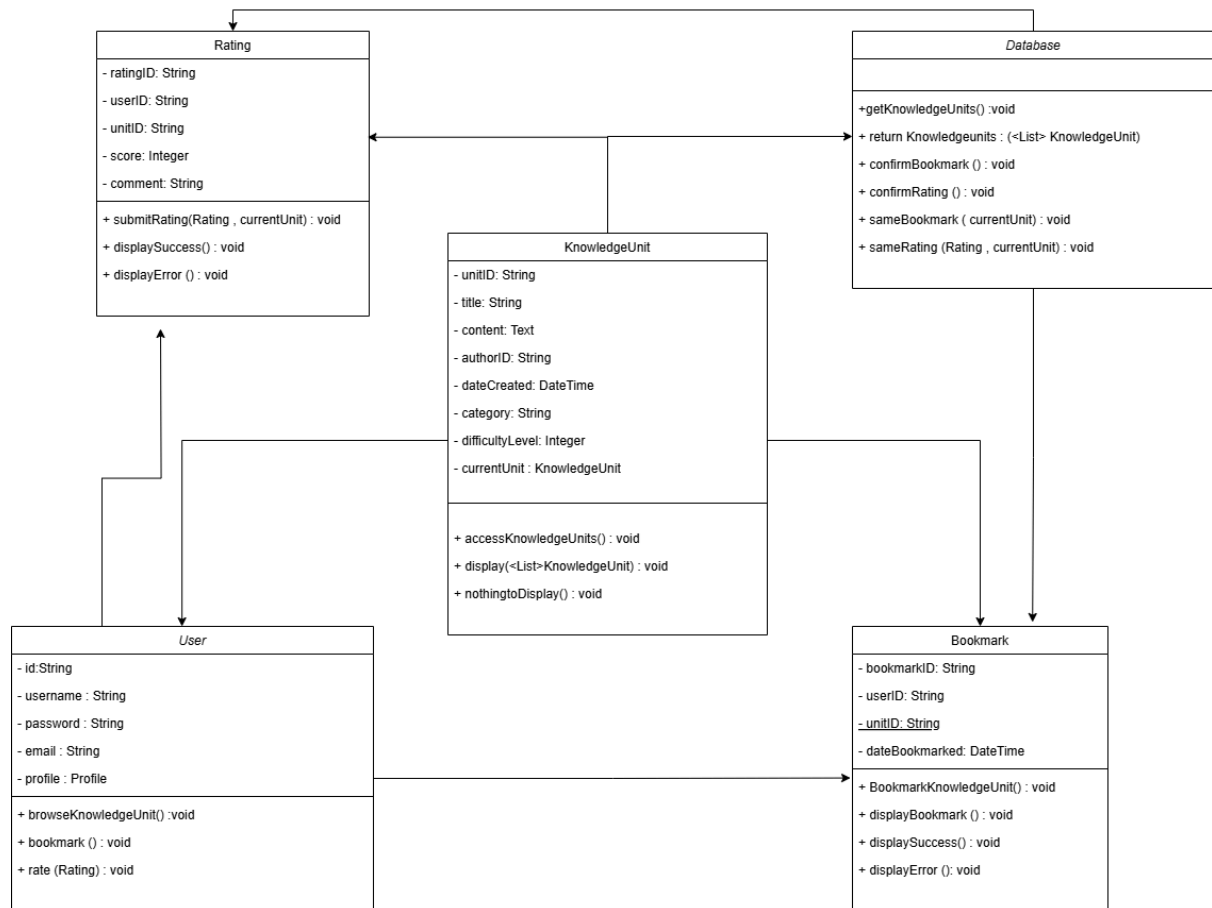


Figure 4.3.2: Class diagram for <Mental Health Literacy> Subsystem

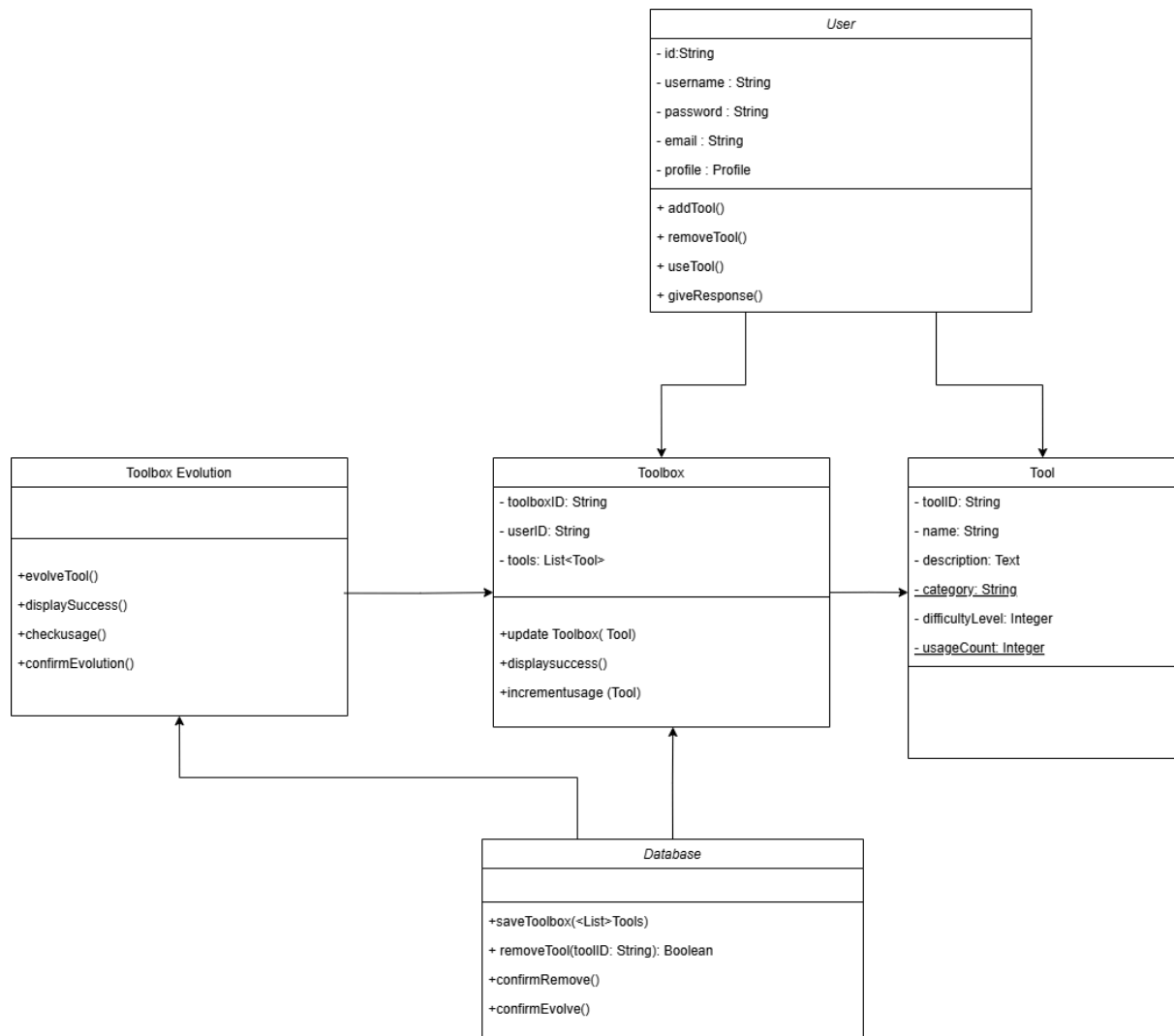


Figure 4.3.3: Class Diagram for <Skill-Building Toolbox > Subsystem

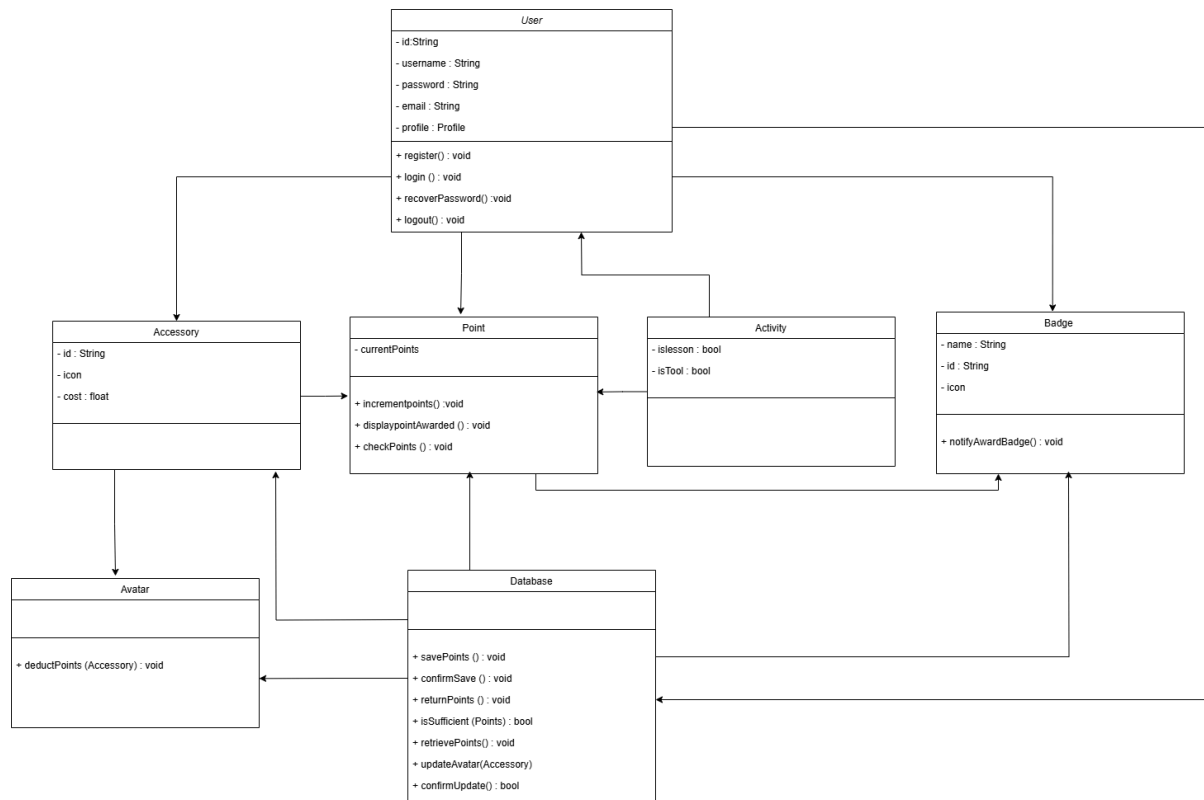


Figure 4.3.4: Class Diagram for <Motivation and Rewards> Subsystem

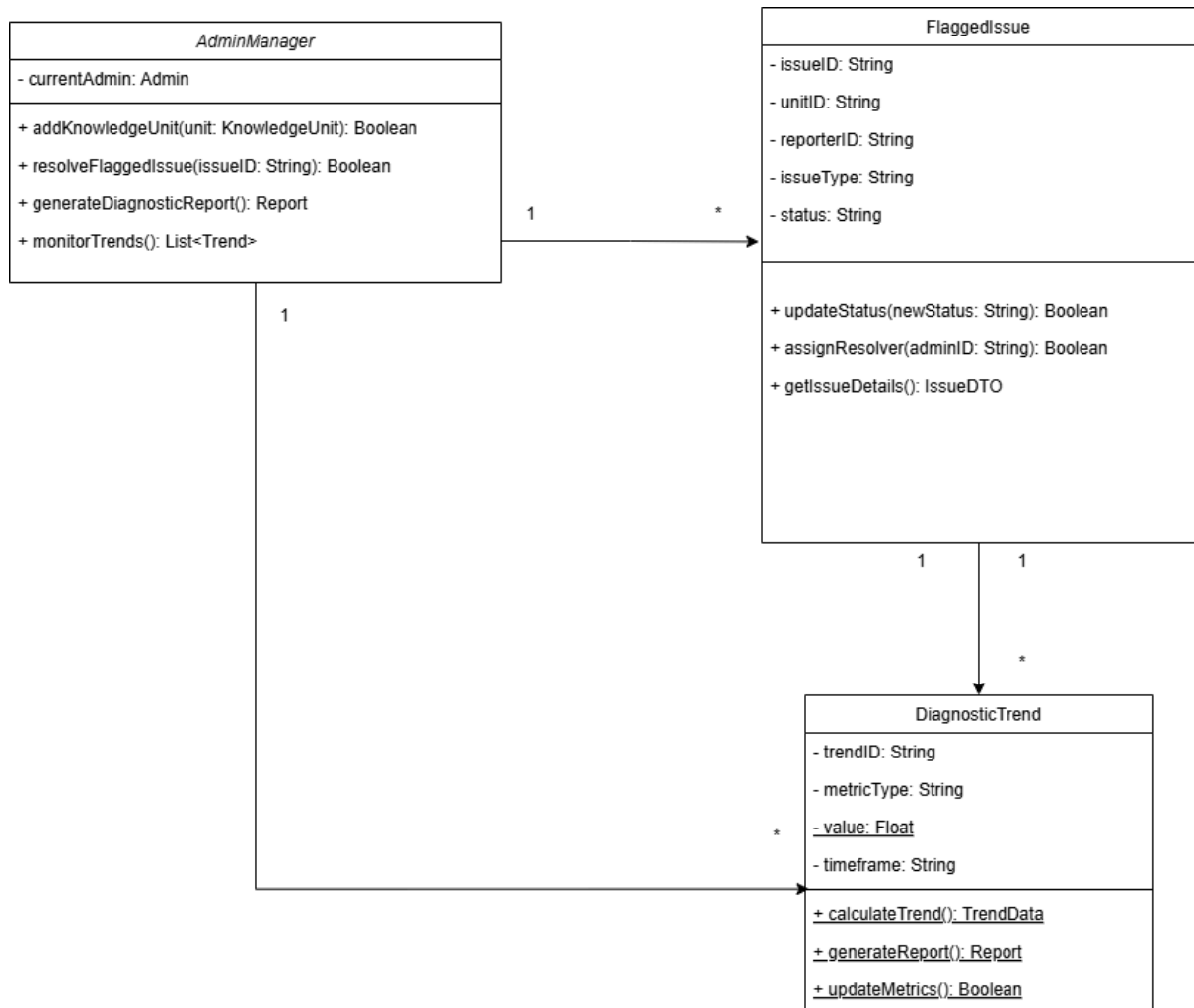


Figure 4.3.5: Class Diagram for <Admin Management> Subsystem

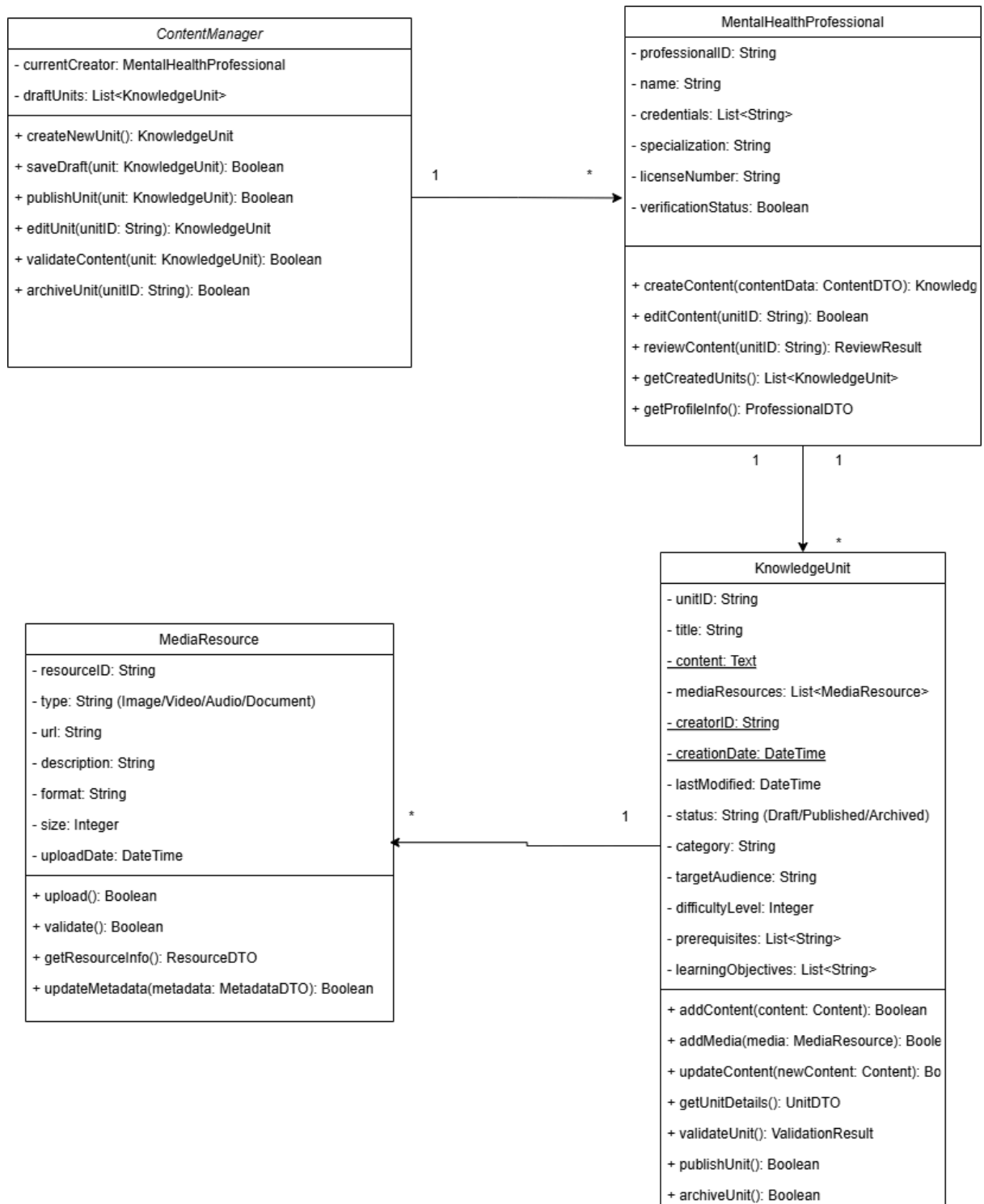


Figure 4.3.6: Class Diagram for <Content Creation> Subsystem

4.4 Process View

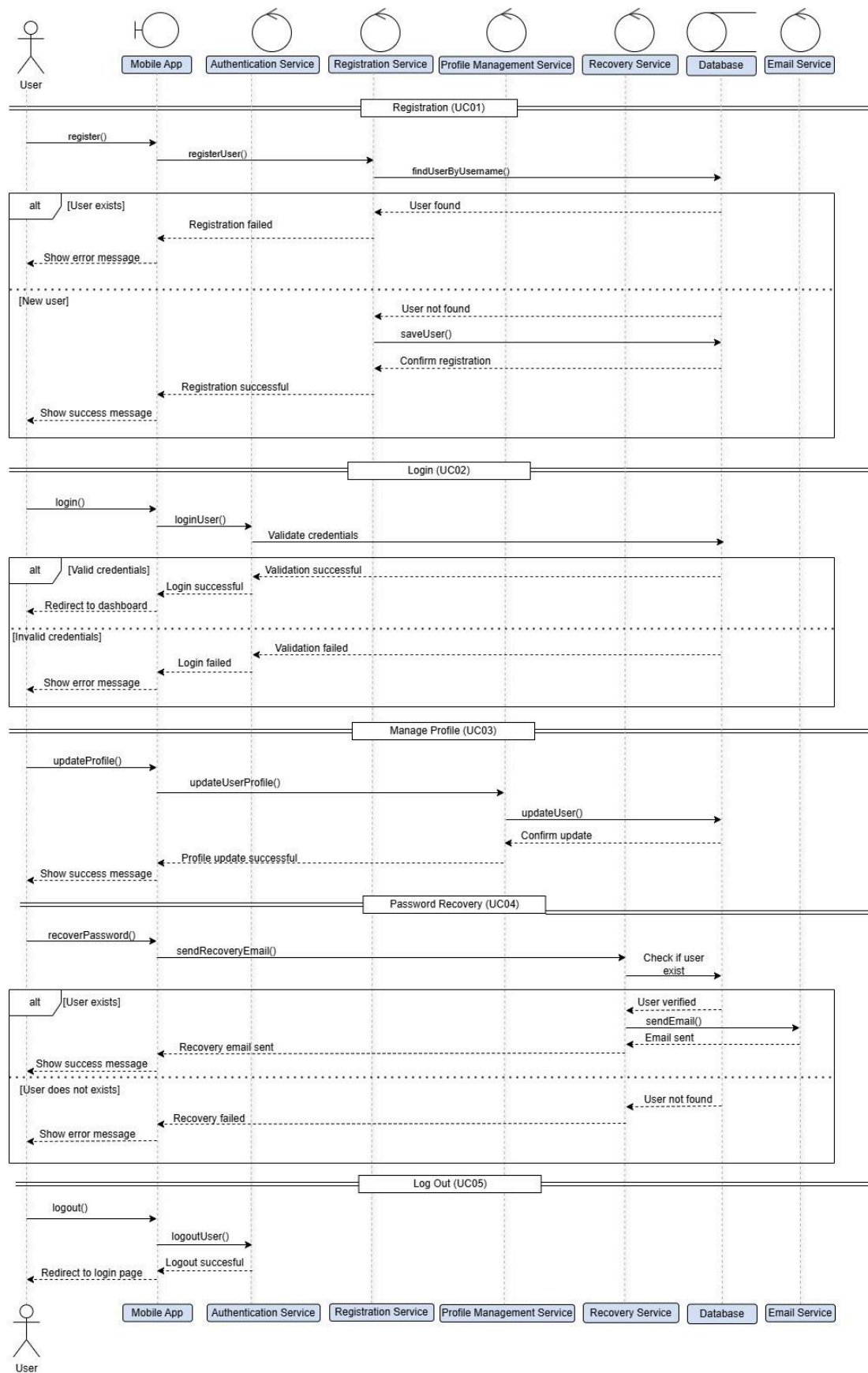


Figure 4.4.1: SD001 Sequence Diagram for <User Module Management>

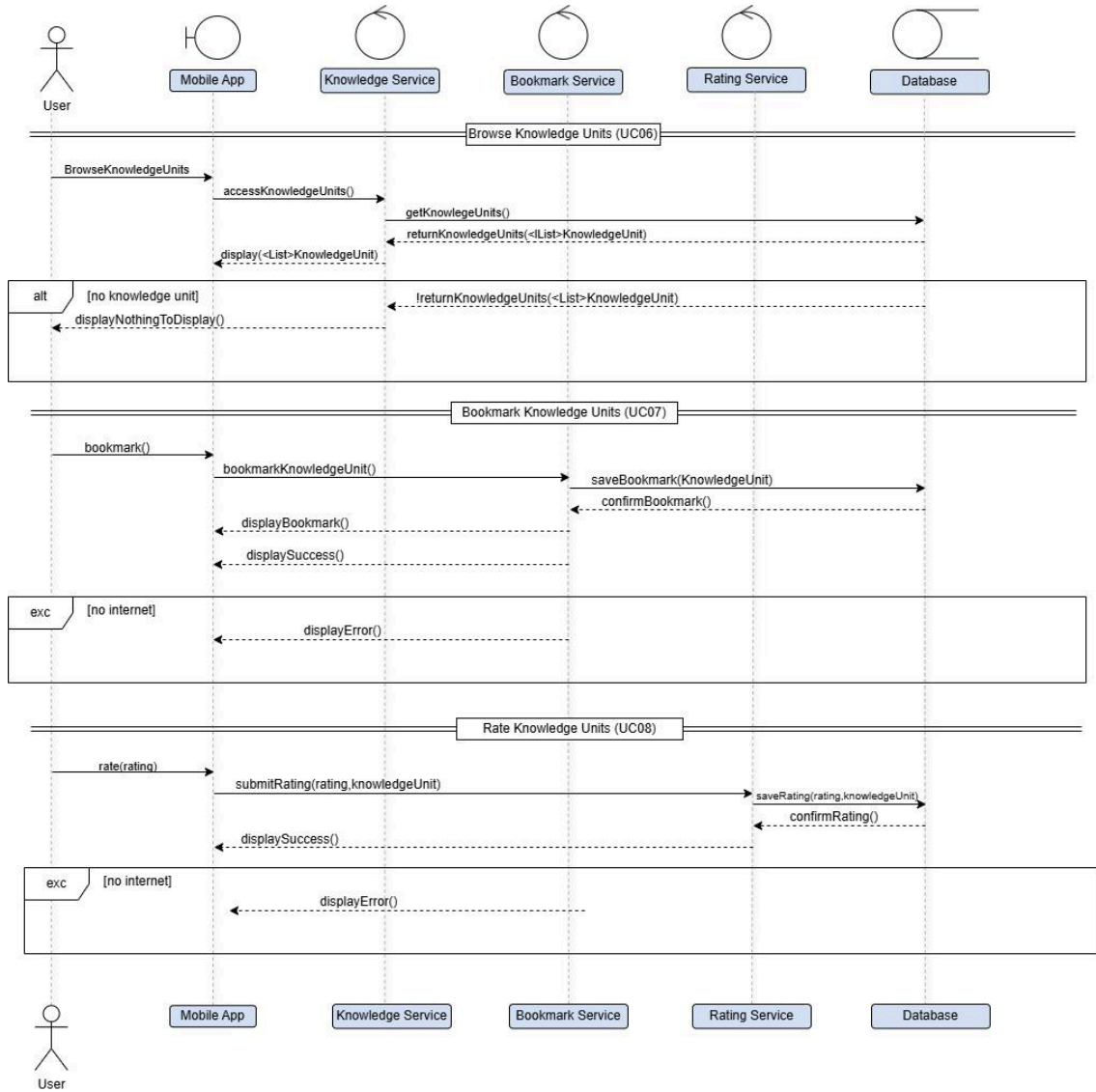


Figure 4.4.2: SD002 Sequence Diagram for <Mental Health Literacy Module>

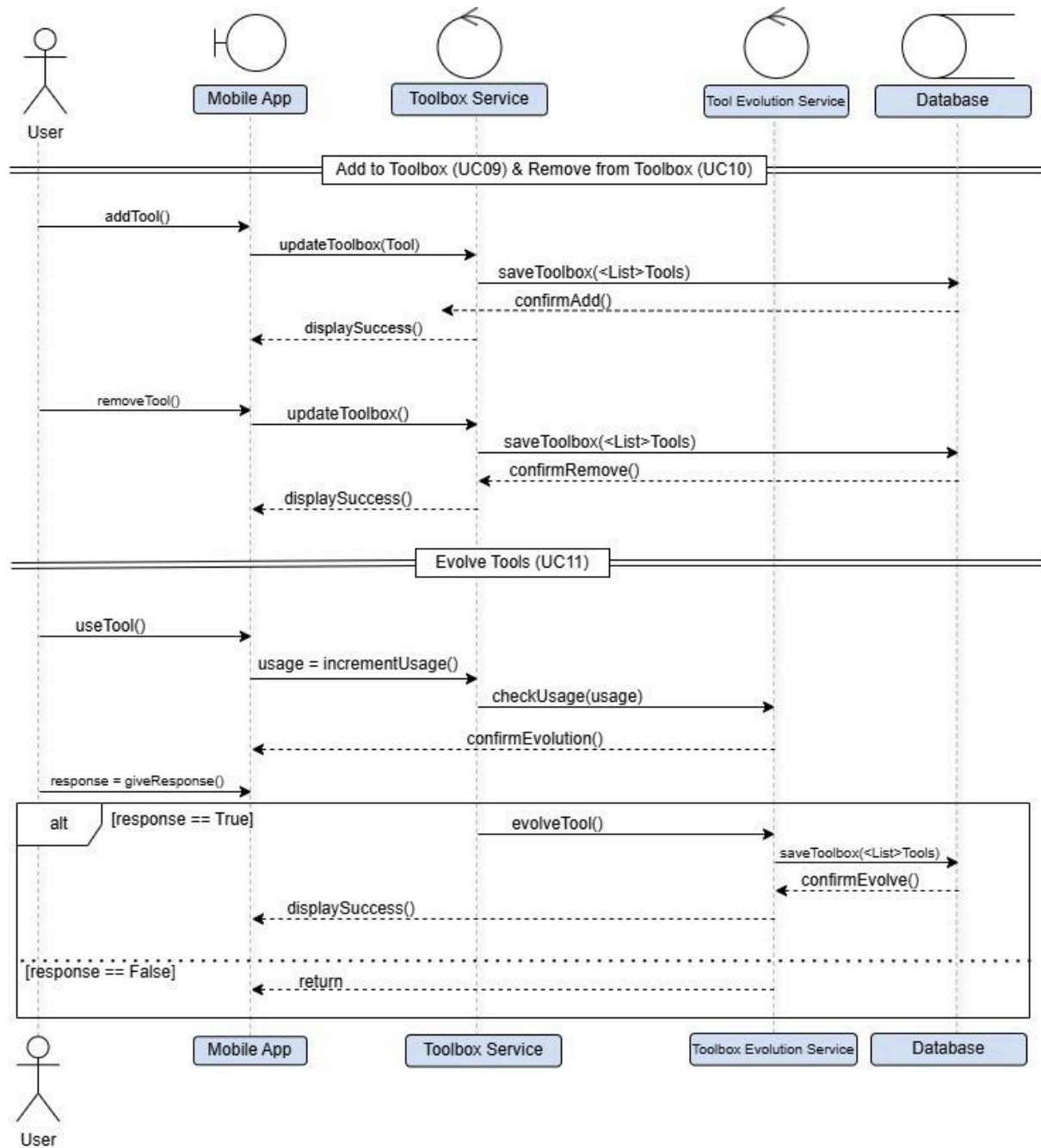


Figure 4.4.3: SD003 Sequence Diagram for <Skill-Building Toolbox Module>

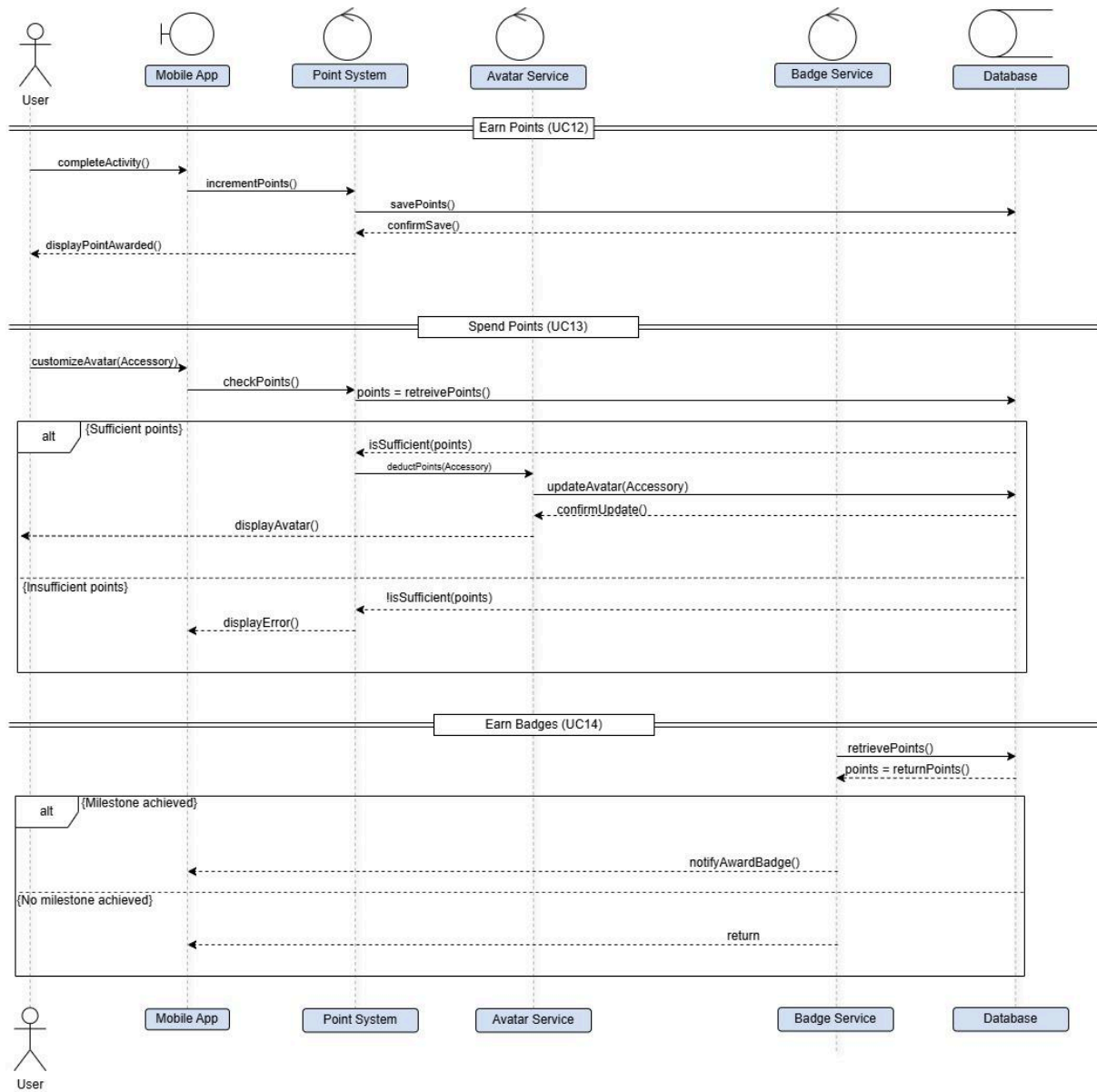


Figure 4.4.4: SD004 Sequence Diagram for <Motivation and Rewards Module>

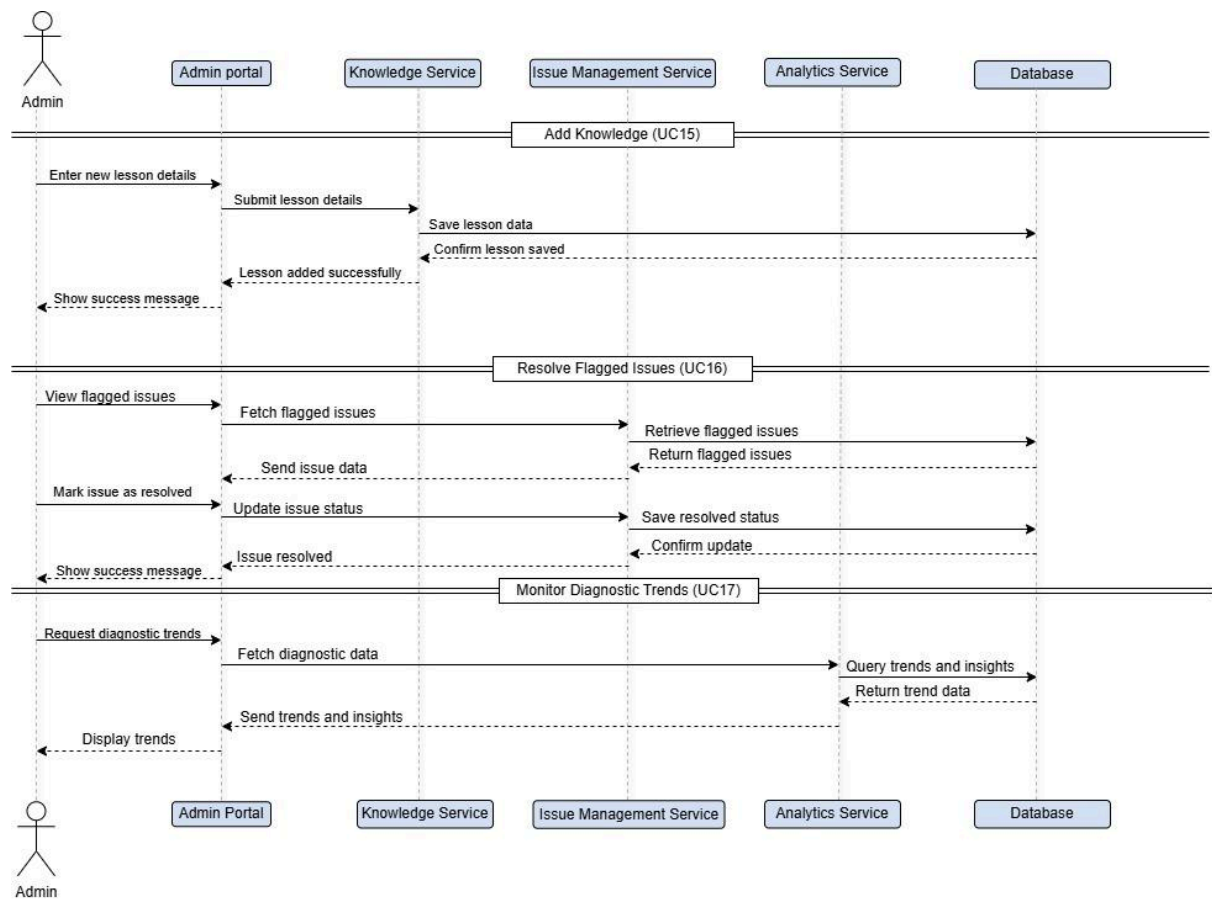


Figure 4.4.5: SD005 Sequence Diagram for <Admin Management Module>

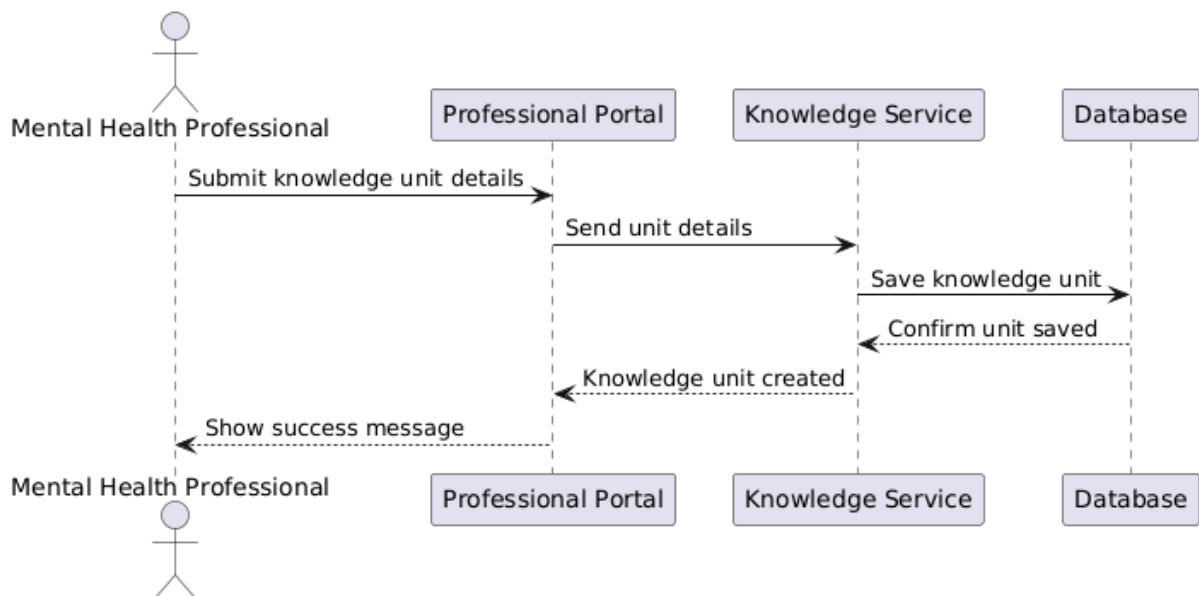


Figure 4.4.6: SD006 Sequence Diagram for <Create Knowledge Units>

4.5 Deployment View

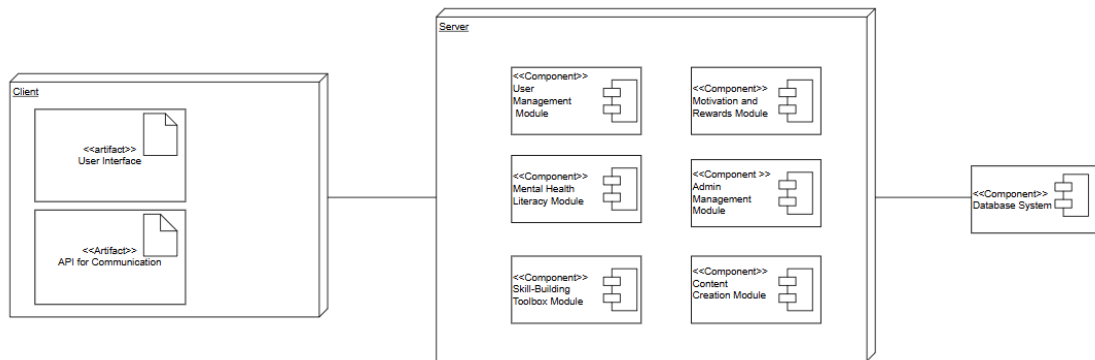


Figure 4.5.1: Deployment diagram for MindWell Self-Help & Diagnostic System

5. Data Design

The major data or systems entities are stored into a relational database named as..., processed and organized into n entities as listed in Table 5.1.

Table 5.1: Description of Entities in the Database for <User Management> Subsystem

No.	Entity Name	Description
1.	User	Represents the core user information such as ID, username, password, email, and profile.
2.	Profile	Stores additional user details such as first name, last name, and avatar information.
3.	Registration Service	Handles user registration and ensures that all new users are properly stored in the database.
4.	Authentication Service	Manages user login and logout processes. Authenticates users based on username and password.
5.	Profile Management Service	Provides functionalities for updating user profiles, linking data with User and Profile entities.
6.	Recovery Service	Facilitates password recovery by generating and sending recovery emails.
7.	Email Service	Manages email operations, including sending emails for recovery or notifications.
8.	Database	The central repository that stores and organizes data related to users, profiles, and services.

Table 5.2: Description of Entities in the Database for <Mental Health Literacy > Subsystem

No.	Entity Name	Description
1.	KnowledgeUnitManager	Manages operations related to knowledge units, including creation, retrieval, and updates.
2.	KnowledgeUnit	Represents a single unit of knowledge, including attributes like title, content, and metadata.
3.	Bookmark	Stores information about knowledge units bookmarked by users for easy access.

4.	Rating	Tracks user ratings of knowledge units, including scores and optional comments.
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Table 5.3: Description of Entities in the Database for <Skill-Building Toolbox > Subsystem

No.	Entity Name	Description
1.	Tool	Represents a specific tool or skill, such as exercises or techniques available to users.
2.	Toolbox	Contains tools selected by a user for personalized use and management.
3.	ToolboxManager	Manages user toolboxes, including adding, removing, and evolving tools.

Table 5.4: Description of Entities in the Database for <Motivation and Rewards > Subsystem

No.	Entity Name	Description
1.	Points	Tracks points earned by users through activities such as completing tasks or milestones.
2.	Rewards	Represents the rewards available to users, such as badges or premium features.
3.	RewardManager	Manages the distribution and redemption of rewards based on user points.

Table 5.5: Description of Entities in the Database for <Admin Management> Subsystem

No.	Entity Name	Description
1.	AdminManager	Manages administrative tasks, including user management and content moderation.
2.	FlaggedIssue	Tracks reported issues with knowledge units, flagged by users for review.
3.	DiagnosticTrend	Stores data related to diagnostic trends, providing insights into user behaviors and needs.

Table 5.6: Description of Entities in the Database for <Content Creation> Subsystem

No.	Entity Name	Description
1.	ContentManager	Oversees the creation and management of content, ensuring relevance and accuracy.
2.	MediaResources	Represents additional resources such as videos, images, or documents linked to content.
3.	KnowledgeUnit	Represents individual knowledge units, including metadata and associated resources.
4.	MentalHealthProfessional	Represents content creators or professionals who contribute knowledge units to the system.

5.1 Data Dictionary

5.1.1 <User Management> Subsystem

5.1.1.1 Entity: User

Attribute Name	Type	Description
id	String	Stores a unique id for the User.
username	String	Stores the username for the User.
password	String	Stores the password for the User.
email	String	Stores the email for the User.
profile	Profile	Creates a Profile instance for the User.

5.1.1.2 Entity: Profile

Attribute Name	Type	Description
firstName	String	Stores the firstName for the Profile.
lastName	String	Stores the lastName for the Profile.
avatar	String	Stores a URL for the avatar for the Profile.

5.1.2 <Mental Health Literacy > Subsystem

5.1.2.1 Knowledge Unit Manager

Attribute Name	Type	Description
currentUnit	KnowledgeUnit	The currently active or selected knowledge unit.

5.1.2.2 Knowledge Unit

Attribute Name	Type	Description
unitID	String	Unique identifier for the knowledge unit.
title	String	The title or name of the knowledge unit.
content	Text	The main content or body of the knowledge unit.
authorID	String	Identifier for the author or creator of the knowledge unit.
dateCreated	DateTime	The date and time when the knowledge unit was created.
category	String	The category or classification of the knowledge unit.
difficultyLevel	Integer	A numerical representation of the difficulty level (e.g., 1 = Easy, 5 = Hard).

5.1.2.3 Bookmark

Attribute Name	Type	Description
bookmarkID	String	Unique identifier for the bookmark.
userID	String	Identifier for the user who created the bookmark.
unitID	String	Identifier for the associated

		knowledge unit.
dateBookmarked	DateTime	The date and time when the bookmark was created.

5.1.2.4 Rating

Attribute Name	Type	Description
ratingID	String	Unique identifier for the rating.
userID	String	Identifier for the user who gave the rating.
unitID	String	Identifier for the associated knowledge unit.
score	Integer	A numerical score given by the user (e.g., 1-5).
comment	String	Additional feedback or comments provided by the user about the knowledge unit.

5.1.3 <Skill-Building Toolbox> Subsystem

5.1.3.1 Entity : Tool

Attribute Name	Type	Description
toolID	String	A unique identifier for the tool
name	String	The name of the tool
description	Text	A detailed description of the tool
category	String	The category to which the tool belongs
difficulty level	Integer	A numerical value indicating the difficulty of using the tool
usageCount	Integer	A counter indicating how many times the tool has been used

5.1.3.2 Entity : ToolboxManager

Attribute Name	Type	Description
currentToolbox	Toolbox	The currently active or selected toolbox.

5.1.3.3 Entity : Toolbox

Attribute Name	Type	Description
toolboxID	String	Unique identifier for the toolbox.
userID	String	Identifier for the user who owns the toolbox.
tools	List<Tool>	A collection of tools contained within the toolbox.

5.1.4 <Motivation and Rewards> Subsystem

5.1.4.1 Entity : RewardManager

Attribute Name	Type	Description
currentUser	User	Represents the user currently using the reward system

5.1.4.2 Entity : Points

Attribute Name	Type	Description
pointID	String	A unique identifier for each point transaction
userID	String	The unique identifier of the user associated with the points
amount	Integer	The number of points involved in the transaction
transactionType	String	The type of transaction

5.1.4.3 Entity : Badge

Attribute Name	Type	Description
badgeID	String	A unique identifier for each badge
name	String	The name of the badge
description	String	A detailed description of the badge and its significance
requirementType	String	Specifies the type of requirement to earn the badge
requirementValue	Integer	The value required to meet the badge criteria

5.1.5 <Admin Management> Subsystem

5.1.5.1 AdminManager

Attribute Name	Type	Description
currentAdmin	Admin	The currently logged-in administrator managing the system

5.1.5.2 FlaggedIssue

Attribute Name	Type	Description
issueID	String	A unique identifier for each flagged issue in the system
unitID	String	The identifier of the knowledge unit associated with the flagged issue
reporterID	String	The identifier of the user or admin who reported the issue
issueType	String	Specifies the type of issue
status	String	The current status of the issue

5.1.5.3 DiagnosticTrend

Attribute Name	Type	Description
trendID	String	Unique ID identifying trends.
metricType	String	Description of the type of metric used to diagnose trends.
value	float	Value storing the magnitude of a trend.
timeFrame	String	Description of the time frame where diagnostic trends appear within.

5.1.6 <Content Creation> Subsystem

5.1.6.1 ContentManager

Attribute Name	Type	Description
currentCreator	MentalHealthProfessional	Attribute to store details about the mental health professional that authors the content.
draftUnits	List<KnowledgeUnit>	List that stores drafts of knowledge units that are not published yet.

5.1.6.2 MentalHealthProfessional

Attribute Name	Type	Description
professionalID	String	Unique ID storing a string of the mental health professional's ID.
name	String	Stores the name of the mental health professional.
credentials	List<String>	Stores a list of credentials that the professional has.
licenseNumber	String	Stores the license number of the mental health professional.
verificationStatus	boolean	Stores a boolean of the status of verification for the mental health professional.

5.1.6.3 KnowledgeUnit

Attribute Name	Type	Description
unitID	String	String containing the unique ID of the knowledge unit.
title	String	String containing the title of the knowledge unit.
content	Text	Text containing the complete content of a particular knowledge unit.

mediaResources	List<MediaResource>	A list of media resources that are used inside the knowledge unit.
creatorID	String	A unique ID identifying the creator of the knowledge unit.
creationDate	DateTime	Stores the date and time of knowledge unit creation.
lastModified	DateTime	Stores the date and time when the knowledge unit was last modified.
status	String	Stores information about the status of publication of the knowledge unit.
category	String	Stores information about the category of the knowledge unit.
targetAudience	String	Stores information about the target audience of the knowledge unit.
difficultyLevel	integer	Stores a number signifying the difficulty of the knowledge unit.
prerequisites	List<String>	Stores a list specifying the prerequisites needed to take the knowledge unit.
learningObjectives	List<String>	Stores a list specifying the learning objectives of a knowledge unit.

5.1.6.4 MediaResource

Attribute Name	Type	Description
resourceID	String	Unique ID identifying the media resource.
type	String	Stores information about the type of media resource is stored, e.g. image, video.
url	String	Stores the URL to access the media resource.

description	String	Stores a short description of the media resource.
format	String	Stores information about the format of the media resource, e.g. .jpg, .mov
size	Integer	Stores information about the size of the media resource.
uploadDate	DateTime	Stores information about the date and time the media resource was uploaded.

6. User Interface Design

The interface includes...

6.1 Screen Images

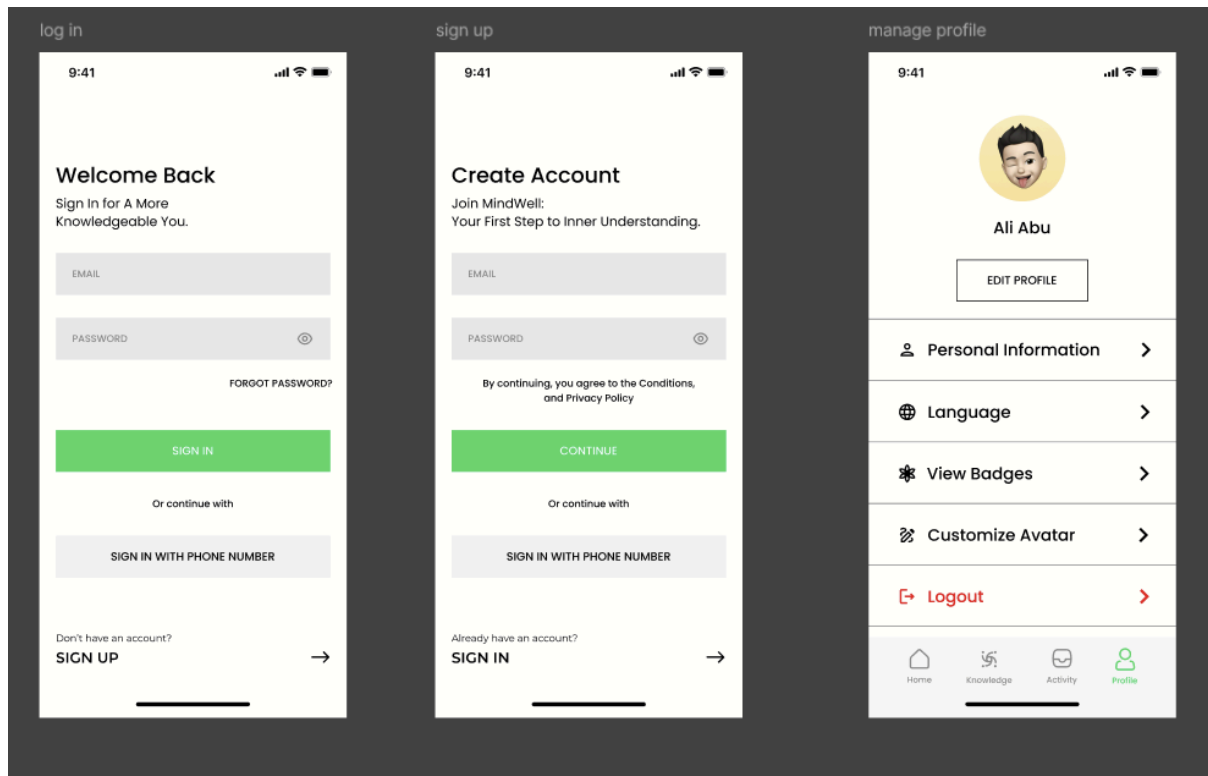


Figure 6.1: Interface for User Module with Log In, Sign Up and Manage Profile

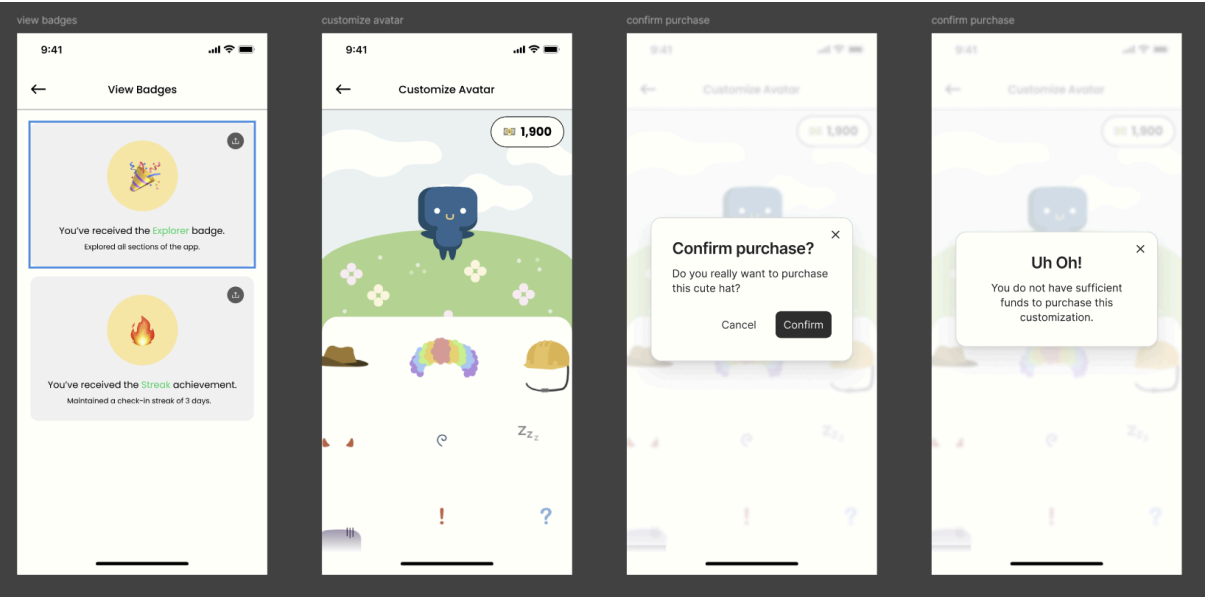


Figure 6.2: Interface for Rewards and Motivation Subsystem

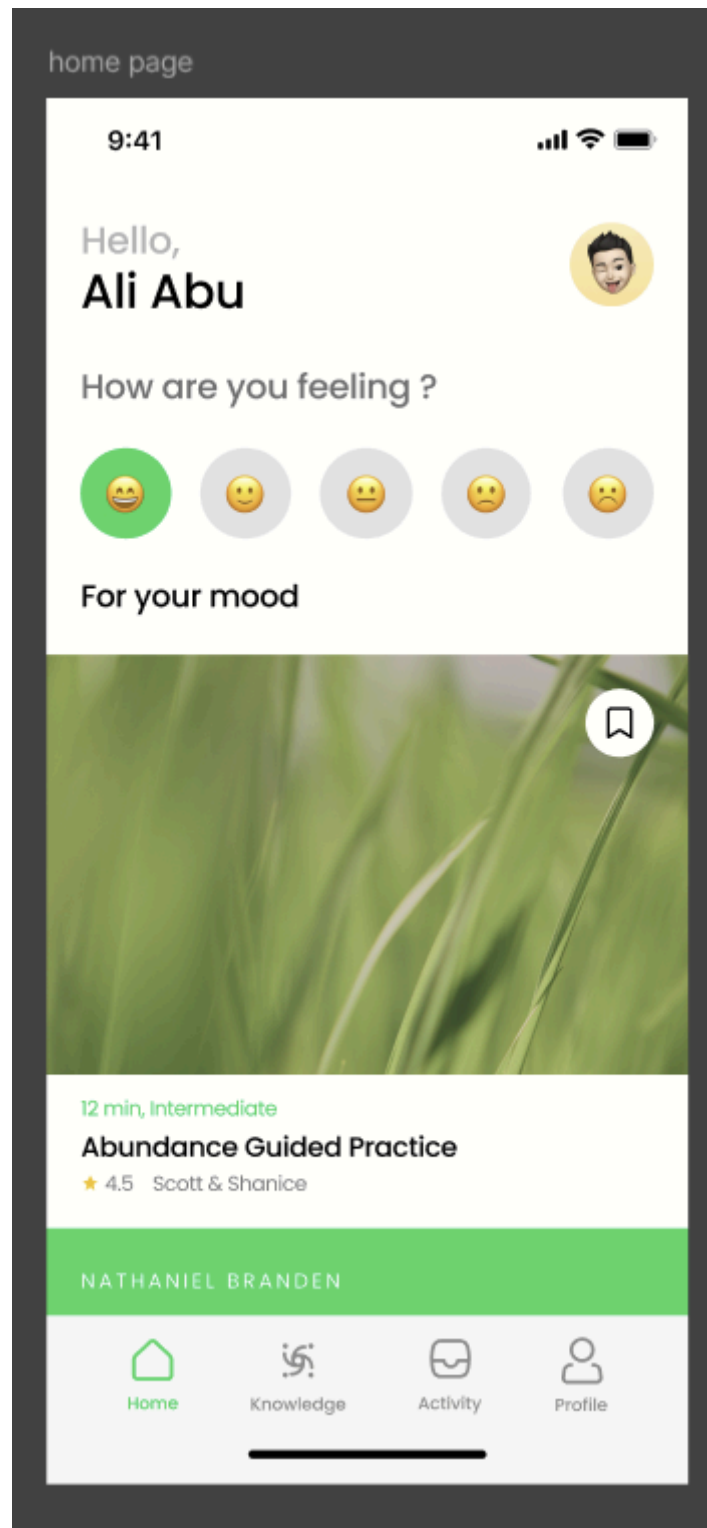


Figure 6.3: Interface for Home Page

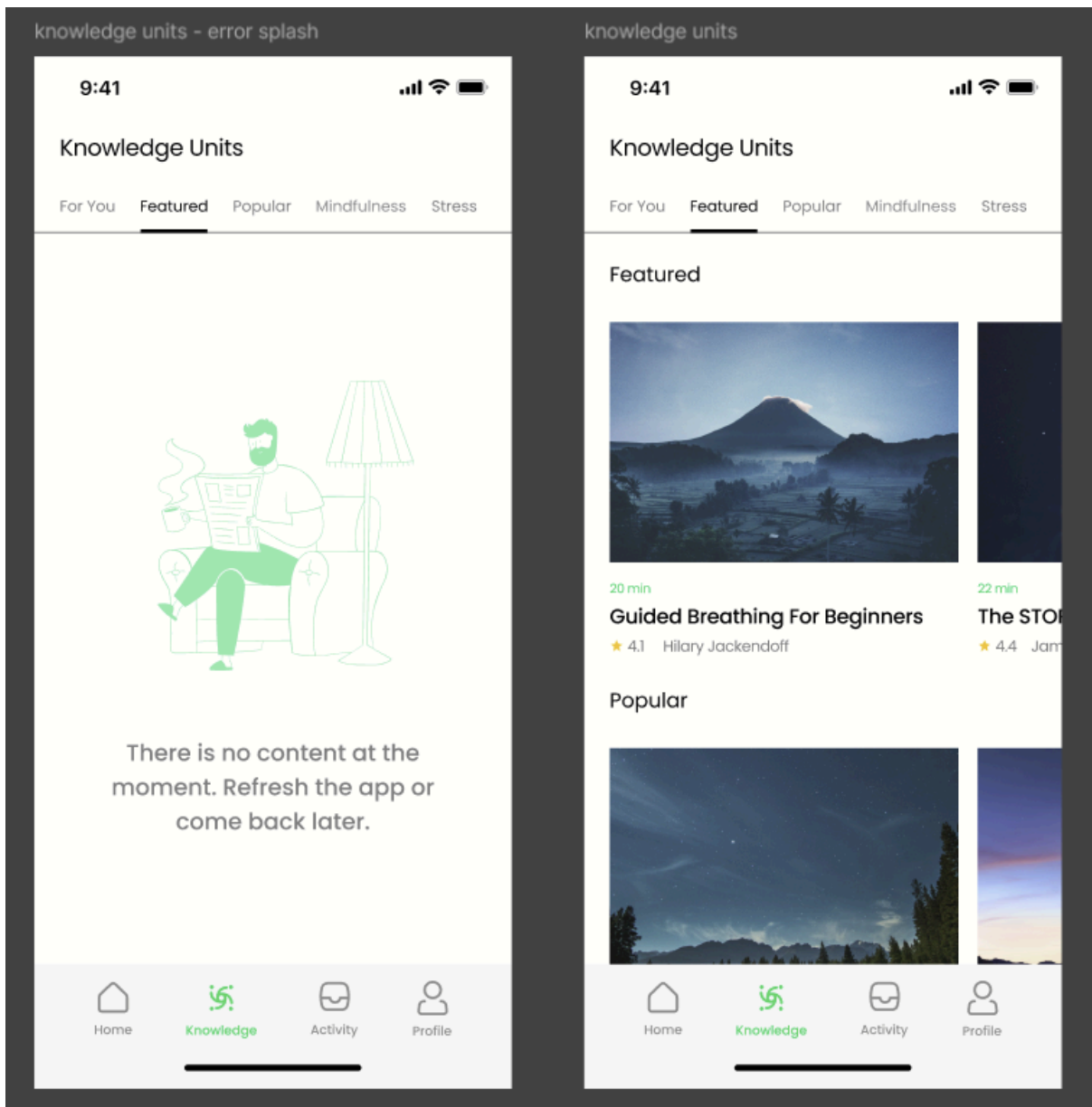


Figure 6.4: Interface for Knowledge Unit Subsystem (Browse Knowledge Units)

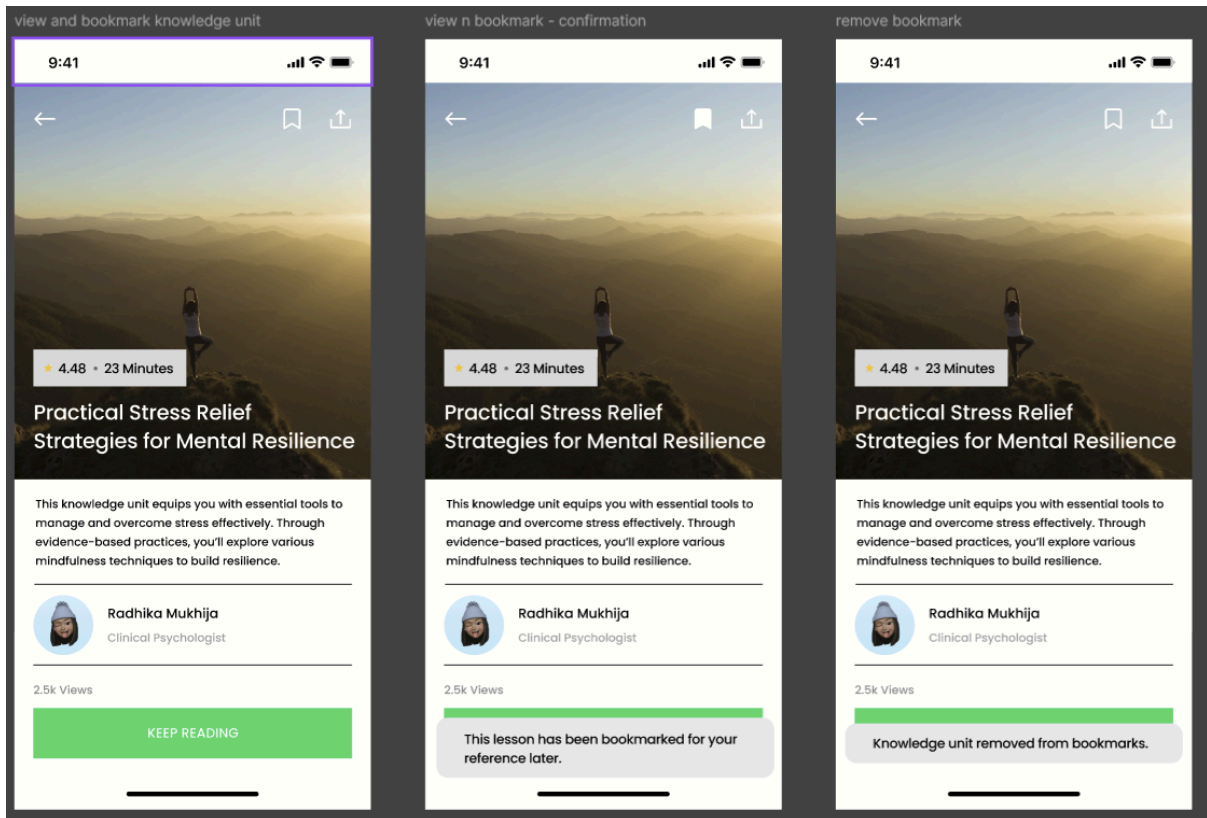


Figure 6.5: Interface for Bookmark knowledge unit with bookmark knowledge unit and remove bookmark

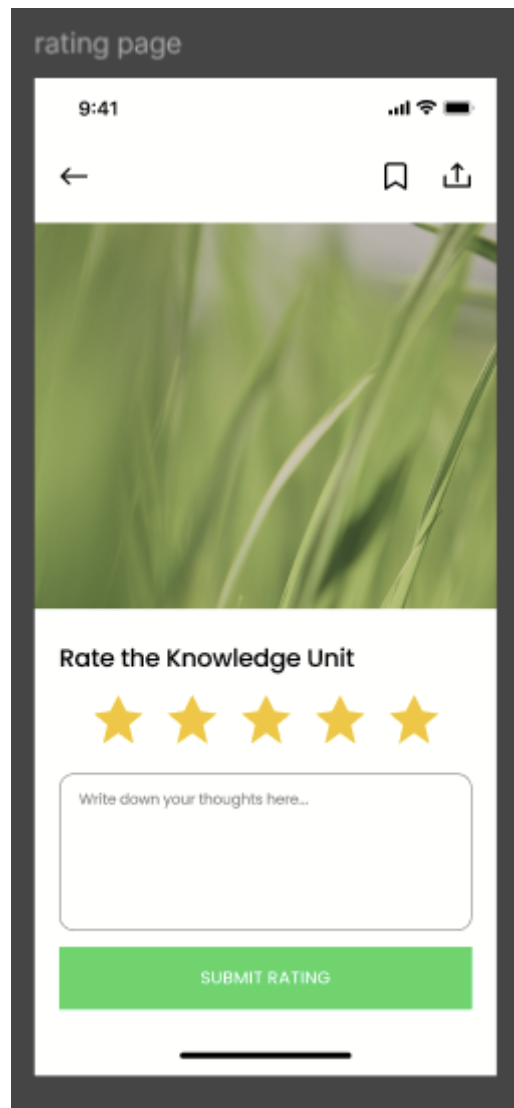


Figure 6.6: Interface for Knowledge Unit with Rate Knowledge Unit

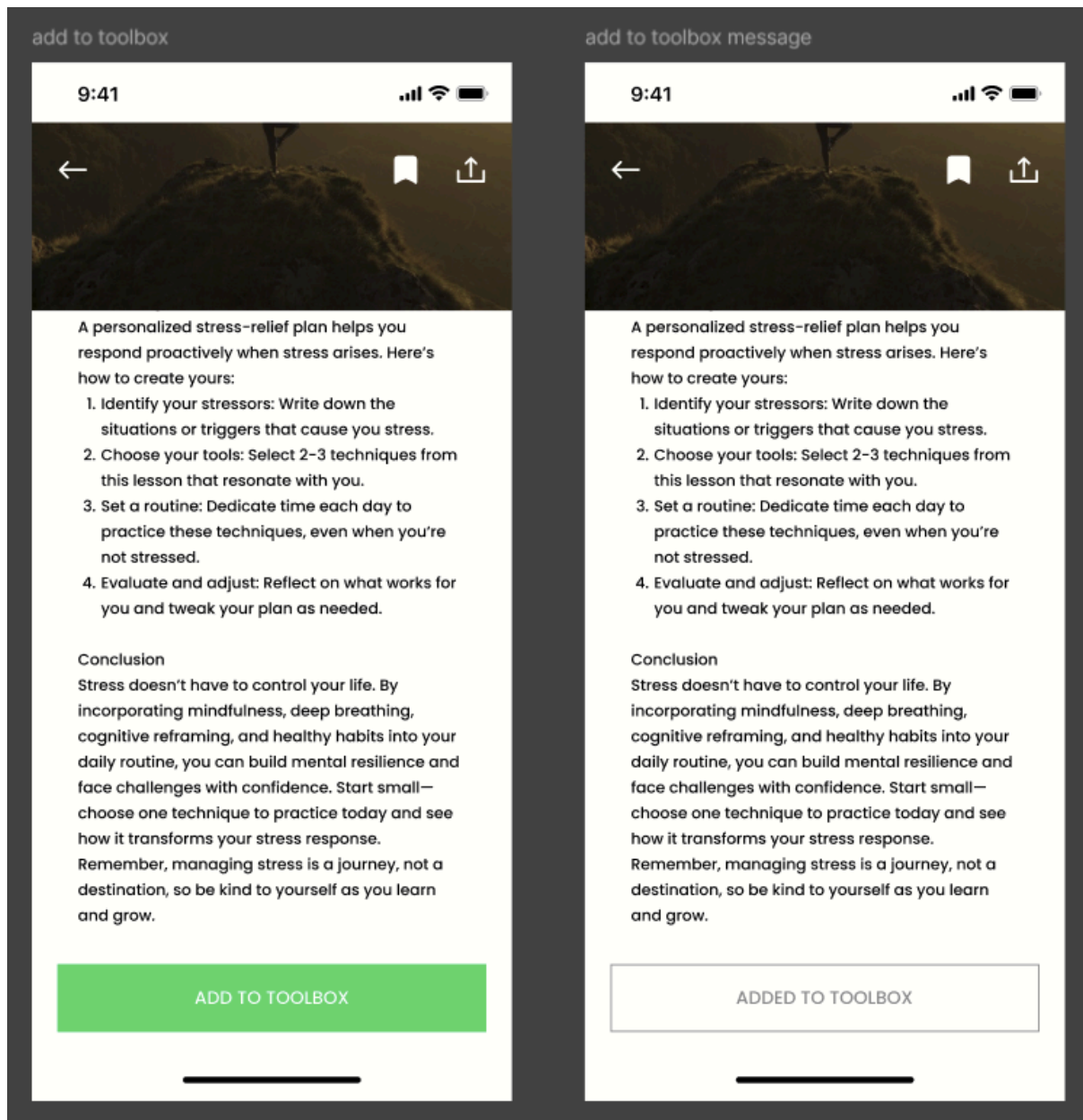


Figure 6.6: Interface for Toolbox with Add to Toolbox and confirmation display

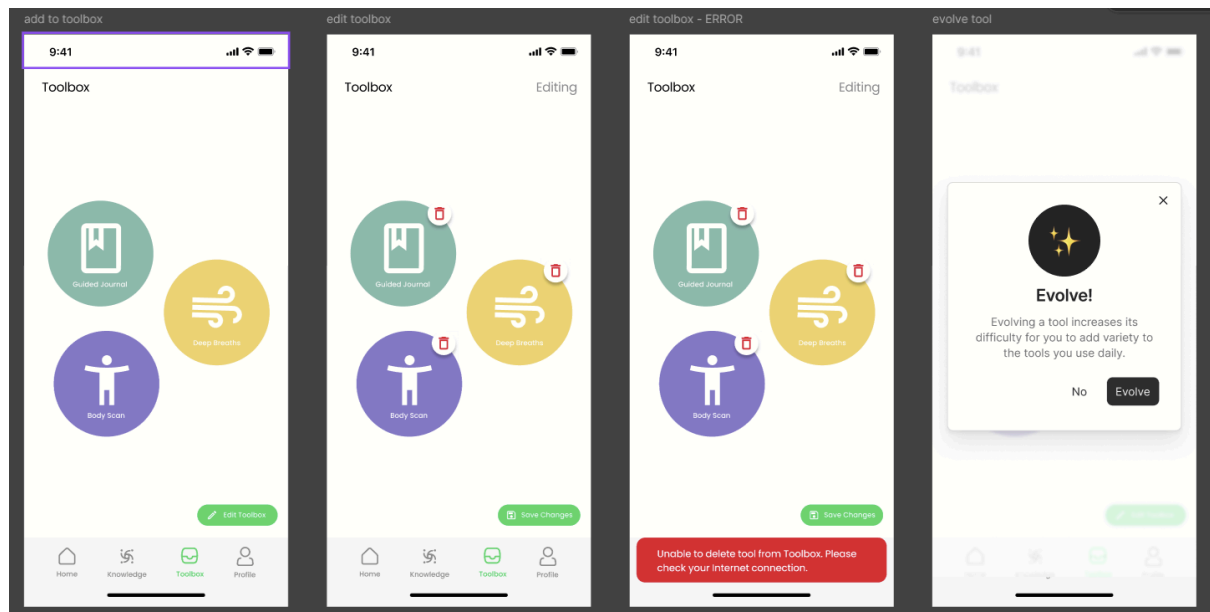


Figure 6.2: Interface for Toolbox subsystem with Add to Toolbox, Edit Toolbox, Delete Items from Toolbox and Evolve Tools

Figma Link: [Figma](#)

7. Traceability

Sprint #	Subsystem/ Package	User Story ID	
Sprint 1	User Management	US001: User registration	Registration system for users which enables new users to register and store their information securely in the database.
		US002: User login	Authentication system, which allows users to log in using their credentials and access their accounts..
		US003: Manage profile	Authentication system which allows users to log in using their credentials and access their accounts.
		US004: Password recovery	Password recovery system which supports users in recovering their accounts through secure recovery processes.
		US005: User logout	Logout system which ensures users can securely log out of the platform.
Sprint 2	Mental Health Literacy	US006: Browse Knowledge Units	Content browsing system which facilitates browsing of mental health knowledge units by users.
		US007: Bookmark Knowledge Units	Bookmark system which allows users to save knowledge units for quick future access.
		US008: Rate Knowledge Units	Rating system which enables users to rate knowledge units, providing feedback for content improvement.
Sprint 3	Skill-Building Toolbox	US009: Add to Toolbox	Tool addition system which allows users to add skill-building tools to their personalized toolbox
		US010: Remove from Toolbox	Tool removal system which provides functionality for users to remove unwanted tools from their toolbox.

		US011: Evolve Tools	Tool evolution system which enables users to upgrade or enhance their tools as they advance.
Sprint 4	Motivation and Rewards	US012: Earn Points	Point earning system which allows users to earn points for completing activities, promoting engagement.
		US013: Spend Points	Point redemption system which facilitates the redemption of earned points for rewards or benefits.
		US014: Earn Badges	Badge system which rewards users with badges for achieving milestones, encouraging continued participation.
Sprint 5	Content Creation	US015: Add Knowledge Unit	Knowledge unit addition system which supports content creators in adding knowledge units to the platform.
		US018: Create Knowledge Units	Knowledge unit creation system which provides tools for content creators to design and structure new knowledge units.
Sprint 6	Admin Management	US016: Resolve Flagged Issues	Issue resolution system which enables admins to review and resolve flagged issues or content effectively.
		US017: Monitor Diagnostic Trends	Diagnostic monitoring system which allows admins to monitor user trends and insights for strategic improvements.

Table 7.1: Traceability

8. Test Cases

A test case is a detailed set of conditions, inputs, and expected outcomes designed to assess the functionality and behavior of a specific feature, module, or system. It serves as a specification for the steps to be followed and the expected results to be observed during the testing process. In this document, organize your test cases according to the package, followed by the related use case and their sequence diagram.

8.1 TC001: Test <User Management Basics> Subsystem: <User Story Register (US001)>

List all test cases before providing the details for each under each package/subsystem and use story.

This test contains the following test cases:

(a) TC001_01: Test <Scenario of sequence diagram1 - register (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test the Register Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile app and navigates to the registration page.	Click to open the app	The registration page should be open	As Expected			
2	Enter personal details: full name, already registered email, and password.	Full Name: User Email: user@example.com Password: Password@123	System accepts inputs, validates format, and checks for mandatory fields. Error messages appear for invalid inputs.	As Expected			

3	Agrees to the terms and conditions by checking the provided box	Tick the agreement box	System allows user to proceed	As Expected			
4	Submit the registration form.	Tick the submit button	Account is created, and a confirmation message is displayed: "Registration successful."	As Expected			

(b) TC001_02: Test <Scenario of sequence diagram2- login (SD002)>

DESCRIPTION	VERSION	TEST DATE
Test the Login Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_02_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile application and navigates to the login page.	Click to open the app	The login page should be open	As Expected			
2	Enters registered email address and password.	Registered email: user@example.com Password: Password@123	Email is verified.	As Expected			
3	Login into app	Clicks the "Login" button	Login is successful. Redirected to the user dashboard and displays a	As Expected			

			welcome message or dashboard with personalized content.				
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(c) TC001_03: Test <Scenario of sequence diagram3- manage profile (SD003)>

DESCRIPTION	VERSION	TEST DATE
Test the Manage Profile Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_03_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Manage Profile" section from the dashboard.	Click manage profile	"Manage Profile" page is displayed successfully.	As Expected			
2	Users view their existing profile details (e.g., name, email, profile picture).		Details of user displayed successfully	As Expected			
3	User selects the specific field(s) they wish to edit	Edit changes	Allow user to edit details	As Expected			
4	User clicks the "Save Changes" button.	Click the "Save Changes" button.	Successfully save changes and update changes on the profile page, system displays a confirmation	As Expected			

			message: "Profile updated successfully."				
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(d) TC001_04: Test <Scenario of sequence diagram4- password recover(SD004)>

DESCRIPTION	VERSION	TEST DATE
Test the Password Recover Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_04_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User clicks the "Forgot Password" link on the login page.	Click on Forgot Password link	Redirected to the password recovery page.	As Expected			
2	User enters the registered email address and submit the recovery request.	Registered email: user@example.com	Email is verified.	As Expected			
3	User opens the recovery email and follows the link to reset the password.	Click link given from email	Redirected to the password reset page.	As Expected			
4	User enters and confirms the new password. Submit the form.	Click the "Submit" button.	Password is reset successfully. Confirmation message is displayed.	As Expected			

(e) TC001_05: Test <Scenario of sequence diagram5- log out(SD005)>

DESCRIPTION	VERSION	TEST DATE
Test the LogOut Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_05_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the menu or settings section of the application.	Open settings page	Settings page is displayed successfully.	As Expected			
2	User selects the "Log Out" option.	Click on "Logout" button	System prompts the user for confirmation: "Are you sure you want to log out?"	As Expected			
3	User confirms the log-out action.	Click "yes" on the confirmation	System terminates the user session and clears temporary data associated with the session and redirected to the login page or home screen.	As Expected			

8.1.1 TC001_01: Test <state scenario of sequence diagram1 (SD001)>

This test contains the following scenarios:

(a) TC001_01_01: Test <normal scenario of sequence diagram register (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test the normal flow of Register Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile app and navigates to the registration page.	Click to open the app	The registration page should be open	As Expected			
2	Enter personal details: full name, already registered email, and password.	Full Name: User Email: user@example.com Password: Password@123	System accepts inputs, validates format, and checks for mandatory fields. Error messages appear for invalid inputs.	As Expected			
3	Agrees to the terms and conditions by checking the provided box	Tick the agreement box	System allows user to proceed	As Expected			
4	Submit the registration form.	Tick the submit button	Account is created, and a confirmation message is displayed:	As Expected			

			"Registration successful."				
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(b) TC001_01_02: Test <alternate scenario of sequence diagram register (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test alternate flow 1 of Register Functionality in the system	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_01_02)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile app and navigates to the registration page.	Click to open the app	The registration page should be open	As Expected			
2	Enter personal details: full name, already registered email, and password.	Full Name: User Email: wrongemail.com Password: Wrongpassword	A1: Invalid input during registration, System highlights the invalid fields	As Expected			
2	Enter personal details: full name, already registered email, and password.	Full Name: User Email: user@example.com Password: Password@123	System accepts inputs, validates format, and checks for mandatory fields. Error messages appear for invalid inputs.	As Expected			

3	Agrees to the terms and conditions by checking the provided box	Tick the agreement box	System allows user to proceed	As Expected			
4	Submit the registration form.	Tick the submit button	Account is created, and a confirmation message is displayed: "Registration successful."	As Expected			

(c) TC001_01_03: Test <alternate scenario of sequence diagram register (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test alternate flow 2 of Register Functionality in the system	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_01_03)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile app and navigates to the registration page.	Click to open the app	The registration page should be open	As Expected			
2	Enter personal details: full name, already registered email, and password.	Full Name: User Email: user@example.com Password: Password@123	A2: Email already exists in the system, System displays an error message: "Email is already registered." User is directed to the login page or password	As Expected			

			recovery option.				
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(d) TC001_01_04: Test <exception scenario of sequence diagram register (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test exception flow of Register Functionality in the system	4.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_01_04)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile app and navigates to the registration page.	Click to open the app	The registration page should be open E1: No internet connection, System displays an error message : “No internet connection. Please check your network and try again. User is unable to proceed with registration until the connection is restored. E2: Server error during registration, System displays an error message: “Unable to process your registration at this time. Please try again later.” Registration is aborted, and the user is prompted to retry later.	As Expected			

2	Enter personal details: full name, already registered email, and password.	Full Name: User Email: user@example.com Password: Password@123	System accepts inputs, validates format, and checks for mandatory fields. Error messages appear for invalid inputs.	As Expected			
3	Agrees to the terms and conditions by checking the provided box	Tick the agreement box	System allows user to proceed	As Expected			
4	Submit the registration form.	Tick the submit button	Account is created, and a confirmation message is displayed: "Registration successful." E1: No internet connection, System displays an error message : "No internet connection. Please check your network and try again. User is unable to proceed with registration until the connection is restored." E2: Server error during registration, System displays an error message: "Unable to process your registration at this time. Please try again later." Registration is aborted, and the user is prompted to retry later.	As Expected			

8.1.2 TC001_02: Test <Scenario of sequence diagram Login (SD002)>

(a) TC001_02_01: Test <normal scenario of sequence diagram login (SD002)>

DESCRIPTION	VERSION	TEST DATE
Test normal flow of Login Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_02_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile application and navigates to the login page.	Click to open the app	The login page should be open	As Expected			
2	Enter registered email address and password.	Registered email: user@example.com Password: Password@123	Email is verified.	As Expected			
3	Login into app	Clicks the "LogIn" button	Login is successful. Redirected to the user dashboard and displays a welcome message or dashboard with personalized content.	As Expected			

(b) TC001_02_02: Test <alternate scenario of sequence diagram login (SD002)>

DESCRIPTION	VERSION	TEST DATE
Test alternate flow 1 of Login Functionality in the system	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_02_02)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile application and navigates to the login page.	Click to open the app	The login page should be open	As Expected			
2	Enter registered email address and password.	Email: wrongemail.com Password: Wrongpassword	A1 :Incorrect email and password entered, System displays an error message: "Invalid email or password. Please try again."Users are prompted to re-enter their credentials.	As Expected			
3	Enter registered email address and password.	Registered email: user@example.com Password: Password@123	Email is verified.	As Expected			
4	Login into app	Clicks the "Login" button	Login is successful. Redirected to the user dashboard and displays a	As Expected			

			welcome message or dashboard with personalized content.				
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(c) TC001_02_03: Test <alternate scenario of sequence diagram login (SD002)>

DESCRIPTION	VERSION	TEST DATE
Test alternate flow 2 of Login Functionality in the system	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_02_03)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile application and navigates to the login page.	Click to open the app	The login page should be open	As Expected			
2	Enter registered email address and password.	Click forgot password	A2: Forgot password. User clicks the "Forgot Password" link. System redirects the user to the password recovery page.	As Expected			
3	Enter registered email address and password.	Registered email: user@example.com	Email is verified.	As Expected			

		Password: Password@ 123					
4	Login into app	Clicks the “Login” button	Login is successful. Redirected to the user dashboard and displays a welcome message or dashboard with personalized content.	As Expected			

(d) TC001_02_04: Test <exception scenario of sequence diagram login(SD002)>

DESCRIPTION	VERSION	TEST DATE
Test exception flow of Login Functionality in the system	4.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_02_04)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Opens the mobile application and navigates to the login page.	Click to open the app	The login page should be open E1: No internet connection, System displays an error message : “No internet connection. Please check your network and try again. User is unable to proceed with registration until the connection is restored. E2: Server error during registration, System displays an error message: “Unable to	As Expected			

			process your registration at this time. Please try again later.” Registration is aborted, and the user is prompted to retry later.				
2	Enters registered email address and password..	Registered email: user@example.com Password: Password@123	Email is verified. E1: No internet connection, System displays an error message : “No internet connection. Please check your network and try again. User is unable to proceed with registration until the connection is restored. E2: Server error during registration, System displays an error message: “Unable to process your registration at this time. Please try again later.” Registration is aborted, and the user is prompted to retry later.	As Expected			
3	Login into app	Clicks the “Login” button	Login is successful. Redirected to the user dashboard and displays a welcome message or dashboard with personalized content. E1: No internet connection, System displays an error message : “No internet connection. Please check your network and try again. User is unable to proceed with registration until the connection is restored. E2: Server error during registration, System displays an error message: “Unable to process your registration at this time. Please try	As Expected			

			again later.” Registration is aborted, and the user is prompted to retry later.				
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8.1.3 TC001_03: Test <state scenario of sequence diagram manage profile (SD003)>

(a) TC001_03_01: Test <normal scenario of sequence diagram manage profile (SD003)>

DESCRIPTION	VERSION	TEST DATE
Test normal flow of Manage Profile Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_03_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Manage Profile" section from the dashboard.	Click manage profile	"Manage Profile" page is displayed successfully.	As Expected			
2	Users view their existing profile details (e.g., name, email, profile picture).		Details of user displayed successfully	As Expected			
3	User selects the specific field(s) they wish to edit	Change profile picture	Allow user to change profile picture	As Expected			
4	User clicks the "Save Changes" button.	Click the "Save Changes" button.	Successfully save changes and update changes on the profile page, system displays a confirmation message: "Profile updated successfully."	As Expected			

(b) TC001_03_02: Test <alternative scenario of sequence diagram manage profile (SD003)>

DESCRIPTION	VERSION	TEST DATE
Test alternative flow 1 of Manage Profile Functionality in the system	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_03_02)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Manage Profile" section from the dashboard.	Click manage profile	"Manage Profile" page is displayed successfully.	As Expected			
2	Users view their existing profile details (e.g., name, email, profile picture).		Details of user displayed successfully	As Expected			
3	User selects the specific field(s) they wish to edit	Change email: wrongemailformat@mail.	A1 : Invalid input during profile update. System highlights the incorrect fields (e.g., invalid email format). User is prompted to correct the errors and resubmit.	As Expected			
4	User clicks the "Save Changes" button.	Click the "Save Changes" button.	Successfully save changes and update changes on	As Expected			

			the profile page, system displays a confirmation message: "Profile updated successfully."				
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(c) TC001_03_03: Test <alternative scenario of sequence diagram manage profile (SD003)>

DESCRIPTION	VERSION	TEST DATE
Test alternative flow 2 of Manage Profile Functionality in the system	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_03_03)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Manage Profile" section from the dashboard.	Click manage profile	"Manage Profile" page is displayed successfully.	As Expected			
2	Users view their existing profile details (e.g., name, email, profile picture).		Details of user displayed successfully	As Expected			
3	User selects the specific field(s) they wish to edit	Edit changes	A2: User cancels profile update. User clicks the "Cancel" button.No changes are saved, and the user is returned to the	As Expected			

			previous screen.				
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(d) TC001_03_04: Test <exception scenario of sequence diagram manage profile (SD003)>

DESCRIPTION	VERSION	TEST DATE
Test exception flow of Manage Profile Functionality in the system	4.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_03_04)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Manage Profile" section from the dashboard.	Click manage profile	"Manage Profile" page is displayed successfully.	As Expected			
2	Users view their existing profile details (e.g., name, email, profile picture).		Details of user displayed successfully	As Expected			
3	User selects the specific field(s) they wish to edit	Edit changes	Allow user to edit details	As Expected			
4	User clicks the "Save Changes" button.	Click the "Save Changes" button.	E1: No internet connection. System displays an error message: "Unable to connect. Please check your internet connection." Users cannot save changes until the connection is restored.	As Expected			

			E2: System timeout or server error. System displays an error message: “Unable to update profile at this time. Please try again later.” Changes are not saved, and the user is prompted to retry.				
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8.1.4 TC001_04: Test <state scenario of sequence diagram Password Recover (SD004)>

(a) TC001_04_01: Test <normal scenario of sequence diagram Password Recover (SD004)>

DESCRIPTION	VERSION	TEST DATE
Test normal flow of Password Recover Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_04_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User clicks the "Forgot Password" link on the login page.	Click on Forgot Password link	Redirected to the password recovery page.	As Expected			
2	User enters the registered email address and submit the recovery request.	Registered email: user@example.com	Email is verified.	As Expected			
3	User opens the recovery email and follows the link to reset the password.	Click link given from email	Redirected to the password reset page.	As Expected			
4	User enters and confirms the new password. Submit the form.	Click the "Submit" button.	Password is reset successfully. Confirmation message is displayed.	As Expected			

(b) TC001_04_02: Test <alternate scenario of sequence diagram Password Recover (SD004)>

DESCRIPTION	VERSION	TEST DATE
Test alternative flow 1 of Password Recover Functionality in the system	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_04_02)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User clicks the "Forgot Password" link on the login page.	Click on Forgot Password link	Redirected to the password recovery page.	As Expected			
2	User enters the registered email address and submit the recovery request.	Invalid email: user@mail.	A1 : Invalid email address entered. System displays an error message: "Email address not found. Please try again." User is prompted to re-enter a valid email.	As Expected			
3	User opens the recovery email and follows the link to reset the password.	Click link given from email	Redirected to the password reset page.	As Expected			
4	User enters and confirms the new password. Submit the form.	Click the "Submit" button.	Password is reset successfully. Confirmation message is displayed.	As Expected			

(c) TC001_04_03: Test <alternate scenario of sequence diagram Password Recover (SD004)>

DESCRIPTION	VERSION	TEST DATE
Test alternative flow 2 of Password Recover Functionality in the system	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_04_03)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User clicks the "Forgot Password" link on the login page.	Click on Forgot Password link	Redirected to the password recovery page.	As Expected			
2	User enters the registered email address and submit the recovery request.	User cancels the recovery process.	A2 : User exits the process, and no action is taken.	As Expected			

(d) TC001_04_04: Test <exception scenario of sequence diagram Password Recover (SD004)>

DESCRIPTION	VERSION	TEST DATE
Test exception flow of Password Recover Functionality in the system	4.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_04_04)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
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1	User clicks the "Forgot Password" link on the login page.	Click on Forgot Password link	Redirected to the password recovery page.	As Expected			
2	User enters the registered email address and submit the recovery request.	Registered email: user@example.com	E1: Email delivery failure. System displays an error message: "Unable to send password recovery email. Please try again later." User is prompted to retry.	As Expected			
3	User opens the recovery email and follows the link to reset the password.	Click link given from email	E3: Password reset link expired. System displays an error message: "The password reset link has expired. Please initiate a new password recovery process."	As Expected			
4	User enters and confirms the new password. Submit the form.	Click the "Submit" button.	E2: No internet connection. System displays an error message: "Unable to connect. Please check your internet connection." User cannot proceed with password recovery until the connection is restored.	As Expected			

8.1.5 TC001_05: Test <state scenario of sequence diagram logout (SD005)>

(a) TC001_05_01: Test <normal scenario of sequence diagram logout (SD005)>

DESCRIPTION	VERSION	TEST DATE
Test the normal flow of LogOut Functionality in the system	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_05_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the menu or settings section of the application.	Open settings page	Settings page is displayed successfully.	As Expected			
2	User selects the "Log Out" option.	Click on "Logout" button	System prompts the user for confirmation: "Are you sure you want to log out?"	As Expected			
3	User confirms the log-out action.	Click "yes" on the confirmation	System terminates the user session and clears temporary data associated with the session and redirects to the login page or home screen.	As Expected			

(b) TC001_05_02: Test <alternative scenario of sequence diagram logout (SD005)>

DESCRIPTION	VERSION	TEST DATE
Test the LogOut Functionality in the system	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_05_02)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the menu or settings section of the application.	Open settings page	Settings page is displayed successfully.	As Expected			
2	User selects the "Log Out" option.	Click on "Logout" button	System prompts the user for confirmation: "Are you sure you want to log out?"	As Expected			
3	User confirms the log-out action.	Click cross on the confirmation	A1: User cancels the log-out action. User chooses "Cancel" on the confirmation prompt. System retains the user's session, and no changes are made.	As Expected			

(c) TC001_05_03: Test <exception scenario of sequence diagram logout (SD005)>

DESCRIPTION	VERSION	TEST DATE
Test the normal flow of LogOut Functionality in the system	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Koo Xuan	

Test Case ID (TC001_05_03)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the menu or settings section of the application.	Open settings page	Settings page is displayed successfully.	As Expected			
2	User selects the "Log Out" option.	Click on "Logout" button	System prompts the user for confirmation: "Are you sure you want to log out?"	As Expected			
3	User confirms the log-out action.	Click "yes" on the confirmation	E1: System error during log out. System displays an error message: "Unable to log out at this time. Please try again later." User session remains active until the issue is resolved. E2: No internet connection (if relevant). System stores the log-out request locally and terminates the session when connectivity is restored.	As Expected			

8.2 TC002: Test <Name of Package 2> Subsystem: <Name of Use Case (UC002)>

This test contains the following test cases:

- (a) TC002_01: Test <Scenario of sequence diagram4 (SD004)>**
- (b) TC002_02: Test <Scenario of sequence diagram5 (SD005)>**
- (c) ...**

8.2.1 TC002_01: Test Browse Knowledge Units

(a) TC002_01_01: Test normal scenario Browse Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to browse available knowledge units.	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Knowledge Units" section from the dashboard or main menu.	User clicks on the "Knowledge Units" section in the main menu.	The "Knowledge Units" section is displayed.	As Expected			
2	System displays a list of knowledge units, categorized by topics (e.g., stress management, mindfulness, coping strategies)	None (system action).	Knowledge units categorized by topics are displayed.	As Expected			
3	User scrolls through the list or uses filters/search functionality to find specific knowledge units.	User scrolls or applies filters (e.g., "Mindfulness").	Filtered or scrolled list of knowledge units is displayed.	As Expected			
4	User selects a knowledge unit to view details, such as:	User taps on a specific	Details of the selected knowledge	As Expected			

	• Title and description of the unit. • Content type (e.g., article, video, guided exercise).	knowledge unit.	unit are displayed.				
5	User clicks on the selected unit to access the full content.	User clicks "View Content" button.	The full content of the knowledge unit is displayed.*	As Expected			
6	System loads the content and displays it in a readable/viewable format	None	Content is displayed in a readable/viewable format.	As Expected			

(b) TC002_01_02: Test alternative scenario Browse Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to browse available knowledge units.	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	No knowledge units available.	None	System displays a message: "No knowledge units are currently available. Please check back later."	As Expected			
2	Redirects user to the dashboard or previous screen.	None	User is redirected to the dashboard or previous screen.	As Expected			

(c) TC002_01_03: Test exceptional scenario Browse Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to browse available knowledge units.	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	No internet connection.	None	System displays an error message: "Unable to load knowledge units. Please check your internet connection."	As Expected			
2	User cannot browse knowledge units until connectivity is restored.	None	User is prevented from browsing knowledge units until internet connectivity is restored.	As Expected			

8.2.2 TC002_02: Test Bookmark Knowledge Units

(a) TC002_02_01: Test normal scenario Bookmark Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to bookmark knowledge units	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Knowledge Units" section from the dashboard or main menu.	User clicks on the "Knowledge Units" section in the main menu.	The "Knowledge Units" section is displayed.	As Expected			
2	User selects a specific knowledge unit from the displayed list or search results.	User selects a knowledge unit by clicking on its title	The selected knowledge unit's details are displayed.	As Expected			
3	System displays details of the selected knowledge unit.	None	Details such as title, description, and content type are displayed.	As Expected			
4	User clicks the "Bookmark" button/icon associated	User clicks the "Bookmark" button.	The system marks the unit as bookmarked.	As Expected			

	with the knowledge unit						
5	System marks the unit as bookmarked and saves it under the user's account.	None	Bookmarking status is updated, and the unit is added to the user's bookmarks.				
6	System provides confirmation feedback and displays "Knowledge unit bookmarked successfully."	None	A confirmation message is displayed.				
7	The bookmarked unit is added to the "Bookmarked Units" section for quick access.	None	The knowledge unit appears in the "Bookmarked Units" section.				

(b) TC002_02_02: Test alternative scenario Bookmark Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to bookmark knowledge units	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User un-bookmarks a knowledge unit.	User clicks the "Unbookmark" button/icon on a previously bookmarked unit.	The system removes the unit from the user's bookmarks and updates the status.	As Expected			
2	System removes the unit from the user's bookmarks and updates the status.	None	The unit is no longer displayed in the "Bookmarked Units" section.	As Expected			
3	System provides confirmation feedback, e.g., "Knowledge unit removed from bookmarks."	None	A confirmation message is displayed to the user	As Expected			

(c) TC002_02_03: Test exceptional scenario Bookmark Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to bookmark knowledge units	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	No internet connection.	None	System displays an error message: "Unable to bookmark knowledge unit. Please check your internet connection."	As Expected			
2	The unit is not bookmarked until connectivity is restored.	None	Bookmarking is unsuccessful, and the user must retry once the internet connection is restored.	As Expected			

8.2.3 TC002_03: Test Rate Knowledge Units

(a) TC002_03_01: Test normal scenario Rate Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to rate knowledge units	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Knowledge Units" section.	User opens the app and navigates to the "Knowledge Units" section.	The system displays a list of available knowledge units	As Expected			
2	User selects and opens a specific knowledge unit.	User clicks on the desired knowledge unit.	The system retrieves and displays the content of the selected knowledge unit.	As Expected			
3	System displays the content of the knowledge unit along with a rating feature.	The selected knowledge unit is loaded.	The content is displayed, including a rating mechanism (e.g., star-based system or like/dislike buttons).	As Expected			
4	User selects a rating (e.g., 1–5 stars or thumbs	User chooses a specific rating.	The system processes the rating input.	As Expected			

	up/thumbs down).						
5	System records the user's rating and saves it to the database.	User's selected rating (e.g., 4 stars).	The rating is successfully stored in the database for the specific knowledge unit.				
6	System displays confirmation feedback.	User completes the rating action.	A message such as "Your rating has been submitted" is displayed.				

(b) TC002_03_02: Test alternative scenario Rate Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to rate knowledge units	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User changes their rating	User navigates back to the rated knowledge unit. User selects a new rating.	The system updates the rating in the database. Confirmation feedback, e.g., "Your rating has been updated," is displayed.	As Expected			

2	User cancels rating:	User exits the knowledge unit without selecting a rating.	No rating is recorded in the database.	As Expected			
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(c) TC002_03_03: Test exceptional scenario Rate Knowledge Units

DESCRIPTION	VERSION	TEST DATE
Test to rate knowledge units	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	No internet connection	User submits a rating without an active internet connection	System displays an error message: "Unable to submit your rating. Please check your internet connection." The rating is not saved until the connection is restored.	As Expected			

8.3 TC003: Test <Skill-Building Toolbox> Subsystem: <Add to Toolbox>

List all test cases before providing the details for the first use case in module2. Include the sub-sections accordingly.

This test contains the following test cases:

(a) TC003_01: Test < Normal Scenario Add to Toolbox

DESCRIPTION	VERSION	TEST DATE
Test to add useful tools into their Toolbox	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	An option that allows users to add the skill to their Toolbox is displayed on the screen.	The system identifies a skill the user has read about and displays the "Add to Toolbox" option.	The "Add to Toolbox" option appears on the screen with a message prompting the user to take action.	As Expected			
2	Users are prompted to select the option to add the tool to their Toolbox or dismiss the message.	The user sees the prompt and makes a decision (Add or Dismiss).	The system registers the user's action: either adds the tool or dismisses the prompt.	As Expected			
3	If the user selects to dismiss the message, AF1 occurs.	The user clicks on the "Dismiss" option.	The tool is not added to the user's Toolbox.	As Expected			

4	The tool is added to the user's Toolbox.	The user selects the "Add to Toolbox" option.	The tool is saved in the user's Toolbox. The system confirms the addition, e.g., "Tool added to your Toolbox."	As Expected			
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(b) TC003_01_02: Test Alternative Scenario Add to Toolbox

DESCRIPTION	VERSION	TEST DATE
Test to add useful tools into their Toolbox	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	The tool is not added to the user's Toolbox	The user chooses to dismiss the "Add to Toolbox" option.	No changes are made to the user's Toolbox	As Expected			

(c) TC003_01_03: Test Exceptional Scenario Add to Toolbox

DESCRIPTION	VERSION	TEST DATE
Test to add useful tools into their Toolbox	3.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	No internet connection	The user attempts to add the tool to their Toolbox without internet access.	The system displays an error message: “Unable to add tool to Toolbox. Please check your internet connection.” The tool is not added until connectivity is restored	As Expected			

(a) TC003_02_01: Test Normal Scenario Remove from Toolbox

DESCRIPTION	VERSION	TEST DATE
Test to remove tools from my Toolbox	1.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	User navigates to the "Toolbox" page.	User selects the "Toolbox" option from the navigation menu	The system displays the "Toolbox" page, listing all the tools added by the user.	As Expected			
2	User selects the "edit Toolbox" option.	User clicks on the "Edit Toolbox" button.	The system displays the editable version of the Toolbox, allowing users to manage their tools.	As Expected			
3	User removes tools that are no longer relevant to them by clicking the "delete tool" button.	User clicks the "Delete Tool" button next to a specific tool they want to remove.	The tool is removed from the Toolbox. The system confirms the deletion with a message such as "Tool has been removed from your Toolbox."	As Expected			

(b) TC003_02_02: Test Exception Scenario Remove from Toolbox

DESCRIPTION	VERSION	TEST DATE
Test to remove tools from my Toolbox	2.0	9-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	HAFIZ	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	No internet connection	User tries to remove a tool without an active internet connection.	The system displays an error message: "Unable to remove tool from Toolbox. Please check your internet connection." The tool is not removed until the internet connection is restored.	As Expected			

8.3.3 TC003_03: Test Evolve Tools

(a) TC003_03_01: Test normal scenario of Evolve Tools use case.

DESCRIPTION	VERSION	TEST DATE
Test the evolve tools functionality	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC003_03_01)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FA IL	Teste d By	Tester Commen ts
1	User navigates to the "Toolbox" section from the navigation bar.	User uses "Simple Breathing" 5 times.	After the 5th try, the system shows a prompt to evolve the tool.	As Expecte d			
2	Agree to evolve the tool by clicking the relevant button.	User taps on "Evolve" to evolve the tool.	The tool is evolved and the difficulty is increased the next time the user uses the tool.	As Expecte d			

(b) TC003_03_02: Test <normal scenario of sequence diagram1 (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test the flow when the evolve tools message is dismissed.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC003_03_02)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FAIL	Tested By	Tester Comments
1	User navigates to the "Toolbox" section from the navigation bar.	User uses "Simple Breathing" 5 times.	After the 5th try, the system shows a prompt to evolve the tool.	As Expected			
2	Disagree to evolve the tool by clicking the relevant button.	User taps on "No" to dismiss tool evolution.	The tool is not evolved.	As Expected			

(c) TC003_03_03: Test <normal scenario of sequence diagram1 (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test the flow when the evolve tools message is dismissed with no internet connection.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC003_03_03)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FA IL	Teste d By	Tester Commen ts
1	User navigates to the "Toolbox" section from the navigation bar.	User uses "Simple Breathing" 5 times.	After the 5th try, the system shows a prompt to evolve the tool.	As Expecte d			
2	Connectivity is turned off, attempt to evolve tool.	User turns off any connectivit y and taps on "Evolve".	Error message displays: "Unable to evolve tool. Please check your Internet connection."	As Expecte d			

8.4 TC004: Test Motivation and Rewards

List all test cases before providing the details for the first use case in module2. Include the sub-sections accordingly.

This test contains the following test cases:

- (a) TC004_01: Test Earn Points**
- (b) TC004_02: Test Spend Points**
- (c) TC004_02: Test Receive Badges**

8.4.1 TC004_01: Test Earn Points

(a) TC004_01_01: Test normal sequence of Earn Points use case

DESCRIPTION	VERSION	TEST DATE
Test that points are correctly earned after a single activity.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC004_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FAIL	Tested By	Tester Comments
1	User navigates to the "Toolbox" section from the navigation bar.	User uses "Simple Breathing".	System adds 10 points to user's total points.	As Expected			
2	User navigates to the "Knowledge Unit" section from the navigation bar.	User taps on "Simple Anxiety Relief Techniques" knowledge unit and completes the knowledge unit.	System adds 5 points to user's total points.	As Expected			

(b) TC004_01_02: Test exception sequence of Earn Points use case

DESCRIPTION	VERSION	TEST DATE
Test that error checking occurs when there is no connectivity.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC004_01_02)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FAIL	Tested By	Tester Comments
1	User navigates to the "Toolbox" section from the navigation bar while there is no connectivity.	User uses "Simple Breathing".	System is unable to add the points; points are stored locally and updated when connection is restored.	As Expected			
2	User navigates to the "Knowledge Unit" section from the navigation bar while there is no connectivity.	User taps on "Simple Anxiety Relief Techniques" knowledge unit and completes the knowledge unit.	System is unable to add the points; points are stored locally and updated when connection is restored.	As Expected			

(c) TC004_01_03: Test alternate flow of Earn Points use case

DESCRIPTION	VERSION	TEST DATE
Test that points are correctly earned after multiple back-to-back activities.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC004_01_03)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FAIL	Tested By	Tester Comments
1	User navigates to the "Toolbox" section from the navigation bar while there is no connectivity.	User use "Simple Breathing" and "Guided Journaling" .	System adds 10 points to user's total points. System adds another 10 points to user's total points.	As Expected			
2	User navigates to the "Knowledge Unit" section from the navigation bar while there is no connectivity.	User taps on "Simple Anxiety Relief Techniques" knowledge unit and completes the knowledge unit.	System adds 5 points to user's total points. System adds another 5 points to user's total points.	As Expected			

8.4.2 TC004_02: Test Spend Points

(a) TC004_02_01: Test normal scenario of Spend Points use case

DESCRIPTION	VERSION	TEST DATE
Test that users can spend points to customize their avatar.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC004_02_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	User tries to open avatar customization.	User taps "Profile" from navigation bar and selects "Customize Avatar".	Avatar customization screen displayed.	As Expected			
2	Try to purchase a customization.	User taps on "Colourful Hat" and confirms purchase by clicking on "Purchase Item".	System deducts points and customization is displayed on the avatar.	As Expected			

(b) TC004_02_02: Test exception case for Spend Points use case

DESCRIPTION	VERSION	TEST DATE
Test that there is an error message when there are insufficient points for customization.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC004_02_02)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FA IL	Tested By	Tester Comments
1	User tries to open avatar customization.	User taps "Profile" from navigation bar and selects "Customize Avatar".	Avatar customization screen displayed.	As Expected			
2	Try to purchase a customization with insufficient points.	User taps on "Colourful Hat" and confirms purchase by clicking on "Purchase Item".	System displays error: "Insufficient points".	As Expected			

8.4.3 TC004_03: Test Receive Badges

(a) TC004_03_01: Test normal scenario of Receive Badges use case

DESCRIPTION	VERSION	TEST DATE
Test that badges are awarded upon milestone completion.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FA IL	Teste d By	Tester Commen ts
1	User completes 3 different lessons to receive the 3 lesson milestone award.	User taps on "Knowledge" from the navigation bar and completes 3 different lessons.	User receives a display message for new badge awarded; badge can be viewed from profile.	As Expecte d			

8.5 TC005: Test Admin Management Module

This test contains the following test cases:

- (a) TC003_01: Test Add Knowledge Unit**
- (b) TC003_02: Test <Scenario of sequence diagram7 (SD007)>**
- (c) ...**

8.5.1 TC005_01: Test Add Knowledge Unit

(a) TC005_01_01: Test normal scenario of Add Knowledge Unit

DESCRIPTION	VERSION	TEST DATE
Verify that admins can successfully add new knowledge units.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC005_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FAIL	Tested By	Tester Comments
1	Input knowledge unit details.	Administrator completes knowledge unit details so that the fields are populated correctly.	Submission can be completed; confirmation is shown.	As Expected			

(b) TC005_01_02: Test exception scenario of Testing Knowledge Unit

DESCRIPTION	VERSION	TEST DATE
Test that there is error checking when submitting a knowledge unit.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC005_01_02)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FA IL	Teste d By	Tester Commen ts
1	Input knowledge unit details incompletely.	Administra tor fills in knowledge unit details incomple te ly.	System highlights missing fields and displays “This field is required”.	As Expecte d			

(c) TC005_01_03: Test exception scenario of Add Knowledge Unit

DESCRIPTION	VERSION	TEST DATE
Test that there is error checking when submitting knowledge unit.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC005_01_03)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FA IL	Teste d By	Tester Commen ts
1	Input knowledge unit details completely and upload invalid file format.	Administra tor fills in knowledge unit details completely and upload an “.exe” file.	System rejects the upload and displays: “Invalid file format. Please upload a valid file type.”	As Expecte d			

(d) TC005_01_04: Test exception scenario of Add Knowledge Unit

DESCRIPTION	VERSION	TEST DATE
Test that there is error checking when submitting a knowledge unit.	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Phavanee	

Test Case ID (TC005_01_04)	Test Steps	Input Data	Expected Results	Actual Results	PAS S/FA IL	Teste d By	Tester Commen ts
1	Input knowledge unit details completely with no connectivity.	Administra tor fills in knowledge unit details completely .	System displays: "Unable to process your request. Please check your Internet connection."	As Expecte d			

8.5.2 TC005_02: Test Resolve flagged issue

(a) TC005_02_01: Test normal scenario of Resolve Flagged Issue use case

DESCRIPTION	VERSION	TEST DATE
Verify successful resolution of a flagged issue	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Admin navigates to the "Resolve Flagged Issues" section.	Admin logs into the system and accesses the "Resolve Flagged Issues" section through the system menu.	System displays a list of flagged issues.	As Expected			
2	Admin selects an issue and reviews its details	Admin clicks on a flagged issue to open the details page for review.	Details of the flagged issue are displayed.	As Expected			
3	Admin edits the content or marks the issue as resolved	Admin modifies the flagged content or changes the issue's status to "Resolved"	System updates the issue status to "resolved" in the database	As Expected			

		using the provided interface options.					
4	Admin is notified of the successful resolution	System automatically triggers a notification after the admin confirms the resolution action.	Notification is displayed: "Issue successfully resolved"	As Expected			

b) TC005_02_02: Test <normal scenario of sequence diagram16 (SD016)>

DESCRIPTION	VERSION	TEST DATE
Verify dismissal of an invalid flagged issue	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Admin navigates to the "Resolve Flagged Issues" section.	Admin logs into the system and selects "Resolve Flagged Issues" from the navigation menu.	System displays a list of flagged issues.	As Expected			

2	Admin selects a flagged issue marked as invalid.	Admin clicks on the flagged issue labeled to open its details.	Details of the flagged issue are displayed.	As Expected			
3	Admin clicks the "Dismiss Flag" button.	Admin clicks on the "Dismiss Flag" option to change the status of the flagged issue to "Dismissed."	System updates the flagged issue status to "dismissed" in the database.	As Expected			
4	Admin is notified of the successful resolution	Admin confirms the dismissal action and receives a system notification about the update.	System updates the flagged issue status to "dismissed" in the database.	As Expected			

c) TC005_02_03: Test <normal scenario of sequence diagram16 (SD016)>

DESCRIPTION	VERSION	TEST DATE
Verify system behavior with no internet connection	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC00	Test Steps	Input Data	Expected Results	Actual Results	PASS/	Tested By	Tester Comments
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1_01_01)					FAIL		
1	Admin navigates to the "Resolve Flagged Issues" section.	Admin logs into the system and attempts to access the "Resolve Flagged Issues" section while the internet connection is disabled.	System displays an error message indicating no internet connection.	As Expected			
2	Admin attempts to edit or dismiss the flagged issue.	Admin selects a flagged issue and tries to edit or dismiss it while the internet connection is disabled.	The system prevents updates and displays the error	As Expected			

d) TC005_02_03: Test <normal scenario of sequence diagram16 (SD016)>

DESCRIPTION	VERSION	TEST DATE
Verify system behavior during a system error	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC00	Test Steps	Input Data	Expected Results	Actual Results	PASS/	Tested By	Tester Comments
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1_01_01)					FAIL		
1	Admin navigates to the "Resolve Flagged Issues" section.	Admin logs into the system and selects "Resolve Flagged Issues" from the main menu to load the flagged issues list.	System attempts to load the flagged issues list.	As Expected			
2	A system error occurs while loading or updating a flagged issue.	Admin attempts to interact with the flagged issue list when a system error occurs.	System displays an error message prompting the admin to retry.	As Expected			
3	Admin retries the operation.	Admin selects "Retry" after the error message is displayed to reload or update the flagged issue.	System successfully loads or updates the flagged issue.	As Expected			

8.5.3 TC005_03: Test View Diagnostic Trends

a) TC005_03_01: Test normal scenario of View Diagnostic Trends

DESCRIPTION	VERSION	TEST DATE
Verify the successful loading of the diagnostic trends dashboard	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Admin navigates to the "Monitor Diagnostic Trends" section.	Admin logs into the system and selects "Monitor Diagnostic Trends" from the main dashboard menu.	System loads the diagnostic trends dashboard.	As Expected			
2	Admin applies a filter (e.g., time range or demographics).	Admin selects a filter option, such as a specific time range or demographic group	System displays diagnostic trends based on the selected filter.	As Expected			
3	Admin views the displayed trends and insights.	Admin reviews the diagnostic trends and	Insights and trends are displayed	As Expected			

		insights shown on the dashboard, ensuring the data corresponds to the selected filter criteria.	correctly based on the selected filter criteria.				
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b) TC005_03_02: Test alternate scenario of View Diagnostic Trends

DESCRIPTION	VERSION	TEST DATE
Verify the system behavior with no data for the selected filters	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Admin navigates to the "Monitor Diagnostic Trends" section.	Admin logs into the system and selects "Monitor Diagnostic Trends" from the main dashboard menu.	System loads the diagnostic trends dashboard.	As Expected			

2	Admin applies filters that do not match any available data..	Admin selects a filters	System displays a "No data available" message.	As Expected			
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c) TC005_03_03: Test exception scenario of View Diagnostic Trends

DESCRIPTION	VERSION	TEST DATE
Verify system behavior with no internet connection	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Admin navigates to the "Monitor Diagnostic Trends" section.	Admin logs into the system and selects "Monitor Diagnostic Trends" from the main dashboard menu.	System displays a "Cannot load dashboard" error message.	As Expected			
2	Admin attempts to retry after	Admin restores the network	System reloads the dashboard	As Expected			

	restoring the connection.	connection and retries loading the dashboard.	successfull y.				
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8.6 TC006: Test Content Creation Subsystem

8.6.1 TC006_01: Test <Create Knowledge Unit>

a) TC006_01_01: Test <normal scenario of sequence diagram18 (SD018)>

DESCRIPTION	VERSION	TEST DATE
Verify successful creation of a knowledge unit	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	User navigates to the "Create Knowledge Units" section.	User logs into the system and selects "Create Knowledge Units" from the main menu.	System displays the knowledge unit creation form.	As Expected			
2	User inputs all required details	User fills in all required fields on the knowledge unit creation form.	System accepts the details without error.	As Expected			
3	User previews the knowledge unit and submits it.	User previews the knowledge unit and clicks "Submit" to finalize it.	System displays a confirmation message and saves the knowledge unit.	As Expected			

4	System notifies the admin for approval (if required).	System triggers a notification to the admin regarding the submitted knowledge unit for approval.	Admin receives a notification about the submitted knowledge unit.				
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b) TC006_01_02: Test <normal scenario of sequence diagram18 (SD018)>

DESCRIPTION	VERSION	TEST DATE
Verify system behavior when required fields are missing	1.0	16-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Athirah	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	User navigates to the "Create Knowledge Units" section.	User logs into the system and selects "Create Knowledge Units" from the main menu.	System displays the knowledge unit creation form.	As Expected			
2	User leaves one or more required fields blank	User fills in the form and leaves one or more required fields blank.	System highlights the incomplete fields and prompts	As Expected			

			the user to make corrections.				
3	User fills in the missing fields and submits the form	.User enters the missing information in the required fields and clicks "Submit."	System successfully accepts and saves the knowledge unit.	As Expected			

9. Traceability Matrix

Table 9.1: Example of Traceability Matrix for Self-Help System

Sprint No	Package Item	User Story ID	User Story Description	Sequence Diagram ID	Sequence Diagram Description	Test Case ID
Sprint 1	Package 1	US_001	User Registration	SD_001	Illustrates the steps for user registration.	TC_001
		US_002	User Login	SD_002	Depicts the user login process.	TC_002
		US_003	Manage Profile	SD_003	Shows the process for managing a user profile.	TC_003
		US_004	Password Recovery	SD_004	Details the password recovery process	TC_004
		US_005	Log Out	SD_005	Illustrates the steps to log out of the system.	TC_005
Sprint 2	Package 2	US_006	Browse Knowledge Units	SD_006	Displays the process of browsing knowledge units.	TC_006
		US_007	Bookmark Knowledge Units	SD_007	Illustrates bookmarking knowledge units.	TC_007
		US_008	Rate Knowledge Units	SD_008	Details the process of rating knowledge units.	TC_008
Sprint 3	Package 3	US_009	Add to Toolbox	SD_009	Describes adding tools to the	TC_009

					user's toolbox	
		US_010	Remove from Toolbox	SD_010	Shows removing tools from the user's toolbox.	TC_010
		US_011	Evolve Tools	SD_011	Details how users evolve tools in their toolbox.	TC_011
Sprint 4	Package 4	US_012	Earn Points	SD_012	Explains earning points through activities	TC_012
		US_013	Spend Points	SD_013	Describes spending points for customization	TC_013
		US_014	Earn Badges	SD_014	Details the process for earning badges.	TC_014
Sprint 5	Package 5	US_015	Add Knowledge Unit	SD_015	Illustrates adding a knowledge unit.	TC_015
		US_016	Resolve Flagged Issues	SD_016	Explains resolving flagged issues in the system.	TC_016
		US_017	Monitor Diagnostic Trends	SD_017	Shows monitoring diagnostic trends and patterns.	TC_017
Sprint 6	Package 6	US_018	Create Knowledge Units	SD_018	Describes creating and curating knowledge	TC_018

					units.	
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